# Notes on Price Theory

Scott Ashworth

September 21, 2017

# Contents

# Motivating Example

How will a person change the number of hours she works in response to a change in taxes and transfer payments?

Ask an economist a question like this, and she will automatically think in terms of models. She might, for example, offer the following analysis:

> Consider an agent endowed with 24 hours of time who can earn a wage of $w$ per hour of work. The government takes a fraction $\tau$ of her labor income in taxes, and also pays her a fixed transfer $T \geq 0$. She spends her post-tax-and-transfer income on a single consumption good. So, if she works $\ell$ hours, she can consume any amount $c$ with $c \leq (1-\tau)w\ell + T$.
>
> The agent chooses hours $\ell$ and consumption $c$ to maximize a function $u(c, \ell)$. For concreteness, let's take $u(c, \ell) = \log(c) + \log(24 - \ell)$. The choices solve:
>
> $$\max_{c,\ell} \ \log(c) + \log(24 - \ell) \quad \text{st} \ \ c \leq (1-\tau)w\ell + T.$$
>
> Since $u$ is strictly increasing in $c$, we know that the inequality in the constraint will hold with equality. Thus we can substitute into the maximand to get:
>
> $$\max_{\ell} \ \log\left((1-\tau)w\ell + T\right) + \log(24 - \ell).$$
>
> The solution depends on the parameters $(w, \tau, T)$:
>
> $$\ell^*(w, \tau, T) \quad \text{and} \quad c^*(w, \tau, T).$$

Being able to solve problems like this on your own is the most essential prerequisite for this course. Give it a shot before turning the page.

I hope you found:

$$\ell^*(w,\tau,T) = 12 - \frac{T}{2(1-\tau)w} \quad \text{and} \quad c^*(w,\tau,T) = 12(1-\tau)w + \frac{1}{2}T.$$

Now, back to our economist's answer:

> Increased transfer payments lead to fewer hours worked, since $\frac{\partial \ell^*}{\partial T} < 0$. If transfers are strictly positive, then a higher tax rate leads to fewer hours worked, since $\frac{\partial \ell^*}{\partial \tau} < 0$ in that case. But if transfers are 0, then the tax rate has no effect on hours worked.

Maybe you wanted to know about hours worked out of simple curiosity. But maybe not. Maybe your question was motivated by the hope that the answer would help you know what tax rate would be best. Our economist can help there as well.

She will define a new function $v$ by:

$$v(w,\tau,T) = u(c^*(w,\tau,T), \ell^*(w,\tau,T)).$$

And then she use this new function as a part of another maximization problem, one in which the choice variables include the tax and transfer.

All of this should raise several questions in your mind.

1. Why model the worker's decisions as the result of maximizing some function?

2. Granting that we should model decisions that way, why should the value of that function play a role in policy decisions?

3. How sensitive are the conclusions to the specific functional form we chose for $u$?

4. If they are sensitive, can we get useful conclusions with weaker assumptions?

Answering questions like these is the major goal of this course. Not surprisingly, this will involve doing a lot of math. On the one hand, we will make extensive use of the mathematical theory of optimization, using techniques both for characterizing solutions and for exploring how solutions change as parameters of the problem change. On the other hand, we will use the axiomatic method to explore how optimization problems are related to more intuitively acceptable descriptions of the problems that face economic agents. By doing both, we will pursue an important subsidiary goal, to show how all of the calculus we do while practicing economics can be interpreted in terms of human agency.

# Part I

# Foundations

# Chapter 1

# Rational Choice Theory

We want to consider a decision maker (DM for short) who chooses "rationally". The standard approach to modeling rational choice can be presented three ways:

1. DM chooses an alternative that is best according to a binary relation $\succsim$, with the interpretation "$a \succsim b$ if and only if DM likes $a$ at least as much as $b$".

2. DM chooses an alternative that maximizes a function $u$, called the utility function.

3. DM chooses according to some choice rule that satisfies a consistency assumption across different choice problems.

And we standardly impose assumptions on preference relations and choice rules that render all three approaches equivalent.

But they are not equivalent in terms of how we think about them. Utility maximization feels like a strange assumption, but is easy (relatively!) to work with. Consistent choice is just the opposite. And that's the value of having three different representations.

Here is how I think of the relationships of the three approaches.

We all have an intuitive account of human action—philosophers call it "folk psychology". A corner-stone is a claim like:

> People usually act to, more or less, satisfy their preferences, in light of their beliefs.

Decision theory formalizes an idealized version of this:

> People ~~usually~~ act to, ~~more or less,~~ satisfy their preferences, in light of their beliefs.

Beliefs will be mostly in the background until we talk about uncertainty explicitly.

## 1.1 The Standard Approach

The formalization starts with a decision maker and a set of alternatives $A$. DM has preferences described by a binary relation $\succsim$, with the interpretation that $a \succsim b$ means DM weakly prefers $a$ to $b$. From $\succsim$, we can derive two other relations on $A$:

1. The *strict preference* relation, $\succ$

$$a \succ b \iff a \succsim b \text{ but not } b \succsim a$$

   interpreted as "DM likes $a$ better than $b$."

2. The *indifference* relation, $\sim$

$$a \sim b \iff a \succsim b \text{ and } b \succsim a$$

   interpreted as "DM is indifferent between $a$ and $b$."

Now for the big assumptions. A preference relation is **rational** if satisfies two axioms

1. **Completeness**: For all $a, b \in A, a \succsim b$ or $b \succsim a$ (or both)

2. **Transitivity**: For all $a, b, c \in A,$ if $a \succsim b$ and $b \succsim c,$ then $a \succsim c$.

A *choice problem* is a nonempty subset $B \subset A$ of alternatives that the DM believes to be feasible. The **preference maximizing choices** are:

$$C^* (B, \succsim) = \{a \in B \mid a \succsim b, \text{ for all } b \in B\}$$

In words, $C^*$ takes the elements of $B$ and returns the subset that are most-preferred by DM. If $C^*(B, \succsim)$ is a singleton, then we assume DM chooses the single element. If it contains multiple elements the DM is indifferent among them, but prefers all elements of $C^*$ to any other element of $B$. In this case, we assume only that DM chooses some element of $C^*(B, \succsim)$.

So we see that rational choice theory has two parts: choices are made to (1) maximize a (2) rational preference relation.

The advantage of this approach is that the axioms are clear and easy to think about. Specifically, we can see how strong they are. For example, completeness means DM can always express a preference between two elements (even if that preference is indifference). DM is not permitted to say, "I don't know how to compare these things." That is, DM has done the internal reflection necessary to state a preference over all possible options. Note well, this does *not* mean DM is certain about how different actions will turn out. Rather, she certain about her preferences over the actions.

It's easiest to think about the implications of transitivity if we break it into parts.

**Theorem 1.1.** *If $\succsim$ is transitive, then $\succ$ and $\sim$ are both transitive.*

**Proof.** Exercise 1.3. ∎

Transitivity of strict preference seems quite reasonable. It means it is not possible to present DM with a sequence of pairwise choices that lead her preferences to cycle, that is, there are no $x, y, z \in A$ such that $x \succ y$, $y \succ z$ and $z \succ x$. However, there are simple thought experiments where transitivity is not terribly convincing. Imagine you have three possible marriage partners, $a$, $b$, and $c$, as follows:

1. $a$ sexier than $b$ sexier than $c$
2. $b$ smarter than $c$ smarter than $a$
3. $c$ richer than $a$ richer than $b$

If you prefer whichever potential mate who is better on two out of three dimensions, then your preferences are $a \succ b \succ c \succ a$.

Transitive indifference is more problematic. The key problem is the so-called "problem of just perceptible differences". Compare a cup of black coffee and a cup with one grain of sugar in it. Most people are indifferent. Consider a third cup with two grains of sugar. Again, most people are indifferent between cup 2 and cup 3. And so on. However, everyone certainly has a preference between cup 1 and cup 1,000,000.

While the assumptions of completeness and transitivity are not completely innocuous, we will maintain them for the rest of the course. A consequence of this is that our decision problems will actually have solutions, at least under favorable topological conditions. The next result is the simplest version of this.

**Theorem 1.2.** *Suppose $B$ is finite and $\succsim$ is rational. Then $C^*(B, \succsim)$ is nonempty.*

**Proof.** Exercise 1.6, part 1. ∎

In the preference-based approach, we start with preferences and use them to derive a choice rule. But it is also interesting to go in the other direction: start with a choice rule and derive preferences from it. Doing so lets us see what assumptions on (potentially) observable choices correspond to assuming that the unobservable preferences are rational.

The basic story is this: choices make up the preference maximizing choice rule for some rational preference if and only if those choices satisfy cross-decision-problem consistency conditions. Rubinstein handles the general case; here we will focus on a simple version.

First, we restrict attention to choice rules that always specify a unique alternative. (Such rules are said to be **resolute.**) Write $\mathcal{B}$ for the set of all non-empty subsets of $A$. Then a resolute choice rule is a function $C : \mathcal{B} \to A$ such that $C(B) \in B$ for all $B \in \mathcal{B}$.

A resolute choice rule $C$ is **contraction consistent** if, whenever $B$ and $D$ are subsets with $D \subset B$, we have $C(B) \in D$ implies $C(B) = C(D)$. Intuitively, removing unchosen alternatives does not affect the choice.

The next result shows the first part of a kind of equivalence between the preference-based approach and the choice function-based approach. (Exercise 1.7 asks you to show the second part of the equivalence.) Say that a choice function, $C$, and the choice function derived from preferences $\succsim$, $C^*(\cdot, \succsim)$, **agree for finite** $B$ if $C(B) = C^*(B, \succsim)$ whenever $B \subset A$ has only finitely many elements. (Note: this does not presuppose that $A$ is finite.)

**Theorem 1.3.** *Suppose $C$ is a contraction consistent, resolute choice rule. Then there is a rational preference $\succsim_C$ such that $C$ and $C^*(\cdot, \succsim_C)$ agree for finite $B$.*

**Proof.** The first step is to come up with the preference relation. Define $a \succsim_C b$ if $C(\{a, b\}) = a$.

Next, we show that this preference is rational. $C(\{a, b\})$ is either $a$ or $b$. In the first case, we have $a \succsim_C b$. In the second, we have $b \succsim_C a$. Since at least one obtains, $\succsim_C$ is complete.

Next we show that this preference is transitive. Suppose $x \succsim_C y$ and $y \succsim_C z$. From the definition, this means $C(\{x, y\}) = x$ and $C(\{y, z\}) = y$. Consider $C(\{x, y, z\})$. It can't be $y$, because if it were, contraction consistency would force $C(\{x, y\}) = y$. And it can't be $z$, because if it were, contraction consistency would force $C(\{y, z\}) = z$. Thus $C(\{x, y, z\}) = x$. But then contraction consistency gives $C(\{x, z\}) = x$, and that implies $x \succsim_C z$.

Finally, we show that $C = C^*(\cdot, \succsim_C)$. We argue by contradiction: suppose there is a finite subset $B$ such that $C(B) \neq C^*(B, \succsim_C)$. By Theorem 1.2, this means that $C(B) = x$ and $C^*(B, \succsim_C)$ contains some $y \neq x$. The second of these implies $y \succsim_C x$, but the first and contraction consistency imply $C(\{x, y\}) = x$, contradicting the definition of $\succsim_C$. ■

The preference relation defined in the proof, $\succsim_C$, is an example of a *revealed* preference. This idea will come back in Section 4.3, where is will even be related directly to empirical work.

The last result linked preferences (which are good for thinking) and choices (which are good for observing). The next one relates preferences to utility (which is good for calculating).

A function $u : A \to \mathbb{R}$ **represents** $\succsim$ if, for every $a, b \in A$, $u(a) \geq u(b)$ if and only if $a \succsim b$. Such a function is sometimes called a **utility function.** It assigns a numerical value to each element in $A$, ranking them numerically in accordance with the individual's preferences. This is useful because maximizing a real-valued function is an easy way to determine most preferred elements of $A$.

Utility functions that represent a preference relation $\succsim$ are not unique—any strictly increasing transformation still represents same preference relation. Formally, consider any strictly increasing function $f : \mathbb{R} \to \mathbb{R}$. Then $v(x) = f(u(x))$ is new utility function representing the same preferences as $u(\cdot)$. To see this, note that

$$\begin{aligned} & v(a) \geq v(b) \\ \Leftrightarrow \quad & f(u(a)) \geq f(u(b)) \\ \Leftrightarrow \quad & u(a) \geq u(b). \end{aligned}$$

Properties of utility functions that are invariant to any strictly increasing transformation are called **ordinal.** The ordinal properties are exactly the ones that are meaningful in terms of preferences. Properties that are *not* ordinal include magnitude (or intensity) of preference—there is no difference between the comparison of 100 to 0 and the comparison of 1 to 0. This is very different from the classical utilitarian concept of utility from Bentham and Mill.

**Theorem 1.4.** *Suppose $\succsim$ is represented by the function $u$. Then $\succsim$ is rational*

**Proof.** Exercise 1.4. ■

To further developing the formal relationship between preferences and utility, we need a technical result. Say that an element $a \in X$ is **$\succsim$-minimal in** $X$ if $x \succsim a$ for all $x \in X$.

**Lemma 1.1.** *Suppose $A$ is finite and $\succsim$ is rational. Then every non-empty subset $X \subset A$ has a $\succsim$-minimal element.*

**Proof.** We proceed by induction on the number of elements of $X$. If $X$ has a single element, then the claim follows from completeness. So assume that the claim is true for any $X'$ with $n$ elements, and let $X$ have $n+1$ elements. Choose an arbitrary $x \in X$, and consider the set $X' = X \setminus \{x\}$. This set has $n$ elements, so the inductive hypothesis tells us that it has a $\succsim$-minimal element—call it $y$. By completeness, we have either $x \succsim y$ or $y \succsim x$. In the first case, $y$ is $\succsim$-minimal in $X$. In the second case, transitivity implies $x$ is $\succsim$-minimal in $X$. Either way, $X$ has a $\succsim$-minimal element. ■

**Theorem 1.5.** *Suppose $A$ is finite. A preference relation, $\succsim$, can be represented by a utility function if $\succsim$ is rational.*

**Proof.** We start by iteratively constructing subsets of $A$. Let $X_1$ be the set of $\succsim$-minimal elements of $A$. This is non-empty by Lemma 1.1. Now assume we have constructed sets $X_1, \ldots, X_k$. If $X_1 \cup \cdots \cup X_k = A$, then we are done. Otherwise, the set $A \setminus (X_1 \cup \cdots \cup X_k)$ is non-empty, and by the Lemma, has a $\succsim$-minimal element. Let $X_{k+1}$ be the set of all these $\succsim$-minimal elements. Notice that this construction will stop after at most $n$ steps, where $n$ is the number of elements of $A$.

Define $u(x) = k$ if $x \in X_k$. I claim that $u$ represents $\succsim$. To see this, consider $a \succ b$. Transitivity implies $a \notin X_1 \cup \cdots \cup X_{u(b)}$. Thus $u(a) > u(b)$. And if $a \sim b$, then $a$ is $\succsim$-minimal if and only if $b$ is, so $u(a) = u(b)$. ■

You'll notice that our result on the existence of a utility representation required $A$ to be finite. This is disappointing, since in applications we want to use calculus, which doesn't even make sense on finite $A$. But we really do need to go beyond just rationality to handle all of the $A$ that we would like.

For an example of what can go wrong, let $A = [0, 1] \times [0, 1]$. The **lexicographic preference** on $A$ is defined as follows:

$$(x_1, x_2) \succsim (y_1, y_2) \Leftrightarrow \begin{cases} x_1 > y_1 \\ \text{or} \\ x_1 = y_1 \ \text{ and } \ x_2 \geq y_2 \end{cases}$$

You can verify that these preferences are rational. But:

**Proposition 1.1.** *There does not exist any utility representation of lexicographic preferences.*

**Proof.** We will use two facts from mathematics:

1. If $x$ and $y$ are any real numbers with $x > y$, then there is a rational number $q$ with $x > q > y$.

2. There does not exist any function $f$ from $[0, 1]$ to the rational numbers such that $x \neq y$ implies $f(x) \neq f(y)$.

The proof of the proposition is by contradiction: we assume there is a utility representation and use it along with fact 1 to construct a function $q : [0,1] \to \mathbb{Q}$ such that $x \neq y$ implies $q(x) \neq q(y)$. Since fact 2 says that is impossible, we know that there cannot be a utility representation after all.

Now for the details. Suppose $u$ is a utility representation of lexicographic preferences. For every $x \in [0,1]$, we have $u(x,1) > u(x,0)$. Fact 1 tells us there is a rational number $q(x)$ such that $u(x,1) > q(x) > u(x,0)$. If $x > y$, we have

$$q(x) > u(x,0) > u(y,1) > q(y),$$

so $q(x) \neq q(y)$. ■

Preferences are **continuous** if, for any sequences of bundles $x^n \to x$ and $y^n \to y$, if $x^n \succsim y^n$ for all $n$, then $x \succsim y$. A useful way to see what this means is to see that it rules out lexicographic preferences. Consider the bundle $x = (0,1)$ and the sequence $(y^n) = (1/n, 0)$. For every $n$, we have $y^n \succ x$, because $1/n > 0$. But the sequence has limit $y = (0,0)$, and $x \succ y$. In this case, the preference switched from strict one way to strict the other way without ever passing through indifference along the way. Continuity rules that out.

This is just what we need to get a utility representation.

**Theorem 1.6** (Debreu)**.** *Preferences are complete, transitive, and continuous if and only if they are represented by a continuous utility function.*

## 1.2 Problems and Alternatives

The standard approach to preferences, choices, and utility has been extremely fruitful for economics and other social sciences. But not everything that we might want to model can be captured in the standard framework.

Here is an example, from an experiment conducted by Kahneman and Tversky. In one arm of the experiment, subjects were told:

> Imagine that the U.S. is preparing for the outbreak of an unusual Asian disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Assume that the exact scientific estimate of the consequence of the program are as follows:
>
> - If program A is adopted, 200 people will be saved.
> - If program B is adopted, there is 2/3 probability that no one will be saved, and 1/3 probability that 600 people will be saved.

72% of subjects reported that program A was best.

In the second arm, subjects were told:

> Imagine that the U.S. is preparing for the outbreak of an unusual Asian disease, which is expected to kill 600 people. Two alternative programs to combat the disease have been proposed. Assume that the exact scientific estimate of the consequence of the program are as follows:
>
> - If program C is adopted, 400 people will die with certainty.
> - If program D is adopted, there is 2/3 probability that 600 people will die, and 1/3 probability that no one will die.

78% of subjects reported that program D was best.

Why is this a problem for the standard approach? Well, A and C are the *same* program, as are B and D. So the experimental results say that identical alternatives will be treated differently depending on how they are described. (Kahneman and Tversky call this a **framing effect.**) The problem for the standard approach is that that approach ignores descriptions altogether.

Psychologists are good at cooking up experiments like this to falsify just about any assumption you might want to make about decision making. Opinions are divided about how to respond. One camp holds that the deviations from the standard approach are not a big deal, and the standard approach is a good approximation for applications in the social sciences. Another camp holds that the deviations are important for applications, and that we need improved models that can accommodate them. Much recent work in behavioral economics has tried to assess the importance of deviations in field, rather than lab, data.

The standard approach can be called into question by thought experiments just as much as by actual experiments. Suppose a choice function defined on $\{x, y, z\}$ fails contraction consistency:

$$C(\{x, y, z\}) = \{x\} \quad \text{but} \quad C(\{x, y\}) = \{y\}.$$

Before you declare this choice function irrational, two people speak up.

Ann says:

> That is my choice function, and it is rational. The alternatives are $x =$ duck, $y =$ chicken, and $z =$ frog legs. I prefer duck to chicken exactly when the chef is well trained. And I learn about the quality of the chef from the menu. If she cooks frog legs, she is probably well trained. But if she doesn't, I'd rather be safe and order chicken.

Ann learns information relevant to her decision from the feasible set.

Bob says:

> That is my choice function, and it is rational. The alternatives are cookies: $x$ = chocolate-chip, $y$ = oatmeal-raisin, and $z$ = peanut-butter. My parents taught me that it's rude to take the best desert when others have to choose latter, so I always choose my second favorite.

Bob has a complete and transitive order on alternatives, but chooses the second best from any feasible set.

These examples suggest allowing preferences to depend on the feasible set, or even on the way it is described. But we don't want to go too far with this—we still want to say that some observations are ruled out by the model. Coming up with disciplined ways to incorporate choice-set dependence into preferences is an active area of current research. Here, we'll just look quickly at one approach.

The model we are about to develop is motivated by empirical evidence that default options often influence choices. One example comes from Sweden's public pension reform, passed in 1998. One provision of the plan was that, from 2000 on, proceeds of a 2.5% payroll tax were put into individual investment accounts. Individual could choose up to five funds from an approved list. There was close to free entry to the list, and there were 456 funds available at the start. One was a default fund for those who made no choice. At the start of the plan, against the background of extensive advertising campaign to promote choice of funds, 33.1% of investors allocated to the default. By 2003, 91.6% of new entrants allocated to the default. Experiences like this have made setting defaults an important part of policy design in the "nudge" approach to policy reform.

Here is a model that gives a role to defaults.[1] As before, $A$ is the set of alternatives. But now, a choice problem is a pair $(B, d)$, where $B \subseteq A$ is the feasible set and $d \in B$ is the default. The decision maker is characterized by two functions. The first is a utility function $u : A \to \mathbb{R}$ that represents DM's "true" preferences. The second function, $b : A \to \mathbb{R}_+$, gives a bonus to the default.

For simplicity assume that $x \neq y$ implies $u(x) \neq u(y)$. Choices are given by the function

$$C^{u,b}(B, d) = \begin{cases} d & \text{if } u(d) + b(d) \geq u(x) \text{ for all } x \neq d \\ x & \text{if } u(x) > u(d) + b(d) \text{ and } u(x) > u(y) \text{ for all } y \neq x, d \end{cases}$$

For an example, take $A = \{x, y\}$, with the functions $u(x) = 1$, $u(y) = 0$ and $b(x) =$

[1] This model is from chapter 3 of Rubinstein.

$b(y) = 2$. Then $C^{u,b}(A, x) = x$ but $C^{u,b}(A, y) = y$. Thus choices are sensitive to the default, as we intended.

And the model does restrict how choices change across decision problems. Let $B$ and $D$ be feasible sets with $d \in D \subset B$, and suppose $C^{u,b}(B, d) \in D$. There are two possibilities:

1. $C^{u,b}(B, d) = d$.

   The definition of $C^{u,b}$ tells us that $u(d) + b(d) > u(x)$ for all $x \in B$ with $x \neq d$. Since $x \in D$ implies $x \in B$, this means $u(d) + b(d) > u(x)$ for all $x \in D$ with $x \neq d$, and $C^{u,b}(D, d) = d$.

2. $C^{u,b}(B, d) = x \neq d$.

   The definition of $C^{u,b}$ tells us that there is an $x$ with $u(x) > u(d)+b(d)$ and $u(x) > u(y)$ for all $y \in B$ other than $x$ and $d$. Since $x \in D$ implies $x \in B$, this means $u(x) > u(d) + b(d)$ and $u(x) > u(y)$ for all $y \in B$ other than $x$ and $d$, and $C^{u,b}(D, d) = x$.

Thus $C^{u,b}$ satisfies a restricted version of contraction consistency.

This is far from the last word on the subject of choice-set dependence, but it is enough to illustrate that the phenomenon can be captured in a model that both builds on the decision theoretic tradition and has enough bite to avoid vacuousness.

The strategy for the next four chapters is this. We will apply the standard approach of decision theory to specific decision-making environments: choice under uncertainty, consumer choice in markets, choice of production plans, and normatively good choice by a policy maker. In each case, we will exploit the special structure of the applied problem to motivate assumptions about preferences, show how those assumptions on preferences give useful special structure to utility functions, and use that structure to learn about choices.

## Exercises

**Exercise 1.1.** Consider a DM with preferences

$$a \sim b \succ c \succ d \succ e \sim f.$$

1. What is $C^*(\{a, b, c\}, \succsim)$?

2. What is $C^*(\{d, e, f\}, \succsim)$?

3. Construct two different utility representations for these preferences.

**Exercise 1.2.** Kahneman and Tversky (1984) asked experimental subjects to consider the three following choice problems.

> You are about to buy a stereo for \$125 and a calculator for \$15.
>
> You learn there is a \$5 calculator discount at another store branch, ten minutes away. Do you make the trip?
>
> You learn there is a \$5 stereo discount at another store branch, ten minutes away. Do you make the trip?
>
> You learn both items are out of stock. You must go to the other branch, but as compensation you will get a \$5 discount. Do you care which item is discounted?

1. What are your answers?

2. Most people answer yes to the first question, no to the second question, and are indifferent in the third case. Let $x$ be traveling to the other store and getting a calculator discount, $y$ be traveling to the other store and getting a stereo discount, and $z$ be staying at the first store. Are the usual preferences over $\{x, y, z\}$ rational?

**Exercise 1.3.** Prove Theorem 1.1.

**Exercise 1.4.** Prove Theorem 1.4.

**Exercise 1.5.** Consider two people. Let $\succsim_1$ be 1's (complete and transitive) preferences on a finite set $A$, and let $\succsim_2$ be 2's. For their "joint" preference $\succsim^*$they define

$$x \succsim^* y \text{ if } x \succsim_1 y \text{ and } x \succsim_2 y$$

In words, as a pair they weakly prefer $x$ to $y$ if both of them weakly prefer $x$ to $y$. Prove that $\succsim^*$ is transitive. Show by example that it need not be complete.

**Exercise 1.6.** We saw that infinite $A$ can cause problems for the existence of a utility representation of rational preferences. Infinite $A$ can also cause problems for $C^*$.

1. Suppose $A$ is finite and $\succsim$ is complete and transitive. Show that $C^*(B, \succsim)$ is nonempty for all $B \subseteq A$. (Hint: Mimic the argument from Lemma 1.1.)

2. Let $A = [0, 1]$ and define $\succsim$ by $x \succsim y$ if and only if $x \geq y$. Find a subset $B \subset A$ such that $C^*(B, \succsim)$ is empty.

**Exercise 1.7.** This exercise will provide a converse to Theorem 1.3. To avoid the problem pointed out in the previous exercise, assume throughout that $A$ is finite.

Suppose $\succsim$ is complete and transitive, and that there is no pair $x \neq y$ with $x \sim y$.

1. Show that $C^*(\cdot, \succsim)$ is resolute.
2. Show that $C^*(\cdot, \succsim)$ is contraction consistent.

# Chapter 2

# Optimization and Concavity

## 2.1 How to Think About Derivatives

In calculus, you learned the following definition of the derivative. Let $f : \mathbb{R} \to \mathbb{R}$, and let $x^0 \in \mathbb{R}$. If the limit

$$\lim_{h \to 0} \frac{f(x^0 + h) - f(x^0)}{h}$$

exists and is a finite number $L$, then the derivative of $f$ at $x^0$ is $f'(x^0) = L$.

A slightly different way of thinking about this yields more insight into the generalization to multiple dimensions. If the limit above is equal to $L$, then the function given by

$$\eta(h) = \frac{f(x^0 + h) - f(x^0)}{h} - L \tag{2.1}$$

satisfies $\lim_{h \to 0} \eta(h) = 0$. Rearrange Equation 2.1 to get

$$f(x^0 + h) = f(x^0) + L \cdot h + \eta(h) \cdot h.$$

This motivates a different definition of the derivative. If there is a number $L$ and a function $\eta : \mathbb{R} \to \mathbb{R}$ such that $\lim_{h \to 0} \eta(h) = 0$ and, for all $h$,

$$f(x^0 + h) = f(x^0) + L \cdot h + \eta(h) \cdot h,$$

then $L$ is the derivative of $f$ at $x^0$, denoted $f'(x^0)$.

Taking $x = x^0 + h$, this reformulation says that, near $x^0$, the function $f$ is well approximated by the affine function $x \mapsto f(x^0) + f'(x^0) \cdot (x - x^0)$. The sense of "well-approximated" is that the approximation error, $(x - x^0)\eta(x - x^0)$ goes to 0 "faster than" $x - x^0$.

This idea of approximation by affine maps is just what we need to generalize to higher dimensions. A function $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $x^0$ if there is a linear map $L : \mathbb{R}^n \to \mathbb{R}$ and a function $\eta : \mathbb{R}^n \to \mathbb{R}$ such that $\lim_{\|h\| \to 0} \eta(h) = 0$ and, for all $h$,

$$f(x^0 + h) = f(x^0) + L(h) + \|h\|\eta(h).$$

In a slight abuse of notation, we write $\mathbf{D}f(x^0)$ both for the linear map $L$ and for its matrix representation with respect to the standard basis of $\mathbb{R}^n$. This matrix turns out to be in terms of the *partial derivatives* you studied in calculus:

$$\mathbf{D}f(x^0) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x^0) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x^0) \end{pmatrix}.$$

This way of thinking about derivatives provides the best way to understand the role of derivatives in optimization.

## 2.2 Optimization Problems

We are now going to spend some time studying optimization problems of the form

$$\max_x f(x) \tag{2.2}$$

$$\text{st } x \in X. \tag{2.3}$$

Here, $f$ is a function from some domain $D \subset \mathbb{R}^n_+$ to $\mathbb{R}$, and $X \subset D$. $f$ is the **objective function** and $X$ is the **feasible set.**

We typically describe $X$ in terms of functions: for $m$ functions $h_i : D \to \mathbb{R}$, write

$$X = \{x \in D \mid h_i(x) \geq 0 \ \forall i\}.$$

Stack the constraint functions to simplify this to

$$X = \{x \in D \mid h(x) \geq 0\}.$$

**Theorem 2.1** (Extreme Value Theorem)**.** *If $f$ is continuous and $X$ is closed and bounded, then the problem 2.2 has a solution.*

(Outside of $\mathbb{R}^n$, that $X$ is closed and bounded is not sufficient. The more general notion is *compactness*. In $\mathbb{R}^n$, compactness is equivalent to being closed and bounded.)

Note that $X$ will be closed if $h$ is continuous.

For some purposes, this is all we need. But many important applications require a characterization of the solution. And that is easiest if we have differentiability.

### 2.2.1 Necessary Conditions

Consider the problem:

$$\begin{aligned} &\max_x \ f(x) \\ &\text{st} \ \ x \in X \subset \mathbb{R}^n \end{aligned}$$

If $x^0$ is a point in the interior of $X$ where $\mathbf{D}f(x^0) \neq 0$, then $x^0$ cannot be the solution to the optimization problem. To see this, let $k$ be the least index for which $\frac{\partial f}{\partial x_k}(x^0) \neq 0$ and let $e_k$ be the unit vector in the $k$th direction. At $x = x^0 + \epsilon e_k$, we have

$$f(x) = f(x^0) + \epsilon \frac{\partial f}{\partial x_k}(x^0) + |\epsilon|\eta(\epsilon).$$

If $\frac{\partial f}{\partial x_k}(x^0) > 0$, we can choose $\epsilon$ positive and small enough that $\frac{\partial f}{\partial x_k}(x^0) > \eta(\epsilon)$, in which case $f(x) > f(x^0)$. Similarly, if $\frac{\partial f}{\partial x_k}(x^0) < 0$, we can choose $\epsilon < 0$ negative and small enough in absolute value that $\left|\frac{\partial f}{\partial x_k}(x^0)\right| > \eta(\epsilon)$, in which case again $f(x) > f(x^0)$. Thus a necessary condition for interior $x^0$ to maximize $f$ is that $\mathbf{D}f(x^0) = 0$.

A similar argument works when $x^0$ is on the boundary of $X$, but it gives a bit less. For now, we'll focus on the case where $X = \mathbb{R}^n_+$; a more general case will come latter.

Suppose $x^0$ is a boundary point. If $\frac{\partial f}{\partial x_k}(x^0) \neq 0$ and $x_k > 0$, the preceding argument works without change. But if $x_k = 0$, only the part with $\epsilon > 0$ is valid. So we get the weaker condition that $\frac{\partial f}{\partial x_k}(x_0) \leq 0$.

We have proved:

**Theorem 2.2.** *Suppose that $f$ is differentiable and $x^0$ solves*

$$\begin{aligned} &\max_x \ f(x) \\ &\text{st} \ \ x \in \mathbb{R}^n_+. \end{aligned}$$

*Then $\mathbf{D}f(x^0) \leq 0$ and $\mathbf{D}f(x^0) \cdot x^0 = 0$.*

**Remark 2.1.** I'm using the following conventions on vector inequalities:

- $x \geq y$ if $x_i \geq y_i$ for all $i$;
- $x > y$ if $x \geq y$ and $x_i > y_i$ for some $i$; and
- $x \gg y$ if $x_i > y_i$ for all $i$.

## 2.3 Concave Optimization

It is rare that we just need a necessary condition of maximization. Sufficient conditions involve assuming more about both $X$ and $f$.

A subset $X \subset \mathbb{R}^n$ is **convex** if, whenever $x$ and $y$ are in $X$, and $\lambda \in [0,1]$, we have $\lambda x + (1-\lambda)y \in X$. The function $f : X \to \mathbb{R}$ is **concave** if $\lambda \in [0,1]$ implies that $f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda)f(y)$. It is **strictly concave** if the inequality is strict for all $\lambda \in (0,1)$. If $f$ is differentiable, we have a particularly useful equivalent condition:

**Theorem 2.3.** *Let $f$ be differentiable. Then $f$ is concave if and only if*

$$f(y) \leq f(x) + \mathbf{D}f(x) \cdot (y - x),$$

*and $f$ is strictly concave if and only if*

$$f(y) < f(x) + \mathbf{D}f(x) \cdot (y - x).$$

This Theorem makes it easy to establish sufficient conditions for maximization.

**Theorem 2.4.** *Suppose $X$ is convex and $f$ is differentiable and concave. Then $x^0$ solves*

$$\max_x \ f(x)$$
$$\text{st} \ \ x \in \mathbb{R}^n_+$$

*if and only if $\mathbf{D}f(x^0) \leq 0$ and $\mathbf{D}f(x^0) \cdot x^0 = 0$.*

**Proof.** Theorem 2.2 established the "only if" direction. So we just need to show that $\mathbf{D}f(x^0) \leq 0$ and $\mathbf{D}f(x^0) \cdot x^0 = 0$ imply that $x^0$ solve the maximization problem.

Consider any $x \in X$. Theorem 2.3 tells us that

$$\begin{aligned} f(x) &\leq f(x^0) + \mathbf{D}f(x^0)(x - x^0) \\ &= f(x^0) + \mathbf{D}f(x^0)x, \end{aligned}$$

where the equality is from $\mathbf{D}f(x^0)x^0 = 0$. Since $\mathbf{D}f(x^0) \leq 0$ and $x \geq 0$, we have $\mathbf{D}f(x^0)x \leq 0$, and thus $f(x) \leq f(x^0)$. ∎

We can also use Theorem 2.3 to give a useful criterion for recognizing when a differentiable function is concave. Start with the case of $n = 1$, so $D$ is just an interval of $\mathbb{R}$. Suppose $f$ is concave, and $a, b \in D$ with $b > a$. From the previous theorem, we know that

$$f(b) \leq f(a) + f'(a)(b - a),$$

which can be rearranged to get

$$f'(a) \geq \frac{f(b) - f(a)}{b - a}.$$

Similarly, rearrange the inequality

$$f(a) \leq f(b) + f'(b)(a - b)$$

to get

$$f'(b) \leq \frac{f(b) - f(a)}{b - a}.$$

Together, these inequalities imply that $f'(b) \leq f'(a)$, so the derivative of a concave function is nonincreasing. A similar argument (exercise!) shows that the derivative of a strictly concave function is decreasing. If $f$ is twice differentiable, then these results imply that $f$ concave implies $f'' \leq 0$. However, they do *not* imply that a strictly concave function has a negative second derivative. After all, a strictly decreasing function can have isolated points where the derivative is zero: consider $-x^3$.

For the case of twice differentiable $f$, it's easy to establish the converse statements. Assume first that $f''(x) \leq 0$ for all $x$. A result called **Taylor's Theorem with remainder** says that, for all $x \leq y$, there is a $z \in [x, y]$ such that

$$f(y) = f(x) + f'(x)(y - x) + \frac{1}{2}f''(z)(y - x)^2.$$

Since $f'' \leq 0$, this implies that

$$f(y) \leq f(x) + f'(x)(y - x),$$

which is concavity. Similarly, $x \neq y$ and $f'' < 0$ imply that

$$f(y) < f(x) + f'(x)(y - x),$$

which is strict concavity.

Similar statements hold for the multidimensional case—the only complication is that the second derivative is now a matrix called the *Hessian* of $f$ at $x^0$:

$$\mathbf{D}^2 f(x^0) = \begin{pmatrix} \frac{\partial^2 f}{\partial^2 x_1}(x^0) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(x^0) \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(x^0) & \cdots & \frac{\partial^2 f}{\partial^2 x_n}(x^0) \end{pmatrix}.$$

The generalization of a negative second derivative is that the Hessian be *negative semidefinite*: $x^\top \mathbf{D}^2 f(x^0) x \leq 0$ for all $x$. (If you ever need to check that a matrix is negative semidefinite, there is a test based on determinants. You can read about it on Wikipedia.)

### 2.3.1 The Kuhn-Tucker Theorem

Let $f$ and $h_i$ (for $i = 1, \ldots, m$) be differentiable functions from $\mathbb{R}^n$ to $\mathbb{R}$. Consider the following problem:

$$\begin{aligned} &\max_x \; f(x) & \\ &\text{st} \;\; h_i(x) \geq 0 & \text{for all } i \\ &\qquad x_j \geq 0 & \text{for all } j \end{aligned}$$

The **Lagrangian** is the function $\mathcal{L} : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ given by

$$\mathcal{L}(x, \lambda) = f(x) + \lambda \cdot h(x).$$

The FOCs for simultaneously maximizing wrt $x$ and minimizing wrt $\lambda$, assuming all of those variables must be non-negative, are

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial x_j} &= \frac{\partial f}{\partial x_j}(\overline{x}) + \sum_{i=1}^{m} \lambda_i \frac{\partial h_i}{\partial x_j}(\overline{x}) \leq 0 \qquad \text{with equality if } \overline{x}_j > 0 \\
\frac{\partial \mathcal{L}}{\partial \lambda_i} &= h_i(\overline{x}) \geq 0 \\
\lambda_i &\geq 0 \\
\lambda_i h_i(\overline{x}) &= 0
\end{aligned}$$

We sometimes condense the last three lines, saying $h_i(\overline{x}) \geq 0$ and $\lambda_i \geq 0$ with **complementary slackness.**

**Theorem 2.5** (Kuhn-Tucker: sufficiency)**.** *Suppose $f$ and each $h_i$ are quasiconcave. If*

1. *the FOCs hold at $\overline{x}$,*
2. *$\mathbf{D}f(\overline{x}) \neq 0$, and*
3. *$\mathbf{D}h_i(\overline{x}) \neq 0$ for each binding constraint $i$,*

*then $\overline{x}$ solves the maximization problem.*

It's straightforward to show that these conditions work well in problems where all the functions are concave.

**Proposition 2.1.** *Assume $f$ and $h_i$ ($i = 1, \ldots, m$) are all concave. If there is an $\overline{x} \geq 0$ and a vector of shadow prices $\lambda \geq 0$ such that $(\overline{x}, \lambda)$ solve the FOC, then $\overline{x}$ solves*

$$\max_{x \geq 0} f(x) \mid h(x) \geq 0.$$

**Proof.** Since $f$ and each $h_i$ are concave,

$$\mathcal{L}(x, \lambda) = f(x) + \lambda \cdot h(x)$$

is concave in $x$. Thus, for all $x$,

$$\mathcal{L}(x, \lambda) \leq \mathcal{L}(\overline{x}, \lambda) + \mathbf{D}\mathcal{L}(\overline{x}, \lambda) \cdot (x - \overline{x}).$$

If $\overline{x}_i > 0$, then $\frac{\partial \mathcal{L}}{\partial x_i}(\overline{x}, \lambda) = 0$. If $\overline{x}_i = 0$, then $\frac{\partial \mathcal{L}}{\partial x_i}(\overline{x}, \lambda) \leq 0$. Either way,

$$\frac{\partial \mathcal{L}}{\partial x_i}(\overline{x}, \lambda)(x_i - \overline{x}_i) \leq 0.$$

Since this works for each $i$, we have

$$\mathbf{D}\mathcal{L}(\overline{x}, \lambda) \cdot (x - \overline{x}) \leq 0.$$

And that ensures $\mathcal{L}(x, \lambda) \leq \mathcal{L}(\overline{x}, \lambda)$.

By the complementary slackness conditions, either the $i$th constraint binds, and $h_i(\overline{x}) = 0$, or the $i$th constraint is slack and $\lambda_i = 0$. Either way, $\lambda_i h_i(\overline{x}) = 0$, so $\mathcal{L}(\overline{x}, \lambda) = f(\overline{x})$.

For any feasible $x$, we have $h(x) \geq 0$. Since $\lambda \geq 0$, that implies $\lambda \cdot h(x) \geq 0$, so $f(x) \leq \mathcal{L}(x, \lambda)$. Putting all this together, we have

$$f(x) \leq \mathcal{L}(x, \lambda) \leq \mathcal{L}(\overline{x}, \lambda) = f(\overline{x}).$$

■

# Part II

# Core Price Theory

# Chapter 3

# Choice Under Uncertainty

Here is a classic economic problem of choice under uncertainty. An investor has wealth $W > 0$. She will do all of her consumption next year. In the meantime, she must decide how to divide her wealth between a money market account that pays no interest, and the a risky stock. With probability 1/2, the stock price increases by 25%, while with probability 1/2, the price falls by 15%. The investor wants to maximize the expected value of the function

$$-\frac{1}{\lambda}e^{-\lambda c},$$

where $c$ is her final consumption. How should she invest?

If she puts $\alpha$ of her wealth in the stock, her final wealth is

$$(W - \alpha) + (1.25)\alpha = W + (.25)\alpha$$

if the stock goes up, and is

$$(W - \alpha) + (.85)\alpha = W - (.15)\alpha$$

if the stock goes down. Thus she chooses $\alpha$ to maximize

$$\frac{1}{2}\left[-\frac{1}{\lambda}e^{-\lambda(W+(.25)\alpha)}\right] + \frac{1}{2}\left[-\frac{1}{\lambda}e^{-\lambda(W-(.15)\alpha)}\right].$$

The first-order condition this maximization problem is

$$\frac{1}{2}(.25)e^{-\lambda(W+(.25)\alpha)} + \frac{1}{2}(-.15)e^{-\lambda(W-(.15)\alpha)} = 0.$$

(The second derivative is negative for all $\alpha$, so the solution to this equation is in fact a maximum.) Factor out $\frac{1}{2}e^{-\lambda W}$ and divide to rewrite the FOC as

$$(.25)e^{-\lambda(.25)\alpha} - (.15)e^{-\lambda(-.15)\alpha} = 0.$$

Solve to get

$$\lambda(.25)\alpha + \lambda(.15)\alpha = \log\left(\frac{.25}{.15}\right)$$

$$\alpha^* = \frac{\log\left(\frac{.25}{.15}\right)}{(.4)\lambda}$$

Some questions:

1. How does this fit into our abstract framework? That is, what are the alternatives?
2. What assumptions on preferences imply DM wants to maximize the expected value of some function?
3. The optimal investment amount was independent of initial wealth. Clearly, that was because of the exponential function. But what is the interpretation of that assumption?
4. The solution is decreasing in $\lambda$. This suggests that $\lambda$ is a measure of how much the investor dislikes the risk inherent in the stock. Is that correct? And how can we make it precise?

To answer these (and other) questions, we need to develop a general theory of *expected utility* and apply it to study *risk* and *risk aversion.*

## 3.1 Expected Utility

Let's start with the simplest setting for choice under uncertainty. DM ultimately cares about which of some set of **consequences** she receives. Write $X$ for the set of all possible consequences.

The environment is such that DM cannot necessarily choose some consequence for sure. Instead, which consequence she receives might be stochastic. The objects of choice are **lotteries**—probability measures on $X$. For our formal development, we will restrict attention to **simple lotteries**—lotteries with countable support. Denote the set of all simple lotteries on $X$ by $\mathcal{L}(X)$.

(The **support** of a probability measure is the smallest event that has probability 1.)

We need to fix some additional notation. Write $p = ((p_i); (x_i))$ for the lottery that gives consequence $x_j$ with probability $p_j$. For example, $(.4, .6; x, y)$ gives consequence $x$ with probability .4. We will abuse notation and also write $p(x)$ for the probability that $p$ assigns to $x$. If a lottery gives consequence $x$ with probability 1, we say it is **degenerate at** $x$, and write $\delta_x$.

Write $\mathrm{supp}(p)$ for the support of lottery $p$. For any two lotteries $p$ and $q$, and any number $\alpha \in [0, 1]$, we can define a new lottery, $\alpha p \oplus (1-\alpha)q$, in the following way: for any $z \in \mathrm{supp}(p) \cup \mathrm{supp}(q)$, the new lottery gives $z$ with probability $\alpha p(z) + (1-\alpha)q(z)$. This new lottery is sometimes called a **compound lottery.**

So far, this is just a special case of our abstract framework from the previous chapter, with $A = \mathcal{L}(X)$. So could follow the development there by, say, imposing continuity to get a continuous function $U$ such that $p \succsim q$ if and only if $U(p) \geq U(q)$. But we can go beyond our results there by taking advantage of the special structure of lotteries, along with the assumption that DM ultimately cares about consequences. Specifically, we will look for a representation of the **expected utility** form—there is a function $u : X \to \mathbb{R}$ such that $p \succsim q$ if and only if $\sum_x p(x)u(x) \geq \sum_x q(x)u(x)$. The function $u$ is called a **Bernoulli utility function.** Finally, notice that this is in fact a special case in that we can write $U(p) = \sum_x p(x)u(x)$.

The key assumption is the **independence axiom:**

$$p \succsim q \quad \text{if and only if, for all } \alpha \in [0,1] \text{ and all } r \in \mathcal{L}(X),\ \alpha p \oplus (1-\alpha)r \succsim \alpha q \oplus (1-\alpha)r.$$

The independence axiom says that, if two lotteries agree with some probability, then the preference between them depends only on what happens on the event that they disagree.

The independence axiom also implies a kind of monotonicity.

**Lemma 3.1.** *Suppose $\succsim$ is a preference on $\mathcal{L}(X)$ that satisfies the independence axiom, and suppose $x$ and $y$ are consequences with $\delta_x \succ \delta_y$. Then, for $1 \geq \alpha > \beta \geq 0$, we have*

$$\alpha\delta_x \oplus (1-\alpha)\delta_y \succ \beta\delta_x \oplus (1-\beta)\delta_y.$$

**Proof.**

$$\begin{aligned}\alpha\delta_x \oplus (1-\alpha)\delta_y &= (\alpha-\beta)\delta_x \oplus (\beta\delta_x \oplus (1-\alpha)\delta_y)\\ &\succ (\alpha-\beta)\delta_y \oplus (\beta\delta_x \oplus (1-\alpha)\delta_y)\\ &= \beta\delta_x \oplus (1-\beta)\delta_y,\end{aligned}$$

where the strict preference is from independence. ■

Independence is not the only axiom we will need, of course. Since we are shooting for a utility representation on an uncountably infinite set, we need $\succsim$ to be complete, transitive, and continuous. To avoid a non-trivial bit of real analysis, I will state the continuity assumption differently than I did before. Preferences $\succsim$ on $\mathcal{L}(X)$ are **continuous** if, for any $p \succ q \succ r$, there is an $\alpha \in (0,1)$ such that

$$q \sim \alpha p \oplus (1-\alpha)r.$$

In the homework, you will show that any preferences that have a representation of the expected utility form satisfy rationality, continuity, and independence.

**Theorem 3.1.** *Suppose $\succsim$ is a preference on $\mathcal{L}(X)$ that satisfies rationality, continuity, and independence. Then there exists a function $u : X \to \mathbb{R}$ such that $p \succsim q$ if and only if $\sum_x p(x)u(x) \geq \sum_x q(x)u(x)$.*

**Proof.** Everything important in the proof already shows up in the case where $X$ has three elements, so I only treat that special case.

If DM is indifferent between all three degenerate lotteries, then the result follows by taking $u$ to be constant. So suppose there is a best consequence $M$ and a worst consequence $m$. Formally, $\delta_M \succ \delta_m$ and, if $z$ is the third member of $X$, $\delta_M \succsim \delta_z \succsim \delta_m$.

Next we construct $u$. Let $u(M) = 1$ and $u(m) = 0$. If $\delta_M \sim \delta_z$, then let $u(z) = 1$. If $\delta_m \sim \delta_z$, then let $u(z) = 0$. Otherwise, continuity implies there is a number $\alpha \in (0,1)$ such that $\delta_z \sim \alpha\delta_M \oplus (1-\alpha)\delta_m$. Let $u(z) = \alpha$.

Now consider any lottery $p$. Independence implies

$$\begin{aligned}p &= p(M)\delta_M \oplus p(z)\delta_z \oplus p(m)\delta_m\\ &\sim p(M)\delta_M \oplus p(z)\left[u(z)\delta_M \oplus (1-u(z))]\delta_m\right] \oplus p(m)\delta_m\\ &= \left[p(M) + p(z)u(z)\right]\delta_M \oplus \left[p(z)(1-u(z)) + p(m)\right]\delta_m.\end{aligned}$$

Then Lemma 3.1 implies $p \succsim q$ if and only if

$$p(M) + p(z)u(z) \geq q(M) + q(z)u(z).$$

But our complete definition of $u$ tells us that this is equivalent to

$$\sum_x p(x)u(x) \geq \sum_x q(x)u(x).$$

■

As in the general case, any monotone transformation of $U$ represents the same preferences over $\mathcal{L}(X)$. But not all monotone transformations will preserve the expected utility property. It should be clear that if $u$ is a Bernoulli utility function for preferences $\succsim$, then so is any positive affine transformation: for each $x$, let $v(x) = au(x) + b$ for real numbers $a > 0$ and $b$. It turns out this is the only class of transformation of Bernoulli utilities that preserve preferences.

**Theorem 3.2.** *Suppose $u$ and $v$ are two Bernoulli utility functions whose expected values represent the same preferences $\succsim$. Then there are numbers $a > 0$ and $b$ such that $v(x) = au(x) + b$ for all $x \in X$.*

**Proof.** Choose $M$ and $m$ in $X$ such that $u(M) > u(m)$. Since $v$ represents the same preferences, we also have $v(M) > v(m)$. Consider the system of two linear equations in two unknowns given by

$$\begin{aligned} v(M) &= au(M) + b \\ v(m) &= au(m) + b. \end{aligned}$$

Solve this to get

$$a = \frac{v(M) - v(m)}{u(M) - u(m)} > 0 \quad \text{and} \quad b = \frac{v(m)u(M) - v(M)u(m)}{u(M) - u(m)}.$$

Now consider an arbitrary $x \in X$. By continuity, there is an $\alpha$ such that $x \sim \alpha\delta_M + (1-\alpha)\delta_m$.

We have

$$
\begin{aligned}
v(x) &= \alpha v(M) + (1-\alpha)v(m) \\
&= \alpha[au(M)+b] + (1-\alpha)[au(m)+b] \\
&= a[\alpha u(M) + (1-\alpha)u(m)] + b \\
&= au(x) + b.
\end{aligned}
$$

■

## 3.2 Difficulties and Extentions

The independence axiom gives a tractable form for utility, making for a powerful theory for applications. But it is a strong assumption, and laboratory experiments can call it into question. The first famous examples were introduced by Maurice Allais. Here is a version of his questions developed by Kahneman and Tversky. Imagine you have to choose between

$$
L_1 = \begin{cases} 3000 & \text{with probability } 0.25 \\ 0 & \text{with probability } 0.75 \end{cases} \quad \text{and } L_2 = \begin{cases} 4000 & \text{with probability } 0.2 \\ 0 & \text{with probability } 0.8 \end{cases}.
$$

Most people prefer $L_2 \succ L_1$.

Now imagine you have to choose between

$$
L_3 = 3000 \text{ with probability } 1 \quad \text{and } L_4 = \begin{cases} 4000 & \text{with probability } 0.8 \\ 0 & \text{with probability } 0.2 \end{cases}.
$$

Most people prefer $L_3 \succ L_4$.

If you have the same preferences as the majority, then your preferences violate the independence axiom:

$$
L_1 = 0.25L_3 \oplus 0.75\delta_0 \quad \text{and} \quad L_2 = 0.25L_4 \oplus 0.75\delta_0.
$$

Another source of difficulties with expected utility arises when DM cares explicitly about randomizing. One way this can arise is motivated by fairness. Imagine that you have two children, Alice and Bob. You also have one indivisible piece of candy. It is reasonable to strictly prefer tossing a coin to decide who gets the candy rather than picking one or the other child deterministically. But if you are indifferent between which child gets the candy deterministically, then independence implies you are indifferent between all lotteries.

A problem with the whole framework of lotteries is that it rules out caring differently about which consequence you receive depending on whatever random factor determines outcomes. Imagine there is a 50% chance of rain. Then an offer of an umbrella if and only if it is raining is the same lottery as an offer of an umbrella if and only if it is not raining. After all, both give you an umbrella with probability 0.5.

One way out of this problem is to redefine consequences—perhaps bring wet versus carrying an umbrella around when it's sunny. Another approach is to use **state-dependent expected utility.**

Imagine there is a set of **states** $\Omega$. The interpretation is that state $\omega \in \Omega$ determines everything not chosen by DM that is relevant to her preferences. An **act** is a map from $\Omega \to X$. Preferences over acts $a$ are represented by the utility function

$$U(a) = \sum_{\omega} p(\omega) u(a(\omega), \omega),$$

where $p$ is a probability measure over $\Omega$ and $u : X \times \Omega \to \mathbb{R}$ is a state-dependent Bernoulli utility.

Finally, we can worry about where the probabilities come from. Sometimes it makes sense to think they are given as part of the problem. Think of gambling at a casino. The more usual case for social science though, is that probabilities are not given. Instead, we use them to represent DM's subjective beliefs. We can formalize this within the context of the model with states.

I won't go through all of the axioms here. I'll just make two points. First, formal developments of this idea typically require the assumption that Bernoulli utilities are constant in the states: $u(x, \omega) = u(x, \omega')$ for all $x \in X$ and $\omega, \omega' \in \Omega$. Second, there is also experimental evidence against the idea of subjective probability. Daniel Ellsberg offered the following example.

You face two urns. Each contains 100 balls, some black and some red. The first urn has 50 black balls and 50 red balls. I'm not telling you the mix in the second. You get to choose an urn and draw a ball. If your ball is red, you win \$100; otherwise you get nothing. Which urn do you prefer?

Now you win if you draw a black ball. Which urn do you prefer?

Many people strictly prefer urn 1 in both cases. This is inconsistent with subjective probability. The first question reveals that you act as if the probability of drawing red from the unknown urn is less than 1/2. But the second question reveals that you act as if that same probability is greater than 1/2.

## 3.3 Utility for Money

For the rest of this chapter, we specialize to the case where prizes are amounts of money. Everything we do works both for simple lotteries and for "continuous" random variables with integrals in place of sums. In the continuous case, it is convenient to identify a lottery with its cdf. We will use both formalizations below.

### 3.3.1 Risk Aversion

Suppose $x > y$ implies $\delta_x \succsim \delta_y$. Then the vN-M representation theorem immediately gives $u(x) > u(y)$.

Notation: For any lottery $p$, write $\mathbb{E}p$ for the expected value of $p$:

$$\mathbb{E}p = \sum_x xp(x).$$

Say that a DM is **risk averse** if, for all lotteries $p$, we have $\delta_{\mathbb{E}p} \succsim p$. DM is **risk loving** if the preference is reversed. And DM is **risk neutral** if $\delta_{\mathbb{E}p} \sim p$.

To determine when a EU maximizer is risk averse, we need one more fact about concave functions. Let $u$ be a concave function on $\mathbb{R}$, and let $p$ be a lottery with finite expected value $\mathbb{E}p$. Then $u(\mathbb{E}p) \geq \sum_x u(x)p(x)$, so DM is risk averse. (The inequality is strict if $u$ is strictly concave.) To see this, let $y = \mathbb{E}p$ in the inequality characterizing concavity to get

$$u(x) \leq u(\mathbb{E}p) + u'(\mathbb{E}p)(x - \mathbb{E}p).$$

Take expected values of both sides to get

$$\begin{aligned}
\sum_x u(x)p(x) &\leq \sum_x \left[u(\mathbb{E}p) + u'(\mathbb{E}p)(x - \mathbb{E}p)\right] p(x) \\
&= u(\mathbb{E}p) \sum_x p(x) + u'(\mathbb{E}p) \sum_x (x - \mathbb{E}p)p(x) \\
&= u(\mathbb{E}p)
\end{aligned}$$

(This inequality is called **Jensen's inequality**.) Thus a EU maximizer is risk averse if and only if her Bernoulli utility function is concave.

We can get a good feel for the way risk aversion manifests in expected utility theory by looking at a very simple problem in the demand for insurance. A consumer's income is subject to the risk of a loss—with probability $\pi$, she will lose $L$. Her initial income is

$Y$. An insurance company is willing to sell insurance against this loss. If the consumer pays $P$, then the company will reimburse $L$ in the event of the loss. The consumer can partially insure: if she pays $\alpha P$, then she is reimbursed $\alpha L$. The consumer maximizes the expectation of a strictly increasing, strictly concave, and differentiable Bernoulli utility of consumption $u$, where consumption is income, plus any reimbursement minus the insurance premium.

The consumer solves

$$\max_{\alpha}(1-\pi)u(Y-\alpha P)+\pi u(Y-L+\alpha L-\alpha P).$$

Differentiate to get the FOC

$$-(1-\pi)Pu'(Y-\alpha P)+\pi(L-P)u'(Y-(1-\alpha)L-\alpha P)=0.$$

Call the insurance contract *actuarially fair* if the expected payout equals the premium: $\pi L = P$. Since $P = \pi P + (1-\pi)P$, we can rewrite the equality for actuarial fairness as

$$\pi(L-P)=(1-\pi)P.$$

Thus if the contract is actuarially fair, the FOC simplifies to

$$u'(Y-\alpha P)=u'(Y-(1-\alpha)L-\alpha P).$$

Strict concavity of $u$ implies that this holds if and only if $\alpha = 1$.

Next consider an insurance contract that is *actuarially unfair* in that $P > \pi L$. I claim that $\alpha = 1$ cannot be a solution in this case. To see why, assume otherwise. Then the FOC would read

$$(1-\pi)Pu'(Y-\alpha P)=\pi(L-P)u'(Y-\alpha P),$$

which is a contradiction.

Together, these results are:

**Proposition 3.1.** *Suppose the consumer is strictly risk averse and has a differentiable Bernoulli utility function. Then the consumer buys full insurance if and only if the insurance is actuarially fair.*

This result generalizes to a much larger set of stochastic processes for income.

The intuition is simple. Actuarially fair insurance allows the consumer to costlessly replace the risky income with it's expected value. Wanting to do so is the very definition of risk aversion. But, with differentiable utility, an expected utility maximizer is approximately risk-neutral for small risks. Since rejecting the last bit of coverage is taking a small bet with positive expected value, the consumer wants to do it.

### 3.3.2 Comparing Risk Aversion

Now we turn to the question of how to compare the risk tolerance of two different decision makers. The first step is a couple of additional definitions.

Let $p$ be a lottery. If $x$ is a sure thing such that $\delta_x \sim p$, then we call $x$ the **certainty equivalent** of $p$. Denote the certainty equivalent of $p$ by $C(p)$, and call $R(p) = \mathbb{E}p - C(p)$ the **risk premium** of $p$. This "functional" notation needs a result that says things are well-defined:

**Theorem 3.3.** *Suppose preferences are represented by the expectation of a Bernoulli utility function that is continuous and strictly increasing. Then every $p$ has exactly one certainty equivalent.*

**Proof.** If $p$ is degenerate, then strict monotonicity directly implies that the only certainty equivalent is the prize that has probability one.

If $p$ is non-degenerate, then there must be two prizes in the support of $p$, say $\overline{x}$ and $\underline{x}$, such that $u(\overline{x}) > \sum_x u(x)p(x)$ and $u(\underline{x}) < \sum_x u(x)p(x)$. Since $u$ is continuous, the intermediate value theorem implies that there is an $\hat{x}$ with $\underline{x} < \hat{x} < \overline{x}$ with $u(\hat{x}) = \sum_x u(x)p(x)$. And strict monotonicity implies that there is only one such. ■

There is a useful approximation to the risk premium if the risk is "small"—that is, if all elements in the support of $p$ are close to $\mathbb{E}p$. By definition, we have

$$u(\mathbb{E}p - R(p)) = \sum_x u(x)p(x).$$

The LHS can be approximated:

$$u(\mathbb{E}p - R(p)) \approx u(\mathbb{E}p) - u'(\mathbb{E}p)R(p).$$

For any $x$ in the support of $p$, write:

$$u(x) \approx u(\mathbb{E}p) + u'(\mathbb{E}p)(x - \mathbb{E}p) + \frac{1}{2}u''(\mathbb{E}p)(x - \mathbb{E}p)^2.$$

Take the expected value to get an approximation of the RHS:

$$\sum_x u(x)p(x) \approx u(\mathbb{E}p) + \frac{1}{2}u''(\mathbb{E}p)\sum_x (x - \mathbb{E}p)^2 p(x) = u(\mathbb{E}p) + \frac{1}{2}u''(\mathbb{E}p)\operatorname{var}(p).$$

Approximately equating the two approximations gives:

$$u(\mathbb{E}p) - u'(\mathbb{E}p)R(p) \approx u(\mathbb{E}p) + \frac{1}{2}u''(\mathbb{E}p)\operatorname{var}(p).$$

Solve for $R(p)$ to get:

$$R(p) \approx -\frac{u''(\mathbb{E}p)}{u'(\mathbb{E}p)} \cdot \frac{\operatorname{var}(p)}{2}.$$

The value of the function $\lambda(x) = -\frac{u''(x)}{u'(x)}$ is called the **coefficient of absolute risk aversion** at $x$. So we can state the approximation as: for small risks, the risk premium is approximately the coefficient of absolute risk aversion times half the variance. This suggests, correctly, that comparing coefficients of absolute risk aversion lets us compare the risk aversion of different decision makers.

Now we can start comparing. Subscript $C$, $R$, and $\lambda$ with the utility function that defines them.

Say that the DM with utility function $u$ is **at least as risk averse as** the DM with utility function $v$ if, for all $p$ and all sure things $x$, if $u$ weakly prefers the lottery $p$, then so does $v$.

**Proposition 3.2.** *Suppose $u$ and $v$ are both strictly increasing and continuously differentiable. Then, the following are equivalent:*

1. *$u$ is at least as risk averse as $v$;*
2. *$C_u(p) \leq C_v(p)$;*
3. *there is an increasing and concave function $h$ such that $u = h \circ v$;*
4. *$\lambda_u(x) \geq \lambda_v(x)$ for all $x$.*

**Proof.** First we show that (1) $\Leftrightarrow$ (2). Each direction will proceed by proving the contrapositive. We have $u$ is *not* at least as risk averse as $v$ if and only if there are $p$ and $x$ such that:

$$\begin{array}{rccc} & \mathbb{E}u(p) \geq u(x) & \text{but} & \mathbb{E}v(p) < v(x) \\ \Leftrightarrow & u(C_u(p)) \geq u(x) & \text{but} & v(C_v(p)) < v(x) \\ \Leftrightarrow & C_u(p) \geq x & \text{but} & C_v(p) < x. \end{array}$$

But the last line holds for some $p$ and $x$ if and only if $C_u(p) > C_v(p)$ for some $p$.

Next we show that (2) $\Leftrightarrow$ (3). Since $v$ is strictly increasing, it has an inverse $v^{-1}$. So let $h = u \circ v^{-1}$. Now we have

$$\begin{array}{rcl} & C_v(p) \geq C_u(p) \text{ for all } p & \\ \Leftrightarrow & u(C_v(p)) \geq u(C_u(p)) \text{ for all } p & \text{monotonicity of } u \\ \Leftrightarrow & h \circ v(C_v(p)) \geq u(C_u(p)) \text{ for all } p & u = h \circ v \\ \Leftrightarrow & h(\mathbb{E}v(p)) \geq \mathbb{E}u(p) \text{ for all } p & \text{definition of } C \\ \Leftrightarrow & h(\mathbb{E}v(p)) \geq \mathbb{E}h(v(p)) \text{ for all } p & u = h \circ v \\ \Leftrightarrow & h \text{ is concave} & \text{Jensen's inequality.} \end{array}$$

Finally, we show (3) $\Leftrightarrow$ (4). Differentiate the identity $u(x) = h(v(x))$ to get

$$u'(x) = h'(v(x))v'(x).$$

Take logs of both sides and differentiate again to get

$$\frac{u''(x)}{u'(x)} = \frac{h''(x)v'(x)}{h'(v(x))} + \frac{v''(x)}{v'(x)},$$

or

$$-\frac{u''(x)}{u'(x)} = -\frac{v''(x)}{v'(x)} - \frac{h''(x)v'(x)}{h'(v(x))}.$$

Thus $\lambda_u(x) \geq \lambda_v(x)$ for all $x$ if and only if $h$ is concave. ■

We can see all of these ideas in action in a more general version of the investment problem we started with. Recall that DM has wealth $W > 0$, and will do all of her consumption in one year. She can divide her wealth between two different securities. We continue to assume that one security is risk-free and the other is risky, but we are less specific about the returns. The risk-free security has gross return $r > 1$, while the risky security pays gross return $\theta$ with simple probability measure $\pi$. DM maximizes the expected value of a Bernoulli utility function $u$ defined over her final wealth $Y$. Assume $u$ is twice differentiable with $u'(x) > 0$ and $u''(x) \leq 0$ for all $x$.

Our first step is to calculate the expected utility of an arbitrary investment plan. If DM puts $\alpha$ in the risky security and the return on that security is $\theta$, then final wealth is

$$Y = \theta\alpha + r(W - \alpha) = \alpha(\theta - r) + rW.$$

This wealth is realized with probability $\pi(\theta)$. Thus the expected utility is

$$\sum_{\theta \in \operatorname{supp} \pi} u(\alpha(\theta - r) + rW)\pi(\theta).$$

Let's assume that DM can borrow money to invest in the risky security, but cannot short-sell the risky security. In that case, she can choose any $\alpha \geq 0$. So her problem is:

$$\max_{\alpha} \sum_{\theta \in \operatorname{supp} \pi} u(\alpha(\theta - r) + rW)\pi(\theta).$$

Concavity of $u$ implies concavity of the entire objective function. This means that $\alpha^*$ is a solution to the optimization problem if and only if it solves the first-order condition:

$$\sum_{\theta \in \operatorname{supp} \pi} (\theta - r)u'(\alpha^*(\theta - r) + rW)\pi(\theta) \leq 0, \tag{3.1}$$

with equality if $\alpha^* > 0$.

There is nothing in our assumptions so far that ensure a solution actually exists. Suppose $\theta > r$ for all $\theta \in \operatorname{supp}(\pi)$. Then, since $u'(x) > 0$ for all $x$, every term in the sum on the LHS of inequality 3.1 is positive, which means the inequality cannot be satisfied. Intuitively, the risky asset pays more than the risk-free asset no matter what. Thus DM would like to borrow arbitrarily large sums to invest in the risky security.

Next suppose DM is risk neutral, so $u$ is linear. That means $u'(x)$ is constant at, say, $k$. The FOC then becomes

$$k \sum_{\theta \in \operatorname{supp} \pi} (\theta - r)\pi(\theta) \leq 0,$$

or

$$k\,(\mathbb{E}\theta - r) \leq 0.$$

If $\mathbb{E}\theta < r$, then this implies $\alpha^* = 0$ is the unique optimum. But if $\mathbb{E}\theta = r$, then *any* $\alpha$ is an optimum. And if $\mathbb{E}\theta > r$, there is again no solution.

From now on, assume that $u$ is strictly concave, and that $\min \operatorname{supp}(\pi) < r < \max \operatorname{supp}(\pi)$. This is not enough to guarantee a solution, so further assume there at least one solution. With these assumptions, we can prove that the solution is unique: differentiate the LHS of inequality 3.1 with respect to $\alpha$ to get

$$\sum_{\theta \in \operatorname{supp}(\pi)} (\theta - r)^2 u''(\alpha(\theta - r) + rW)\pi(\theta) < 0.$$

What can we say about this solution? First consider the case $\mathbb{E}\theta \le r$. I claim that $\alpha^* = 0$ is the solution. Write $\underline{\theta} = \min \operatorname{supp}(\pi)$. Then

$$u'(\alpha(\underline{\theta} - r) + rW) < u'(\alpha(\theta - r) + rW$$

for all $\theta > \underline{\theta}$. Thus

$$\begin{aligned}
\sum_{\theta \in \operatorname{supp}\pi} (\theta - r) u'(\alpha^*(\theta - r) + rW)\pi(\theta) &< \sum_{\theta \in \operatorname{supp}\pi} (\theta - r) u'(\alpha^*(\underline{\theta} - r) + rW)\pi(\theta) \\
&= u'(\alpha^*(\underline{\theta} - r) + rW) \sum_{\theta \in \operatorname{supp}\pi} (\theta - r)\pi(\theta) \\
&= u'(\alpha^*(\underline{\theta} - r) + rW)\,(\mathbb{E}\theta - r) \\
&\le 0.
\end{aligned}$$

Now consider the case $\mathbb{E}\theta > R$. I claim that the solution must have $\alpha^* > 0$. We know that there is a solution $\alpha^* \ge 0$. If we rule out $\alpha^* = 0$, then the claim must be true. Substitute $\alpha^* = 0$ into the FOC to get

$$u'(rW)\,(\mathbb{E}\theta - r) \le 0,$$

which is impossible.

The intuition here is just like in the insurance example—an expected utility maximizer with a differentiable Bernoulli utility function is approximately risk neutral for small risks.

Now we can do some comparative statics. Imagine that two different DM's face this problem, one with utility $u$ and one with utility $v$. And suppose that $u$ is strictly more risk averse as $v$. Finally, assume $\mathbb{E}\theta > r$, so both DMs invest a positive amount in the risky security.

Let $\alpha_u^*$ be the optimal risky investment for utility $u$ and let $\alpha_v^*$ be the optimal investment for utility $v$. The first-order conditions are

$$\sum_{\theta \in \operatorname{supp}\pi} (\theta - r) u'(\alpha_u^*(\theta - r) + rW)\pi(\theta) = 0 \tag{3.2}$$

and

$$\sum_{\theta \in \operatorname{supp}\pi} (\theta - r) v'(\alpha_v^*(\theta - r) + rW)\pi(\theta) = 0. \tag{3.3}$$

Proposition 3.2 implies that the first of these can be written

$$\sum_{\theta \in \text{supp}(\pi)} (\theta - r)h'(v(\alpha_u^*(\theta - r) + rW))v'(\alpha_u^*(\theta - r) + rW)\pi(\theta) = 0$$

for some increasing and strictly concave function $h$.

I claim that DM with utility $v$ invests more in the risky security as does DM with utility $u$—that is, $\alpha_v^* \geq \alpha_u^*$. Intuitively, this is because concavity of $h$ means the LHS of the FOC for $u$ puts more weight on the negative terms in the sum.

First split the FOC for $v$ into a part with $\theta > r$ and a part with $\theta < r$:

$$\sum_{\theta < r}(\theta - r)v'(\alpha_v^*(\theta - r) + rW)\pi(\theta) + \sum_{\theta > r}(\theta - r)v'(\alpha_v^*(\theta - r) + rW) = 0.$$

Next consider a similar splitting in the case of $u = h \circ v$:

$$\sum_{\theta < r}(\theta - r)h'(v(\alpha(\theta - r) + rW))v'(\alpha(\theta - r) + rW)\pi(\theta) + \sum_{\theta > r}(\theta - r)h'(v(\alpha(\theta - r) + rW))v'(\alpha(\theta - r) + rW)\pi(\theta).$$

We can bound this expression from above term-by-term. Let $\tilde{\theta} = \max\{\theta \mid \theta < r\}$. Then

$$\begin{aligned}
&\sum_{\theta < r}(\theta - r)h'(v(\alpha(\theta - r) + rW))v'(\alpha(\theta - r) + rW)\pi(\theta) \\
&\quad \leq \sum_{\theta < r}(\theta - r)h'(v(\alpha(\tilde{\theta} - r) + rW))v'(\alpha(\theta - r) + rW)\pi(\theta) \\
&\quad = h'(v(\alpha(\tilde{\theta} - r) + rW)) \sum_{\theta < r}(\theta - r)v'(\alpha(\theta - r) + rW)\pi(\theta),
\end{aligned}$$

where the inequality is because $h$ concave implies $h'$ is decreasing and each $(\theta - r)$ is negative. (It is not a strict inequality because there might be only one $\theta < r$.)

Similarly, let $\hat{\theta} = \min\{\theta \mid \theta > r\}$. Then

$$\begin{aligned}
&\sum_{\theta > r}(\theta - r)h'(v(\alpha(\theta - r) + rW))v'(\alpha(\theta - r) + rW)\pi(\theta) \\
&\quad \leq \sum_{\theta < r}(\theta - r)h'(v(\alpha(\hat{\theta} - r) + rW))v'(\alpha(\theta - r) + rW)\pi(\theta) \\
&\quad = h'(v(\alpha(\hat{\theta} - r) + rW)) \sum_{\theta < r}(\theta - r)v'(\alpha(\theta - r) + rW)\pi(\theta).
\end{aligned}$$

Together, these bounds give us:

$$\begin{aligned}
& \sum_{\theta<r}(\theta-r)h'(v(\alpha(\theta-r)+rW))v'(\alpha(\theta-r)+rW)\pi(\theta) \\
& \qquad +\sum_{\theta>r}(\theta-r)h'(v(\alpha(\theta-r)+rW))v'(\alpha(\theta-r)+rW)\pi(\theta) \\
< \quad & \qquad h'(v(\alpha(\tilde{\theta}-r)+rW))\sum_{\theta<r}(\theta-r)v'(\alpha(\theta-r)+rW)\pi(\theta) \\
& \qquad +h'(v(\alpha(\hat{\theta}-r)+rW))\sum_{\theta<r}(\theta-r)v'(\alpha(\theta-r)+rW)\pi(\theta).
\end{aligned}$$

By equation 3.3, we have

$$\left|\sum_{\theta<r}(\theta-r)v'(\alpha_v^*(\theta-r)+rW)\pi(\theta)\right| = \sum_{\theta<r}(\theta-r)v'(\alpha_v^*(\theta-r)+rW)\pi(\theta).$$

By strict concavity of $h$ and

$$\alpha_v^*(\tilde{\theta}-r)+rW < \alpha_v^*(\hat{\theta}-r)+rW,$$

we have

$$\begin{aligned}
& \left|h'(v(\alpha(\tilde{\theta}-r)+rW))\sum_{\theta<r}(\theta-r)v'(\alpha(\theta-r)+rW)\pi(\theta)\right| \\
& \quad > \quad h'(v(\alpha(\hat{\theta}-r)+rW))\sum_{\theta<r}(\theta-r)v'(\alpha(\theta-r)+rW)\pi(\theta).
\end{aligned}$$

Since the negative part has greater absolute value, the sum is negative. And since the derivative of the expected utility for the DM with utility function $u$ is less than that sum, we have

$$\sum_{\theta\in\text{supp}(\pi)}(\theta-r)h'(v(\alpha_v^*(\theta-r)+rW))v'(\alpha_v^*(\theta-r)+rW)\pi(\theta) < 0.$$

Since the LHS is strictly decreasing in $\alpha$, this establishes that $\alpha_v^* > \alpha_u^*$.

The way we typically relate this result to something that could be observed in data is to make assumptions about how risk aversion varies with wealth. A plausible assumption is that risk aversion decreases with wealth: $\lambda(x)$ is decreasing in $x$. With this assumption, the previous result implies that if two DMs have the same Bernoulli utility, the one with greater initial wealth invests more in the risky security.

And a last observation is that we now understand why the solution to the motivating example was independent of wealth: risk aversion is constant in wealth, with value $\lambda$ if and only if Bernoulli utility has the form $-\frac{1}{\lambda}e^{-\lambda x}$. (You will provide the details in the problem set.)

### 3.3.3 Stochastic Dominance

So far, our comparative statics results have concerned variation in risk preferences, for a fixed set of lotteries. But it is often useful instead to directly compare lotteries. A standard approach does this through unanimity theorems—giving conditions on two lotteries such that all decision makers with Bernoulli utilities in some class agree that the first lottery is preferred to the second.

To keep things simple, we focus on continuous random variables in this section, and assume that every random variable considered has a density, and has support contained in $[0, \overline{x}]$. We will also assume that all Bernoulli utility functions are continuous and is twice differentiable except possibly at finitely many points. This will let us use one of the most useful facts from calculus: integration by parts. Suppose $U$ and $V$ are differentiable except possibly at finitely many points, with derivatives $U' = u$ and $V' = v$. Then

$$\int_a^b U(x)v(x)\,dx = U(b)V(b) - U(a)V(a) - \int_a^b V(x)u(x)\,dx.$$

This will be our main tool for this section.

Say that lottery $F$ **first-order stochastically dominates** lottery $G$ if every DM with increasing Bernoulli utility function prefers $F$ to $G$. That is, for all increasing functions $u$, we have

$$\int u(x)f(x)\,dx \geq \int u(x)g(x)\,dx.$$

The following theorem makes this definition easier to apply.

**Theorem 3.4.** *$F$ first-order stochastically dominates $G$ if and only if $F(x) \leq G(x)$ for all $x$.*

**Proof.** Integrating by parts with $V(x) = -[1 - F(x)]$, the expected utility of lottery $F$ is

$$\int_0^{\overline{x}} u(x)f(x)\,dx = u(0) + \int_0^{\overline{x}} u'(x)[1 - F(x)]\,dx.$$

Similarly,

$$\int_0^{\overline{x}} u(x)g(x)\,dx = u(0) + \int_0^{\overline{x}} u'(x)[1-G(x)]\,dx.$$

Thus

$$\begin{aligned}\int_0^{\overline{x}} u(x)f(x)\,dx - \int_0^{\overline{x}} u(x)g(x)\,dx &= \int_0^{\overline{x}} u'(x)[1-F(x)]\,dx - \int_0^{\overline{x}} u'(x)[1-G(x)]\,dx \\ &= \int_0^{\overline{x}} u'(x)[G(x)-F(x)]\,dx.\end{aligned}$$

Suppose $F(x) \le G(x)$ for all $x$. Then

$$\int_0^{\overline{x}} u(x)f(x)\,dx - \int_0^{\overline{x}} u(x)g(x)\,dx = \int_0^{\overline{x}} u'(x)[G(x)-F(x)]\,dx \ge 0,$$

where the inequality follows from the supposition $F(x) \le G(x)$ and $u'(x) \ge 0$.

Suppose there is an $x_0$ such that $F(x_0) > G(x_0)$. Since $F$ and $G$ have densities, they are continuous, and there is an interval $(x_0-\epsilon, x_0+\epsilon)$ such that $x_0-\epsilon < x < x_0+\epsilon$ implies $F(x) > G(x)$.

Consider the (weakly) increasing function

$$u(x) = \begin{cases} 1 & \text{if } x \ge x_0+\epsilon \\ \frac{x-(x_0-\epsilon)}{2\epsilon} & \text{if } x_0-\epsilon < x < x_0+\epsilon \\ 0 & \text{if } x \le x_0-\epsilon. \end{cases}$$

This function is differentiable except at $x_0-\epsilon$ and $x_0+\epsilon$, with derivative

$$u'(x) = \begin{cases} 0 & \text{if } x > x_0+\epsilon \\ \frac{1}{2\epsilon} & \text{if } x_0-\epsilon < x < x_0+\epsilon \\ 0 & \text{if } x < x_0-\epsilon. \end{cases}$$

For this $u$, we have

$$\int_0^{\overline{x}} u(x)f(x)\,dx - \int_0^{\overline{x}} u(x)g(x)\,dx = \frac{1}{2\epsilon}\int_{x_0-\epsilon}^{x_0+\epsilon} [G(x)-F(x)]\,dx < 0,$$

where the inequality follows from $F(x) > G(x)$ on the interval. ■

First-order stochastic dominance can be thought of as a "stochastically larger" relationship. It is also interesting to think about ranking random variables in terms of "less risky". To do so, we will restrict attention to comparisons of distributions with the same mean.

Say that random variable $Y$ is a **mean-preserving spread** of random variable $X$ if there is a random variable $Z$ such that:

1. $Y$ has the same distribution as $X + Z$, and
2. $\mathbb{E}(Z \mid X) = 0$ for all $X$.

That is, $Y$ is equal to $X$ plus noise.

**Theorem 3.5.** *Suppose $X$ is a random variable with distribution $F$ and $Y$ is a random variable with distribution $G$, and that $\mathbb{E}(X) = \mathbb{E}(Y)$. Then the following statements are equivalent:*

1. $\int_0^{\overline{x}} u(x)f(x)\,dx \geq \int_0^{\overline{x}} u(x)g(x)\,dx$ *for all concave $u$;*
2. $\int_0^{x} F(x)\,dx \leq \int_0^{x} G(x)\,dx$ *for all $x \in [0, \overline{x}]$; and*
3. *$Y$ is a mean-preserving spread of $X$.*

I'm going to skip the proof, with the following remarks:

1. The equivalence of points 1 and 2 in the theorem has a proof very similar to Theorem 3.4, with the following changes:
   (a) Integrate by parts twice:
   $$\int_0^{\overline{x}} u(x)f(x)\,dx = u(\overline{x}) - u'(\overline{x}) \int_0^{\overline{x}} F(s)\,ds + \int_0^{\overline{x}} u''(x) \int_0^{x} F(s)\,ds\,dx,$$
   and similarly for $\int_0^{\overline{x}} u(x)g(x)\,dx$.
   (b) Integrate by parts in the integral for the expected value of $X$:
   $$\int_0^{\overline{x}} xf(x)\,dx = \overline{x} - \int_0^{\overline{x}} F(s)\,ds,$$
   and similarly for $\int_0^{\overline{x}} xg(x)\,dx$.
   (c) Use the equal means condition and $u''(x) \leq 0$ for all $x$ to compare the expected utilities.
   (d) The clever choice of $u$ in the $2 \Rightarrow 1$ direction has the form $x \mapsto \min(x, x_0)$ for an appropriately chooses $x_0$.
2. The proof of $3 \Rightarrow 1$ is a generalization of the argument in Exercise 3.4.

3. Proofs that either 1 or 2 imply 3 are significantly more involved.

4. Terminology here is a bit confusing. Some references call the condition in point 2 of the Theorem second-order stochastic dominance only when means are equal, while others talk about second-order stochastic dominance without the equal means condition.

We'll conclude with two examples of how useful the idea of mean-preserving spreads can be.

**Example 3.1.** Consider an investor who must divide her wealth $w$ between two assets. These assets have returns $R_1$ and $R_2$, which are independent and identically distributed random variables. The fully diversified portfolio puts half the wealth into each asset; it has return $R = \frac{1}{2}R_1 + \frac{1}{2}R_2$.

Consider some other portfolio, with fraction $\alpha$ in asset 1 and fraction $1 - \alpha$ in asset 2. Its return is

$$\begin{aligned}\alpha R_1 + (1-\alpha)R_2 &= \frac{1}{2}R_1 + \left(\alpha - \frac{1}{2}\right)R_1 + \frac{1}{2}R_2 + \left(1 - \alpha - \frac{1}{2}\right)R_2 \\ &= R + \left(\alpha - \frac{1}{2}\right)R_1 + \left(1 - \alpha - \frac{1}{2}\right)R_2.\end{aligned}$$

By Theorem 3.5, we will know that full diversification is optimal for any risk-averse investor if

$$\mathbb{E}\left(\left(\alpha - \frac{1}{2}\right)R_1 + \left(1 - \alpha - \frac{1}{2}\right)R_2 \mid R\right) = 0$$

for all $R$. By linearity of conditional expectations,

$$\mathbb{E}\left(\left(\alpha - \frac{1}{2}\right)R_1 + \left(1 - \alpha - \frac{1}{2}\right)R_2 \mid R\right) = \left(\alpha - \frac{1}{2}\right)\mathbb{E}(R_1 \mid R) + \left(1 - \alpha - \frac{1}{2}\right)\mathbb{E}(R_2 \mid R).$$

Since $R_1$ and $R_2$ are iid, we must learn the same things about them from observing their sum, so $\mathbb{E}(R_1 \mid R) = \mathbb{E}(R_2 \mid R)$. But then,

$$\mathbb{E}\left(\left(\alpha - \frac{1}{2}\right)R_1 + \left(1 - \alpha - \frac{1}{2}\right)R_2 \mid R\right) = \left(\alpha - \frac{1}{2}\right)\mathbb{E}(R_1 \mid R) + \left(1 - \alpha - \frac{1}{2}\right)\mathbb{E}(R_1 \mid R) = 0.$$

**Example 3.2.** Consider a consumer who lives for two periods, and must decide how much to save from period 1 to period 2. Her income in period 1 is $w_1$ for sure, but her income in period 2 is a random variable $\tilde{w}_2$. For simplicity, assume that the interest rate is 0, so one dollar saved in period 1 yields 1 dollar for period 2, and that no borrowing is allowed. Denoting the amount saved as $s$, consumption in each period is then

$$\begin{aligned} c_1 &= w_1 - s \\ \tilde{c}_2 &= \tilde{w}_2 + s. \end{aligned}$$

(Notice that consumption in period 2 is a random variable.) Finally, suppose that the consumer satisfies the vN-M axioms, and has Bernoulli utility $(c_1, c_2) \mapsto u(c_1) + u(c_2)$, where $u$ is three-times continuously differentiable with $u' > 0$, $u'' < 0$, and $u''' > 0$.

The consumer will choose $s$ to solve

$$\max_{s \geq 0} u(w_1 - s) + \mathbb{E}u(\tilde{w}_2 + s).$$

The first-order condition is

$$-u'\left(w_1 - s\right) + \mathbb{E}u'\left(\tilde{w}_2 + s\right) \leq 0, \quad \text{with equality if } s > 0.$$

This condition is necessary and sufficient for optimization, because $u$ is strictly concave.

Suppose $s^*(\tilde{w}_2)$ is an interior solution, so

$$-u'\left(w_1 - s^*(\tilde{w}_2)\right) + \mathbb{E}u'\left(\tilde{w}_2 + s^*(\tilde{w}_2)\right) = 0, \tag{3.4}$$

and suppose $\hat{w}_2$ is a mean-preserving spread of $\tilde{w}_2$. Then $\hat{w}_2 = \tilde{w}_2 + \epsilon$ for some zero-conditional-mean random variable $\epsilon$. Thus

$$\begin{aligned} \mathbb{E}u'\left(\hat{w}_2 + s^*(\tilde{w}_2)\right) &= \mathbb{E}\left[\mathbb{E}\left(u'\left(\tilde{w}_2 + \epsilon + s^*(\tilde{w}_2)\right) \mid \tilde{w}_2\right)\right] \\ &> \mathbb{E}\left[u'\left(\mathbb{E}\left(\tilde{w}_2 + \epsilon + s^*(\tilde{w}_2) \mid \tilde{w}_2\right)\right)\right] \\ &= \mathbb{E}\left[u'\left(\tilde{w}_2 + s^*(\tilde{w}_2)\right)\right], \end{aligned}$$

where the first equality is the MPS condition and the law of iterated expectations, while the last equality is conditional-mean-zero of $\epsilon$. More interesting is the inequality. It follows from Jensen's inequality and the observation that $u''' > 0$ implies the marginal utility $u'$ is convex.

Combined with Equation 3.4, we see that $s^*(\tilde{w}_2)$ is no longer optimal after the mean-preserving spread. Since $u'$ is decreasing, we need to increase $s$ to compensate. Thus a mean-preserving spread of the second period income leads to an increase in savings.

## Problems

**Exercise 3.1.** One way to construct preferences over monetary lotteries is to evaluate a lottery $L$ by the mean $\mathbb{E}L$ and the variance $\mathrm{var}(L)$. This may or may not be consistent with the von Neumann-Morgenstern axioms.

1. Show that the preferences represented by $U(L) = \mathbb{E}L - \frac{1}{4}\,\mathrm{var}(L)$ do not satisfy the vN-M axioms. (Hint: Consider the mixtures of the lotteries $(1;1)$ and $(1/2, 1/2; 0, 4)$ with the lottery $(1/2, 1/2; 0, 2)$.)

2. Show that the preferences represented by $U(L) = \mathbb{E}L - (\mathbb{E}L)^2 - \mathrm{var}(L)$ do satisfy the axioms.

**Exercise 3.2** (Rubinstein)**.** A decision maker has a preference relation $\succsim$ over the space of lotteries $\mathcal{L}(X)$ on a set of prizes $X$. On Sunday she learns that on Monday she will be told whether she has to choose between $L_1$ and $L_2$ (probability $1 > \alpha > 0$) or between $L_3$ and $L_4$ (probability $1 - \alpha$). She will make her choice at that time.

Here are two possible approaches she can take:

**Approach 1** Delay her decision until Monday ("why bother with the decision now when I can make up my mind tomorrow…").

**Approach 2** Make a contingent decision on Sunday regarding what to do on Monday. That is, she decides what to do if she faces the choice between $L_1$ and $L_2$ and what to do if she faces the choice between $L_3$ and $L_4$ ("On Monday morning I will be so busy …").

1. Formulate Approach 2 as a choice between lotteries.

2. Show that if the preferences of the decision maker satisfy the independence axiom, then her choice under Approach 2 will always be the same as under Approach 1.

**Exercise 3.3.** An investor has wealth $W$ and has to decide how much of it to invest in a risky project that returns $\theta$ per dollar invested, where the expected value of $\theta$ is greater than 0. The balance is invested in a riskless asset that returns $r$ per dollar invested, where $\mathbb{E}\theta > r > 0$. Show that if the investor's utility function is $u(m) = \log(m)$, then the investor will invest a constant fraction of wealth in the risky project.

**Exercise 3.4.** Let lottery $p = (2/3, 1/3; 10, 20)$ and lottery $p' = (1/3, 5/9, 1/9; 5, 15, 30)$.

1. Show that $p'$ corresponds to a random variable that is the sum of the random variable corresponding to $p$ and a conditional-mean-zero random variable.

2. Show that any risk-averse expected utility maximizer prefers $p$ to $p'$. (Show this directly; do not appeal to Theorem 3.5.)

**Exercise 3.5** (MIT)**.** Consider a town not very far from Chicago that is filled with expected utility maximizing citizens, all of whom have Bernoulli utility function $u(w)$. Assume that this utility function is differentiable as many times as you like, and further assume it is strictly increasing and concave.

Every individual has some initial wealth as well as a car which must be parked at metered parking spaces each day. The cost of the meter is given by $m$. If the meter is not paid, it indicates "violation." The town hires police who patrol the area, and with probability $p$ the violation is spotted and a fine $f$ is levied. If the police do not spot the violation, the individual pays nothing.

Consider the problem faced by an individual who maximizes their utility for the day by choosing whether or not to feed the meter (for non-Americans, "feed" is slang for "put money into"). Let $\Delta$ denote the returns to the agent from feeding the meter, that is, the difference between the agent's expected utility from feeding the meter and their expected utility from parking illegally.

1. Suppose that the town council considers funding more police, which increases $p$, versus raising the fine $f$. Compute the elasticity of $\Delta$ with respect to $p$, and compare it to the elasticity of $\Delta$ with respect to $f$. To which policy change are the citizens more responsive?

2. For the remainder of this problem suppose that citizens of this town differ according to their initial wealth, $w_0$, on the range $[\underline{w}, \overline{w}]$. Suppose also that the parameters of the problem are such that $\Delta$ is decreasing in initial wealth, and that we observe some citizens feeding the meters and others parking illegally.

Write a simple expression which determines which group of people feeds meters and which group parks illegally. Show that the assumption that the both groups are nonempty requires assuming that the price of the meter, $m$, is larger than the expected value of the fine, $pf$, from parking illegally.

3. Qualitatively, how does the set of people who feed the meters change with $f$, $m$, and $p$? Prove your answer and interpret your results.

**Exercise 3.6.** Consider an expected utility maximizer with quadratic Bernoulli utility for wealth:

$$u(w) = a + bw + cw^2.$$

1. What restrictions (if any) must be placed on the parameters $a$, $b$, and $c$ for this decision maker to be risk averse?

2. Over what domain of wealth can this Bernoulli utility represent the preferences of a decision maker who prefers more to less?

3. Show that if this function satisfies the restrictions from parts (a) and (b), then the function cannot represent the preferences of a decision maker with decreasing absolute risk aversion.

**Exercise 3.7.** Prove that a Bernoulli utility function $u$ has constant absolute risk aversion $\lambda$ if and only if it has the form

$$u(x) = -\frac{a}{\lambda}e^{-\lambda x} + b$$

for constants $a > 0$ and $b$.

**Exercise 3.8.** Consider the insurance problem from lecture, and assume that insurance is actuarially unfair. Also suppose the buyer has decreasing absolute risk aversion. Show that her demand for insurance is decreasing in her initial wealth.

**Exercise 3.9.**

1. Show that, if $F$ first-order stochastically dominates $G$, then the mean of $x$ under $F$ is at least as large of the mean of $x$ under $G$. That is, that $\int xf(x)\,dx \geq \int xg(x)\,dx$.

2. Give an example where $\int xf(x)\,dx > \int xg(x)\,dx$ but $F$ does not first-order stochastically dominate $G$.

# Chapter 4

# Consumer Theory: A First Look

We are now going to turn to a cornerstone of traditional microeconomics: the problem of a consumer facing linear prices.

## 4.1 The Setting

### 4.1.1 Consumer Preferences

The consumer can conceivably consume any bundle of goods in the set $X \subset R^n$. The subset is *not* defined by budget considerations; in the language of abstract decision theory, it is the set $A$. It instead reflects considerations like "consumption of food must be nonnegative". Indeed, we will often use the special case $X = \mathbb{R}^n_+$. But that assumption is sometimes too restrictive. For example, we can capture restrictions based on location with the following trick: let good 1 be ice cream in Chicago, and good 2 be ice cream in New York (and assume there are no other goods). Then we can define

$$X = \{(x, 0) \mid x \geq 0\} \cup \{(0, x) \mid x \geq 0\}$$

to capture the idea that the consumer cannot simultaneously consume ice cream in two different cities.

By the way, this trick of defining multiple copies of physically similar commodities is useful in other contexts. We can distinguish ice cream today from ice cream tomorrow, and ice cream if it is hot outside from ice cream if it is cold outside. These tricks mean that, even though we are developing the theory as if the world is static with no uncertainty, the results are more broadly applicable.

As always, we will assume that the consumer has preferences $\succsim$ on $X$ that are complete, transitive, and continuous. But the context suggests some additional assumptions.

First, it is often reasonable to assume that the consumer prefers more than less. One version of this is: preferences are **monotonic** if $x \geq y$ implies $x \succsim y$. We can strengthen this to rule out indifference: preferences are **strictly monotonic** if $x > y$ implies $x \succ y$.

These assumptions will often be reasonable, but are sometimes too strong. Is there really no limit on how much ice cream you want to eat? And do you want any number of nickel-alloy-coated electrodes? I doubt it. Nonetheless, I also don't think many consumers are going to be happy stopping short of spending their entire income. (Recall that savings can be handled with dated commodities.) The weakest assumption that captures this is as follows. Preferences are **locally non-satiated** if, for any $x \in X$ and any $\epsilon > 0$, there is a $y \in X$ with $\|x - y\| < \epsilon$ and $y \succ x$.

Strict monotonicity⇒monotonicity⇒local non-satiation. Translation of these concepts to utility representations are immediate.

Another class of assumptions about preferences concern preference for variety, or mixing. In intermediate micro, this captured by the assumption that indifference curves are bowed in the direction of the origin. We can make this precise as follows. Preferences are **convex** if, for all $y \in X$, the set $\{x \in X \mid x \succsim y\}$ is convex. (The set that appears in the definition is called the **upper contour set** of $y$.

Clearly, preferences are convex if and only if every utility representation $U$ has the property that, for each $y$, the set $\{x \mid U(x) \geq U(y)\}$ is convex. In that case, we say that the function is **quasiconcave.** An alternative characterization of quasiconvity is this: for any $x$ and $x'$ and any $\lambda \in [0, 1]$,

$$U(x) \geq U(x') \text{ implies } U(\lambda x + (1 - \lambda)x') \geq U(x').$$

Preferences are **strictly convex** if $x^0 \succsim y$ and $x^1 \succsim y$ (with $x^0 \neq x^1$) imply $\lambda x^0 + (1 - \lambda)x^1 \succ y$ for $0 < \lambda < 1$. In the two variable case, this means the indifference curves have no "flat" segments. In utility terms, this corresponds to **strict quasi concavity**: for any $x$ and $x'$ and any $\lambda \in (0, 1)$,

$$U(x) \geq U(x') \text{ implies } U(\lambda x + (1 - \lambda)x') > U(x').$$

Do not confuse concavity and quasiconcavity. Concave functions must be quasiconcave, but the converse is not true. To see this, observe that any increasing function on the interval $[0, 1]$ is quasiconcave, even the ones that are strictly convex. What is true is that if a monotone transformation $f \circ U$ is concave, then $U$ is quasiconcave.

### 4.1.2 The Consumer's Problem

The classic consumer problem is to choose a consumption bundle to maximize utility given fixed, linear prices:

$$\max_{x \in X} U(x) \tag{4.1}$$

$$\text{st } \ px \leq I \tag{4.2}$$

Write the solution as $x^*(p, I)$. This is the **demand correspondence**.

We will make strong assumptions to take the most direct route to the classical theory:

1. Preferences are continuous.

2. Preferences are locally non-satiated.

3. Preferences are strictly convex.

These give us some useful properties:

**Theorem 4.1.** *1. For any $\lambda > 0$, we have $x^*(p, I) = x^*(\lambda p, \lambda I)$.*

*2. If preferences are continuous and all prices are strictly positive, the consumer's problem has a solution.*

*3. If preferences are locally non-satiated, then at any solution the consumer spends all of her income: $px^*(p, I) = I$.*

*4. If $X$ is convex and preferences are strictly convex, there is at most one solution.*

*5. If $X$ is convex and there is a unique solution for all $(p, I)$, then the demand function is continuous.*

**Proof.**

1. The feasible sets are identical.

2. If prices are positive, the budget set is compact. If preferences are continuous, we can choose a continuous utility representation.

3. Suppose $x$ is a bundle that satisfies the budget with strict inequality: $px < I$. Then there is an $\epsilon > 0$ such that $yp < I$ for all $y$ with $\|x - y\| < \epsilon$. But then local non-satiation implies that there is a $y$ that satisfies both $y \succ x$ and $yp < I$, so $x$ does not maximize utility.

4. If $x$ and $y \neq x$ both solve the consumer's problem, then we must have $x \sim y$. But strict convexity then says $\frac{1}{2}x + \frac{1}{2}y \succ x$, contradicting optimality of $x$.

5. This is immediate from a mathematical result called Berge's Theorem of the Maximum. You can read about it in Riley.

■

The first result says that only relative prices matter.

Since we assume continuity, local non-satiation, and strict convexity, we get a continuous demand function.

We'd like to be able to answer comparative statics questions about demand. For example:

- How does demand vary with income, for fixed prices? (Income effect, Engel curve)

- How does demand vary with price, for fixed income? (Price effect)

## 4.2 Solving the Consumer's Problem

To apply these mathematical tools the consumer's problem, add the assumptions:

4. $U$ is continuously differentiable.

5. For all $x \in \mathbb{R}^k_+$, we have $\mathbf{D}U(x) > 0$.

Assumptions 1 through 5 are enough for us to appeal to the Kuhn-Tucker Theorem.

**Proposition 4.1.** *Assume 1 through 5. Then $x^*$ solves the consumer's problem for prices $p$ and income $I$ if and only if there is a $\lambda > 0$ such that*

$$\frac{\partial U}{\partial x_j}(x^*) \leq \lambda p_j \tag{4.3}$$

$$x_j^* \left( \frac{\partial U}{\partial x_j}(x^*) - \lambda p_j \right) = 0 \tag{4.4}$$

$$px^* = I \tag{4.5}$$

**Proof.** A rewriting of the Kuhn-Tucker sufficiency conditions gives the conclusion with $\lambda \geq 0$.

Since we have $\mathbf{D}U(x) > 0$, we know that $\frac{\partial U}{\partial x_j}(x^*) > 0$ for at least one $j$. If $\lambda = 0$, we would have

$$\frac{\partial U}{\partial x_j}(x^*) > 0 = \lambda p_j,$$

a contradiction. Thus $\lambda > 0$. ■

We also freely assume the following whenever we want to ignore corner solutions:

Let $(x^n)$ be a sequence with limit $x$, where $x_j = 0$. Then

$$\lim_{x \to 0} \frac{\partial U}{\partial x_j}(x) = \infty.$$

If this holds, then the FOC for $x_j$ cannot be satisfied with $x_j = 0$.

Let's look at a couple of examples.

**Example 4.1.** Utility is **Cobb-Douglas** if

$$U(x_1, \ldots, x_n) = \prod_i x_i^{\alpha_i}$$

for coefficients $\alpha_i > 0$. The analysis will be simplest if we apply a log transformation, working with

$$u(x_1, \ldots, x_n) = \sum_i \alpha_i \log x_i.$$

Once we do this, we might as well divide by a constant to get $\sum_i \alpha_i = 1$. This gives a strictly concave objective, and the marginal utility of $x_i$ is $\frac{\alpha_i}{x_i}$, which diverges to infinity as $x_i \to 0$. Thus we know that any solution will be interior, and that the FOCs are sufficient. The FOCs simplify to:

$$\frac{\alpha_i}{x_i^*} = \lambda p_i \quad \text{for all } i$$
$$px^* = I.$$

Substitute $\frac{\alpha_i}{\lambda} = p_i x_i^*$ into the budget constraint to get

$$\sum_i \frac{\alpha_i}{\lambda} = I,$$

or

$$\lambda = \frac{1}{I}.$$

Substitute back into the FOC for $x_i$ to get demand:

$$x_i^*(p, I) = \frac{\alpha_i I}{p_i}.$$

The key implication of Cobb-Douglas demand is that the expenditure share on good $i$, namely $\frac{p_i x_i^*}{I}$, is constant at $\alpha_i$.

**Example 4.2.** Utility is **quasi-linear** in the first good if it has the form

$$U(x_1, \dots, x_n) = x_1 + v(x_2, \dots, x_n)$$

for some function $v$. Let's consider the case of $n = 2$ and $v$ strictly increasing and strictly concave. Also assume that $v$ satisfies the **Inada conditions**: $\lim_{x \to 0} v'(x) = \infty$ and $\lim_{x \to \infty} v'(x) = 0$. The FOCs are

$$\begin{aligned} 1 &\leq \lambda p_1 \\ v'(x_2) &\leq \lambda p_2 \\ px &= I, \end{aligned}$$

with complementary slackness.

The first Inada condition ensures that $x_2^* > 0$. We need to consider two cases for $x_1^*$. First, $x_1^*$ might be positive. In this case, the FOCs become

$$\begin{aligned} 1 &= \lambda p_1 \\ v'(x_2) &= \lambda p_2 \\ px &= I. \end{aligned}$$

Eliminate $\lambda$ to get

$$v'(x_2) = \frac{p_2}{p_1}.$$

Since $v$ is strictly concave, this has a unique solution

$$\hat{x}_2\left(\frac{p_2}{p_1}\right) = (v')^{-1}\left(\frac{p_2}{p_1}\right).$$

If $\hat{x}_2(p_2/p_1)$ is consumed of good 2, then $I - p_2\hat{x}_2(p_2/p_1)$ is left over for good 1. If this is non-negative, then demand is

$$\begin{aligned} x_1^*(p, I) &= \frac{I - p_2\hat{x}_2(p_2/p_1)}{p_1} \\ x_2^*(p, I) &= \hat{x}_2(p_2/p_1). \end{aligned}$$

However, if $I - p_2\hat{x}_2(p_2/p_1) < 0$, the previous solution will violate the lower bound on $x_1$. In that case we must look for a corner solution. The FOCs become:

$$\begin{aligned} 1 &\leq \lambda p_1 \\ v'(x_2) &= \lambda p_2 \\ p_2 x_2 &= I. \end{aligned}$$

Substitute the budget into the FOC for $x_2$ to get

$$v'\left(\frac{I}{p_2}\right) = \lambda p_2.$$

Eliminate $\lambda$ to get

$$v'\left(\frac{I}{p_2}\right) \geq \frac{p_2}{p_1}.$$

Since $\hat{x}_2 > \frac{I}{p_2}$ and $v$ is strictly concave, this is in fact a solution. Demand is

$$\begin{aligned} x_1^*(p, I) &= 0 \\ x_2^*(p, I) &= \frac{I}{p_2}. \end{aligned}$$

Focus on demand for good 2:

$$x_2^*(p,I) = \begin{cases} I/p_2 & \text{if } \hat{x}_2 > \frac{I}{p_2} \\ \hat{x}_2\left(\frac{p_2}{p_1}\right) & \text{if } \hat{x}_2 \leq \frac{I}{p_2} \end{cases}$$

- Demand is (weakly) decreasing in $p_2$.
- If both goods are demanded in positive amounts, demand for good 2 is independent of income.
- Demand fails to be differentiable at $\hat{x}_2 = \frac{I}{p_2}$, even though utility is as differentiable as we could want.

In addition to solving specific examples, we can also use the Kuhn-Tucker conditions to recover the intermediate micro characterization of the consumer's optimum. Suppose two goods, $j$ and $k$, are both consumed in positive amounts. Then the FOCs

$$\frac{\partial U(x^*)}{\partial x_j} = \lambda p_j \quad \text{and} \quad \frac{\partial U(x^*)}{\partial x_k} = \lambda p_k$$

can be combined to give

$$\frac{\frac{\partial U(x^*)}{\partial x_j}}{p_j} = \frac{\frac{\partial U(x^*)}{\partial x_k}}{p_k}.$$

We can get a better understanding of these conditions by thinking about what the derivatives really mean. Consider some consumption bundle $x \gg 0$ that satisfies the budget constraint with equality: $px = I$. Reducing consumption of good $j$ by $\epsilon > 0$ will free up $p_j\epsilon$ dollars, which will buy $\frac{p_j}{p_k}\epsilon$ units of good $k$. This changes consumption from $x$ to $x + \epsilon d$, where $d$ is a vector with $-1$ in the $j$th place, $\frac{p_j}{p_k}$ in the $k$th place, and 0 everywhere else. Taylor's theorem says

$$U(x+\epsilon d) - U(x) \approx \mathbf{D}U(x)\cdot(\epsilon d) = \epsilon\left(-\frac{\partial U(x^*)}{\partial x_j} + \frac{p_j}{p_k}\frac{\partial U(x^*)}{\partial x_k}\right).$$

If the bracketed term is positive, then, for small enough $\epsilon$, the change raises utility. If the bracketed term is negative, then, for small enough $\epsilon$, a change to $x - \epsilon d$ raises utility. At an optimum, neither of these can be true, so the bracketed term must be zero.

Think about this as follows: the gradient $\mathbf{D}U(x)$ tells us the "improving directions" for $U$ starting at $x$. Formally, say that vector $d$ is an **improving direction at** $x$ if there is an $\bar{\epsilon}$ such that $\bar{\epsilon} > \epsilon > 0$ implies $x + \epsilon \cdot d \succ x$. The proceeding Taylor argument can be modified to show that, if preferences have a differentiable utility representation, the set of improving directions is the set $\{d \mid \mathbf{D}U(x) \cdot d > 0\}$.

And that observation tells us what the assumption of differentiable utility really means. The assumption that preferences $\succsim$ can be represented by a differentiable utility function $U$ just means that, for each $x$, there is a vector $v(x)$ such that the set of improving directions $d$ is $\{d \mid v \cdot d > 0\}$. That is, the set of improving directions is a **half-space** defined by the hyperplane normal to $v$.

An example of preferences that are not differentiable are given by **Leontieff** utility: $U(x_1, \ldots, x_n) = \min x_k$. Any improving direction must have all components positive. But that is inconsistent with the set of improving directions being a half-space.

So far, we've seen that $\mathbf{D}U(x)$ tells us the directions in which small changes increase utility. As is so often the case, adding convexity allows a "local to global" translation: If preferences are convex as well as differentiable, then the upper contour set of $x$ lies on one side of the hyperplane normal to $\mathbf{D}U(x)$.

We will show this in a more general setting, so we can use it later to study production and welfare economics.

**Theorem 4.2.** *If $g$ is differentiable and $Y = \{y \mid g(y) \geq g(y^0)\}$ is convex, then*

$$y \in Y \implies \mathbf{D}g(y^0) \cdot (y - y^0) \geq 0.$$

**Proof.** Choose any $y \in Y$. By convexity of $Y$,

$$y^\lambda \equiv \lambda y + (1 - \lambda) y^0 \in Y$$

for all $\lambda \in [0, 1]$. Since $Y$ is an upper contour set, this is equivalent to

$$g(y^\lambda) - g(y^0) \geq 0.$$

Define $h(\lambda) = g(y^\lambda) = g(y^0 + \lambda(y - y^0))$. Then

$$\frac{h(\lambda) - h(0)}{\lambda} = \frac{g(y^0 + \lambda(y - y^0)) - g(y^0)}{\lambda}.$$

For any $\lambda \in (0, 1)$, the RHS is $\geq 0$, since $y^\lambda \in Y$. Thus the limit

$$\lim_{\lambda \to 0} \frac{h(\lambda) - h(0)}{\lambda} \geq 0.$$

But that limit is just the derivative of $h$, so $h'(0) \geq 0$.

By the chain rule,

$$\frac{dh}{d\lambda}(\lambda) = \mathbf{D}g(y^0 + \lambda(y - y^0)) \cdot (y - y^0).$$

Set $\lambda = 0$ then gives

$$\mathbf{D}g(y^0)(y - y^0) \geq 0,$$

as desired. ■

## 4.3 Empirical Implications of CP

Now we turn to a discussion of what the classical theory of the consumer implies for data. Consider the following situation. You observe the bundle that some particular consumer would select for each of a finite set of price-income combinations. (Perhaps this is hypothetical data from a survey; perhaps it is actual choice data from a sequence of decision problems that are sufficiently delinked to treat as separate problems.) What observations would refute the hypothesis that the consumer made her choices according to the model of this chapter?

Start with the case of just two observations. In particular, suppose there are two goods, 1 and 2, and the price-income pairs are $(p^a, I^a) = ((1, 2), 10)$ and $(p^b, I^b) = ((2, 1), 10)$. And suppose the consumer chooses bundle $x^a = (2, 4)$ when facing $(p^a, I^a)$ and chooses $x^b = (4, 2)$ when facing $(p^b, I^b)$. In each case, the consumer spends her entire budget. But $p^a \cdot x^b = 8 < 10$, and similarly for $p^b \cdot x^a$.

These two inequalities are not necessarily a problem for the hypothesis that the consumer is maximizing a rational preference relation. After all, she might be globally indifferent, in which case any choice at any price-income pair is ok. But that response rather trivializes the question. And if we add the assumption that the consumer's preferences are supposed to be locally non-satiated, the two inequalities are a problem for our hypothesis. To see this, we will use the following:

**Lemma 4.1.** *Suppose that a consumer maximizes a complete, transitive, and locally non-satiated preference by choosing $x$ when prices are $p$ and income is $I$. If $x'$ is some bundle such that $px' \leq I$, then $x \succsim x'$. And if $x'$ is some bundle such that $px' < I$, then $x \succ x'$.*

**Proof.** If $px' \leq I$ but $x' \succ x$, then $x$ would not be preference maximizing at prices $p$ and income $I$. Now consider some $x'$ such that $px' < I$. There is an $\epsilon > 0$ such that, for any $z$

with $\|z - x'\| < \epsilon$, we have $pz < I$. But local non-satiation implies that, for at least one of those $z$, we have $z \succ x'$. If $x' \succsim x$, transitivity would imply $z \succ x$, contradicting optimality of $x$. ■

Going back to the example, since $p^a x^b < I^a$, we must have $x^a \succ x^b$. But $p^b x^a < I^b$ implies $x^b \succ x^a$. These two strict preferences can't both hold, so the observations are not consistent with the standard model.

In that example, the violation of the hypothesis was particularly stark. But more subtle violations can be detected using transitivity. Consider the following choices.[1]

- At prices $(10, 10, 10)$ and income 300, the consumer chooses $(10, 10, 10)$.
- At prices $(10, 1, 2)$ and income 130, the consumer chooses $(9, 25, 7.5)$.
- At prices $(1, 1, 10)$ and income 11, the consumer chooses $(15, 5, 9)$.

Applying Lemma 4.1 to the first two bullet points tells us that $(9, 25, 7.5) \succsim (10, 10, 10)$. Applying Lemma 4.1 to the last two bullet points tells us that $(15, 5, 9) \succ (9, 25, 7.5)$. And applying Lemma 4.1 to the first and last bullet points tells us that $(10, 10, 10) \succ (15, 5, 9)$. But those three preferences together are inconsistent with transitivity.

You might fear now that we can keep on creating ever more new problems. But don't worry—we have in fact seen examples of everything that can go wrong. The main technical result of this section is designed to make this reassurance precise.

**Definition 4.1.** *Suppose there is a finite set of demand observation of the form: $x^1$ is chosen at $(p^1, I^1)$, $x^2$ is chosen at $(p^2, I^2)$, and so on.*

1. *If $p^i x^j \leq I^i$, then $x^i$ is* directly revealed preferred to *$x^j$, written $x^i \succsim^D x^j$. If $p^i x^j < I^i$, then $x^i$ is* directly revealed strictly preferred to *$x^j$, written $x^i \succ^D x^j$.*

2. *$x^i$ is* revealed preferred to *$x^j$, written $x^i \succsim^R x^j$, if either $x^i \succsim^D x^j$ or there is a sequence $(x^k)_{k=1}^K$ such that*

$$x^i \succsim^D x^1 \succsim^D \cdots \succsim^D x^K \succsim^D x^j.$$

*If, furthermore, at least one of the directly revealed preferences is strict, then $x^i$ is* revealed strictly preferred to *$x^j$, written $x^i \succ^N x^j$.*

[1] This example is taken from Kreps (2012), p. 67.

3. *If the set of observations satisfy the* generalized axiom of revealed preference, *or* GARP *if there are no cycles in the revealed strictly preferred to relation, that is, if there is no $i$ such that $x^i \succ^R x^i$.*

**Theorem 4.3** (Afriat)**.** *If a finite set of demand data violate GARP, then the data are inconsistent with the maximization of a complete, transitive, and locally non-satiated preference relation. Conversely, if a finite set of demand data satisfy GARP, then the data are consistent with consistent with maximization of a preference relation that is complete, transitive, strictly increasing, continuous, and convex.*

The proof of the first claim is an easy combination of Lemma 4.1 and the argument about transitivity in the second example. The proof of the second claim is much harder; see Kreps (2012) §4.2 if you are interested, though be warned the proof will not teach you any techniques useful elsewhere in microeconomics.

This definition and theorem are a lot to digest all at once, so let me give you some pointers.

1. In the definition, I never say that $i \neq j$. Thus, if $p^i x^i < I^i$, we can conclude that $x^i \succ^D x^i$, which implies $x^i \succ^R x^i$, which is a violation of GARP.

2. Comparing the two statements in the Theorem shows that strengthening local non-satiation to strict monotonicity, adding continuity, or adding convexity to the assumptions of completeness, transitivity, and local non-satiation does nothing to the model's ability to accommodate the data. Said differently, those extra assumptions have no testable implications for market demand data.

3. The last point does not mean that the additional properties have no testable assumptions at all. For example, we could ask the consumer to rank the three bundles $x$, $x'$, and $\frac{1}{2}x + \frac{1}{2}x'$. If she tells us that $x \succ \frac{1}{2}x + \frac{1}{2}x'$ and $x' \succ \frac{1}{2}x + \frac{1}{2}x'$, then know she does not have convex preferences. The point, rather, is that market demand data can never reveal this failure of convexity.

### 4.3.1 Empirical Applications

One way that GARP and Afriat's theorem are useful is in applications with real data. Two prominent examples come from experimental studies of altruism and from the economics of the family.[2]

[2]This subsection borrows heavily from lecture slides by Parag Pathak.

Experimental economists study altruism in the context of the **dictator game:** one subject chooses how to divide a sum of money between herself and a second subject. The second subject makes no decisions—whatever the first subject decides is implemented. If subjects were rational and selfish, maximizing their own wealth, the first subject would simply keep all of the money. That is not what happens—many subjects give away significant amounts.

Selfishness is often a useful assumption, but it is certainly not entailed by rationality. So it is interesting to ask if behavior in the dictator game is consistent with rational and altruistic preferences. A famous paper by Andreoni and Miller (*Econometrica,* 2002) tackles this.

They have subjects make a sequence of decisions of the form:

> Here are $M$ tokens. Divide them between yourself and an anonymous other subject. You will get $x$ cents for each token you keep, and the other subject will get $y$ cents for each token you give them.

Think of the parameters $(M, x, y)$ as determining income and the relative price of money for the self and money for the other—the subject might be thought of as solving

$$\begin{aligned} &\max_{c_s, c_o} u(c_s, c_o) \\ &\text{st } \frac{1}{x}c_s + \frac{1}{y}c_o \leq M. \end{aligned}$$

If this interpretation is a good approximation to what people do in the experiment, their choices should satisfy GARP. Andreoni and Miller find that 156 of their 176 subjects are fully consistent with GARP, and most of the others would have no violations if only a few tokens were reallocated. A couple of subjects had many violations.

Another example concerns the economics of the family. Although the theory we have developed is usually motivated as a theory of the individual, it is often applied at the household level. Whether or not this is a good idea depends on how decision-making inside of families works. One model of the family (introduced by Gary Becker) has a single member (typically the father) making the decisions, usually (hopefully?) with altruistic motivation. If that is how families work, the standard model of the consumer will work well. But it will be more problematic if there is non-trivial bargaining within the family. In that case, the distribution of bargaining power, and thus decision power, can be affected by which member of the family brings certain resources.

Duflo (*World Bank Economic Review,* 2003) found a nice natural experiment to shed light on this question, based on an unexpected grant of a large pension to black South

African retirees. Whether the money was received by the grandmother or grandfather had a big impact on how the funds were spent. More money is spent on female children when grandmothers get the money than when grandfathers do. And the effect is more pronounced for the mother's mother than for the father's mother.

A Beckerian household would satisfy GARP, and thus could only display the effect found by Duflo if the decision maker were indifferent between spending on daughters or other things. That is possible, I guess. But interpreting the regularity of the association as a strict preference for spending more on daughters from female-acquired income than from male-acquired income implies that GARP is violated, and the household is not Beckerian.

### 4.3.2 Downward-Sloping Demand?

You will sometimes hear people talk about the "Law of Demand"—the claim that demand curves slope down. This is a pretty good approximation empirically, but it is not in fact an implication of the model we are considering in this chapter.

From intermediate micro, you know that goods for which the law of demand fails are called Giffen goods. And you have probably seen pictures of indifference curves that generate Giffen behavior. We can use Afriat's Theorem to rigorize such pictures.

Fix prices $p$ and income $I$, and suppose the consumer chooses $x$. Let new prices be $p'$ where $p_j = p'_j$ for all goods $j \neq i$, and $p'_i > p_i$. Let $x'$ be any bundle that satisfies the budget constraint with equality: $p'x' = I$. In particular, we allow for $x'_i > x_i$

Given the construction, we must have $p'x > px = I$. Since the original bundle is not affordable at the new prices, there cannot be any violation of GARP in just these two observations. And Afriat's Theorem then implies that there is a well-behaved utility function that is maximized at $x$ when prices are $p$ and at $x'$ when prices are $p'$.

Another thing that you recall from intermediate micro is that stronger results are possible if we consider price changes accompanied by "compensating" income changes. We can use Afriat's Theorem to study the Slutsky compensation, the one that keeps income fixed at the whatever level makes the original bundle just affordable at the new prices.

**Theorem 4.4.** *Suppose a consumer chooses $x$ when facing prices $p$ and income $I$, and chooses $x^s$ when facing prices are $p'$, where $p_j = p'_j$ for $j \neq i$ and $p'_i > p_i$, and income $p'x$. If the choices maximize a complete, transitive, and locally non-satiated preference relation, then $x^s_i \leq x_i$.*

**Proof.** By local non-satiation, $x^s$ must satisfy the budget constraint with equality: $p'x^s = p'x$. Note that this immediately implies $x^s \succsim^R x$

Writing out the budget constraint, we have

$$\sum_{j\neq i} p'_j x^s_j + p'_i x^s_i = \sum_{j\neq i} p'_j x_j + p'_i x_i.$$

Substitute $p'_j = p_j$ for $j \neq i$ to get

$$\sum_{j\neq i} p_j x^s_j + p'_i x^s_i = \sum_{j\neq i} p_j x_j + p'_i x_i.$$

Suppose, to get a contradiction, that $x^s_i > x_i$. Since we know that $p'_i > p_i$, we can conclude that $(p'_i - p_i)x^s_i > (p'_i - p_i)x_i$. Subtract the larger term from the LHS of the displayed equation to get

$$\sum_{j\neq i} p_j x^s_j + p'_i x^s_i - (p'_i - p_i)x^s_i = \sum_{j\neq i} p_j x^s_j + p_i x^s_i.$$

Subtract the smaller term from the RHS of the display to get

$$\sum_{j\neq i} p_j x_j + p'_i x_i - (p'_i - p_i)x_i = \sum_{j\neq i} p_j x_j + p_i x_i.$$

Since we subtracted the larger term from the LHS of the equality, we have

$$\sum_{j\neq i} p_j x^s_j + p_i x^s_i < \sum_{j\neq i} p_j x_j + p_i x_i,$$

or $px^s < px$. And that implies that $x \succ^R x^s$, so we have a violation of GARP. ■

The name "Slutsky compensation" might have already told you what is coming next. We can use income effects to formally link the (observable) uncompensated demand change with the (signable) compensated demand change. This is the *Slutsky equation.*

As before, write $x$ for demand at prices $p$ and income $I$, and $x^s$ for demand at prices $p'$ and income $p'x$. In addition, write $x'$ for demand at prices $p'$ and income $I$. Since

$$x'_i - x_i = (x^s_i - x_i) - (x^s_i - x'_i),$$

we can divide by $\Delta p_i = p'_i - p_i > 0$ to get

$$\frac{x'_i - x_i}{\Delta p_i} = \frac{x^s_i - x_i}{\Delta p_i} - \frac{x^s_i - x'_i}{\Delta p_i}. \tag{4.6}$$

The Slutsky compensation adjusts the consumer's income by $\Delta I = \Delta p_i \cdot x_i$. Substitute this into equation 4.6 to get

$$\frac{x'_i - x_i}{\Delta p_i} = \frac{x^s_i - x_i}{\Delta p_i} - x_i \frac{x^s_i - x'_i}{\Delta I}.$$

The first term on the right-hand side is the compensated effect that is signed by Theorem 4.4. The second term depends on the difference $x^s_i - x'_i$. These demands arise from the same prices, but with different incomes. Thus the ratio $\frac{x^s_i - x'_i}{\Delta I}$ has a natural interpretation as an income effect. This income effect is weighted by the initial demand for good $i$.

### 4.3.3 Aggregating Demand

There is another gap between what the theory predicts and the usual statement of the "Law of Demand". Namely, the law is usually thought of as holding for market demand, while the theory we have been developing is at the individual level.

If aggregating demand across consumers preserved the restrictions that individual demand has to satisfy, then this would be a reasonable lapse. But it does not.

Afriat's Theorem makes this easy to see. Suppose that, in a two-consumer, two-good economy, each consumer always has income 1000. If prices are $(10, 10)$, consumer 1 chooses $(25, 75)$ and consumer 2 chooses $(25, 75)$. If prices are $(15, 5)$, consumer 1 chooses $(40, 80)$ and consumer 2 chooses $(64, 8)$. You should verify that neither consumer violates GARP, and so each is consistent with our theory.

Now calculate market demands. If prices are $(10, 10)$, market demand is $(100, 100)$. If prices are $(15, 5)$, market demand is $(104, 88)$. These demands violate GARP—$(100, 100)$ equals total income at both price vectors, and $(10, 10) \cdot (104, 88) = 1920 < 2000$.

## Problems

**Exercise 4.1.** Suppose $u$ is a utility representation of preferences $\succsim$ and $f$ is a strictly increasing function. Show that if $f \circ u$ is concave, then preferences are convex.

**Exercise 4.2.** Suppose $u$ is a utility representation of preferences $\succsim$. Show that, if there is a strictly increasing function $f$ such that

$$f \circ u(x) = \sum_{j=1}^{m} v_j(x),$$

where each $v_j$ is concave, then preferences are convex.

**Exercise 4.3** (Rubinstein)**.** Consider the preference relations on the interval $[0,1]$ that are continuous. What can you say about those preferences which are also strictly convex?

**Exercise 4.4.** Consider the following three utility functions (in each case, $\alpha_1, \alpha_2 > 0$):

1. $U(x_1, x_2) = \alpha_1\sqrt{x_1} + \alpha_2\sqrt{x_2}$
2. $U(x_1, x_2) = \alpha_1 x_1 + \alpha_2 x_2$
3. $U(x_1, x_2) = \min(\alpha_1 x_1, \alpha_2 x_2)$

For each, answer the following:

(a) Are the preferences monotone? Strictly monotone?

(b) Are the preferences convex? Strictly convex?

(c) For each, calculate the demands for strictly positive prices $(p_1, p_2) \gg 0$ and income $I$. (For utility functions (1) and (2), use the Kuhn-Tucker conditions. For part (3), explain why the Kuhn-Tucker Theorem does not apply, and then find the demands anyway.)

**Exercise 4.5.** Suppose the consumer's utility function is defined as follows:

$$U(x_1, x_2, x_3, x_4) = \min(x_1 \cdot x_2, x_3 \cdot x_4).$$

Find the demand function.

**Exercise 4.6.** An infinitely lived consumer owns 1 unit of cake that she consumes over her lifetime. The cake is perfectly storable and she will receive no more than she has now. Consumption of cake in period $t$ is denoted $x_t$, and her lifetime utility function is given by

$$U(x_0, x_1, \ldots) = \sum_{t=0}^{\infty} \delta^t \log x_t,$$

where $0 < \delta < 1$.

Calculate her optimal level of cake consumption in each period.

(Note: The statement of the Kuhn-Tucker theorem in class was for finite dimensional problems. But it also applies to this problem, even though there are infinitely many choice variables.)

**Exercise 4.7.** A consumer consumes 2 commodities, wheat and candy. His utility from consuming $w$ units of wheat and $c$ units of candy is

$$3 \log w + 2 \log c.$$

He faces 4 constraints:

- Consumption of each good must be nonnegative.
- The consumer has \$10 to spend, and the price of each good is \$1.
- The consumer is on a diet, and cannot consume more than 1550 calories. A unit of wheat has 150 calories, and a unit of candy has 200 calories.

Follow the following steps to solve this consumer's problem:

1. Derive the Kuhn-Tucker optimality conditions for this consumer's problem.
2. Are the conditions derived above sufficient for this problem? Why or why not?
3. Explain why the conditions from part (1) imply consumption of both commodities must be positive.
4. Explain why the conditions from part (1) imply that at least one of the budget and calorie constraint must bind.
5. Look for a solution the the conditions from part (1) in which only the budget constraint binds.
6. Look for a solution the the conditions from part (1) in which only the calorie constraint binds.
7. Look for a solution the the conditions from part (1) in which both constraints bind.

**Exercise 4.8** (Rubinstein)**.** Consider a consumer with a preference relation in a world with two goods, $X$ (an aggregated consumption good) and $M$ ("membership in a club", for example), which can be consumed or not. In other words, the consumption of $X$ can be any nonnegative real number, while the consumption of $M$ must be either 0 or 1.

Assume that the consumers preferences are strictly monotonic and continuous and satisfy the following property:

| Prices | | | Income | Demand | | |
|---|---|---|---|---|---|---|
| $p_1$ | $p_2$ | $p_3$ | $Y$ | $x_1$ | $x_2$ | $x_3$ |
| 1 | 1 | 1 | 20 | 10 | 5 | 5 |
| 3 | 1 | 1 | 20 | 3 | 5 | 6 |
| 1 | 2 | 2 | 25 | 13 | 3 | 3 |
| 1 | 1 | 2 | 20 | 15 | 3 | 1 |

Table 4.1: Data for Exercise 4.9.

*Property E:* For every $x$, there is a $y$ such that $(y, 0) \succ (x, 1)$ (i.e., there is always some amount of the aggregated consumption good that can compensate for the loss of membership).

1. Show that the consumer's preferences can be represented by a utility function of the form:

$$u(x,m) = \begin{cases} x & \text{if } m = 0 \\ x + g(x) & \text{if } m = 1. \end{cases}$$

2. Explain why continuity and strong monotonicity (without property E) are not sufficient for the result in part 1.

3. Calculate the consumer's demand function.

**Exercise 4.9** (Kreps)**.** In a three good world, a consumer has demands given by Table 4.3.3. Are these choices consistent with the maximization of a complete, transitive, and locally non-satiated preference relation?

**Exercise 4.10.** True, false, or uncertain. Explain your answer.

1. "You cannot derive a demand function for lexicographic preferences."

2. "The behavior of a consumer with lexicographic preferences is empirically indistinguishable from that of a consumer who only gets utility from a single good."

# Chapter 5

# Production

The next important class of decision-maker we will study are *producers*. The intermediate micro treatment of producers goes like this: a firm produces output $Y$ using inputs labor, $L$, and capital, $K$, according to the production function $F$. The firm buys inputs at prices $w$ for labor and $r$ for capital, and sells output at price $p$. The firms wants to maximize profit, so it solves

$$\max_{L,K} pF(L,K) - wL - rK.$$

This problem is easiest to solve if $F$ is strictly concave and differentiable. Then an interior solution is characterized by the first order conditions:

$$p\frac{\partial F}{\partial L}(L,K) = w \quad \text{and} \quad p\frac{\partial F}{\partial K}(L,K) = r.$$

These two equations implicitly define two functions, $L^*(w,r)$ and $K^*(w,r)$, called the factor demands.

We will pay attention to several questions as we generalize this model.

1. How can we handle multiple outputs?

2. Can we treat a collection of profit-maximizing producers as a single, aggregate producer?

3. What is the connection between profit maximization and the efficient allocation of resources?

To give a sense of the importance of these questions, we will take a brief look at the model economists typically use to think about the skill distribution and income inequality.

This model is based on an aggregate technology and is clearly related to normative concerns, so the second and third questions take on an obvious importance.

Our approach to these questions will be to treat production as a special case of the rational choice model from Chapter 1. The set $A$ will be all conceivable production plans, while the feasible plans will be represented by a set $Y$ called a technology.

Our treatment of technologies will be general, but we will restrict attention to a very special case of preferences—maximization of profit at fixed prices. Sometimes this is offered as a good descriptive model of firms in a competitive market. It *is* often a useful first approximation there, but we should note how weak the common justification for the assumption really is. That justification is that profit maximization is what owners of the firm want. But:

1. How do we know owners agree? They typically do not when markets are incomplete or the firm has market power.
2. How relevant are owners' wishes? Modern economies are characterized by a separation of ownership and control. This creates principal-agent problems that complicate any simple relationship between what owners want and what actually happens.

Nonetheless, the profit-maximizing producer is central to microeconomic theory. Partly this is because, in many applications, the conceptual problems mentioned above don't seem to matter much in practice. But this is also because the profit-maximizing producer provides the simplest example of the link between prices and efficiency.

## 5.1 Technology

We will work in $\mathbb{R}^n$, with the interpretation that each dimension $i$ measures the quantity of some good $i$. Think of production as changing the set of stuff in the world—starting from some status quo $\omega \in \mathbb{R}^n$, the act of production changes things so that the collection of goods is $\omega' \in \mathbb{R}^n$. We call the increment $y = \omega' - \omega$ a **production plan**.

Typically, production will use some goods to produce others. That is, some components of $y$ will be positive (net inputs) and others will be positive (net outputs). Reflecting this, production plans are sometimes called **netput vectors**.

A **technology** is a set $Y$ of production plans. Some useful assumptions:

**Nonempty** $Y$ is a nonempty set.

**Closed** $Y$ is a closed set.

**No free lunch** $Y \cap \mathbb{R}^K_{++} = \emptyset$. In words, producing any output requires some input.

**Irreversability** Suppose $y \in Y$ and $y \neq 0$. Then $-y \notin Y$.

These first four assumptions are basically technical, and we won't make a big deal out of them. The next two are substantive.

**Possibility of inaction** $0 \in Y$. Thinking back to intermediate micro, this says there are no sunk costs. It is perfectly consistent with fixed cost.

**Free disposal** If $y \in Y$ and $y' \leq y$, then $y' \in Y$. Throwing things away is costless.

More interesting for us will be the idea of returns to scale.

**Nonincreasing returns to scale** If $y \in Y$ and $\alpha \in [0, 1]$, then $\alpha y \in Y$. That is, any feasible netput vector can be scaled down.

**Nondecreasing returns to scale** If $y \in Y$ and $\alpha \geq 1$, then $\alpha y \in Y$. That is, any feasible netput vector can be scaled up.

**Constant returns to scale** If $y \in Y$ and $\alpha \geq 0$, then $\alpha y \in Y$. That is, any feasible netput vector can be scaled up or down.

A particularly useful assumption is:

**Convexity** $Y$ is a convex set.

Think of this as combing two ideas:

1. Nonincreasing returns. (If inaction is possible and technology is convex, then it has nonincreasing returns.)
2. "Balanced" input combinations are at least as productive as "unbalanced". Similarly, "Balanced" output combinations are no more expensive than "unbalanced".

Sometimes it makes sense to separate the goods into inputs and outputs. This is particularly useful when there is just one output. In that case, we can describe a technology with a **production function**, $f$. If $Y$ is a technology for which good 1 is the output, let

$$f(z) = \sup_x \{x \mid (x, -z) \in Y\}.$$

(Note that $\sup \emptyset = -\infty$.)

Alternatively, if we are given a production function $f$, we can define $Y$ by

$$Y = \{(x, -z) \mid x \leq f(z)\}.$$

(Note the slippage due to free disposal being built into second definition.)

In the single-output case, we can restate the returns to scale assumptions in terms of the production function.

- Nonincreasing returns: $f(\alpha x) \geq \alpha f(x)$ for all $\alpha \in [0, 1]$ and $x$.
- Nondecreasing returns: $f(\alpha x) \geq \alpha f(x)$ for all $\alpha \geq 1$ and $x$.
- Constant returns: $f(\alpha x) = \alpha f(x)$ for all $\alpha \geq 0$ and $x$.
- Convexity: $f$ is concave.

Our definition of a technology is more flexible than one based on production functions, since we can have multiple outputs and don't need a priori distinction of input/output.

## 5.2 Profit Maximization

Let $p$ be a vector in $\mathbb{R}^n_+$ interpreted a prices. Then the **profit** of plan $y$ is $py$. By our sign convention on production plans, this really does correspond to our ordinary definition of profit: outputs are positive and so make a positive contribution, while inputs are negative and so make a negative contribution. To see this a bit more formally, write

$$y^+ = (\max(y_1, 0), \ldots, \max(y_n, 0)) \quad \text{and} \quad y^- = -(\min(y_1, 0), \ldots, \min(y_n, 0)).$$

Then $y = y^+ - y^-$ and

$$py = \underbrace{py^+}_{\text{revenue}} - \underbrace{py^-}_{\text{costs}}.$$

The **profit maximization problem** is

$$\max_y \; py$$
$$\text{st} \;\; y \in Y$$

Call the set of solutions to this problem $y^*(p)$. (As we will see, it is very important that we allow this set to have many elements.)

**Example 5.1.** Consider a technology that transforms a single input, $x$, into a single output, $q$, according to the production function $x \mapsto \sqrt{x}$. Write $z = -x$. Then the technology can be represented by the production possibility set

$$Y = \{(q, z) \mid q \leq \sqrt{(-z)} \text{ and } z \leq 0\}.$$

If prices are $p = (1, \frac{1}{2})$, then the profit-maximization problem is

$$\max_{q,z} \ 1q + \frac{1}{2}z$$
$$\text{st} \ \ q \leq \sqrt{(-z)}.$$

Maximization implies not throwing away output, so this is equivalent to

$$\max_{z} \sqrt{(-z)} + \frac{1}{2}z.$$

The FOC is

$$-\frac{1}{2\sqrt{(-z)}} + \frac{1}{2} = 0.$$

Since the objective function in the simplified problem is strictly concave, the FOC tells us that the unique solution is $z = -1$.

An important contrast between consumer theory and producer theory concerns existence of solutions. In consumer theory, existence is is guaranteed under weak continuity assumptions as long as prices are strictly positive. This is not true for profit maximization.

**Theorem 5.1.** *Suppose $Y$ has nondecreasing returns to scale and some production plan $y \in Y$ has strictly positive profit at prices $p$. Then there is no solution to the profit maximization problem at prices $p$.*

**Proof.** Suppose $y$ is a feasible production plan with $py > 0$. Let $\alpha$ be a number greater than 1. By nondecreasing returns to scale, $\alpha y \in Y$. And

$$p(\alpha y) = \alpha(py) > py,$$

which implies $y$ is not profit maximizing. Since $y$ was arbitrary, there is no profit-maximizing plan. ■

**Corollary 5.1.** *Suppose $Y$ has constant returns to scale. Then either maximized profit is 0 or there is no solution to the profit maximization problem.*

**Proof.** Constant returns implies the possibility of inaction, so profits are at least 0. And the theorem implies that either profits are nonpositive or the profit maximization problem has no solution. ■

**Example 5.2.** Fix a $\beta \in \mathbb{R}_{++}$ and consider the technology

$$Y = \{(q, z) \mid q \leq -\beta z \text{ and } z \leq 0\} \subset \mathbb{R}^2.$$

If the price vector is $(p_1, p_2)$ and the production plan is $(\beta x, -x)$ for some $x > 0$, then profit is

$$p_1(\beta x) - p_2 x = (p_1 \beta - p_2)x.$$

If $(p_1\beta - p_2) > 0$, then there is no solution, and if $(p_1\beta - p_2) < 0$, then the only solution is $x = 0$. Moreover, if $\frac{p_2}{p_1} = \beta$, then profits are 0 for any $x$, and the solution to the profit maximization problem is indeterminate.

**Example 5.3.** Now consider a technology that can transform good 2 into good 1 on a one-for-one basis, but only if a fixed amount of one unit of good 1 is used to start the technology. Formally,

$$Y = \{(q, z) \mid q \leq -z + 1 \text{ and } z \leq 1\} \cup \{(0, z) \mid 0 \leq z < 1\} \subset \mathbb{R}^2.$$

If the price vector is $(p_1, p_2)$ and the production plan is $(x - 1, -x)$ for some $x \geq 1$, then profit is

$$p_1(x - 1)) - p_2 x = (p_1 - p_2)x - p_1.$$

If $(p_1 - p_2) > 0$, then this is increasing without bound and there is no solution. If $(p_1 - p_2) \leq 0$, then this is negative for all $x$. But the feasible plan $(0, 0)$ gives profit 0. Thus there are no prices at which profits are maximized with positive output.

## 5.3 Aggregate Production

We saw in Chapter 4 that the theory of the consumer does not aggregate nicely, in the sense that aggregate demand does not satisfy the restrictions implied by maximizing utility subject to a market budget constraint. Aggregation is much more satisfying for producers. In this section, we will see a formalization of this claim, and then use it as an excuse to look at the standard model economists use to think about skill-based income inequality.

The setting will be an economy with several producers, labeled $i = 1, \ldots, n$. Each of these producers has a technology $Y^i$, which is nonempty and closed. Define the **aggregate production set** by

$$Y^{agg} = \sum_{i=1}^{n} Y^i \equiv \{y \mid \exists (y^1, \ldots, y^n) \text{ with, } \forall i, y^i \in Y^i \text{ and } y = \sum_{i=1}^{n} y^i\}.$$

In words, $Y^{agg}$ is the set of all netput vectors that can be constructed as sums of netput vectors feasible for the individual producers, with exactly one feasible netput vector per producer.

Now we can define the set of aggregate-profit-maximizing netput vectors:

$$y^{agg*} = \arg\max_{y \in Y^{agg}} py.$$

**Theorem 5.2.** $y^{agg*}(p) = \sum_{i=1}^{n} y^{i*}(p)$.

**Proof.** We want to show that two sets are equal, so we must show 1.) $y^{agg*}(p) \subset \sum_{i=1}^{n} y^{i*}(p)$ and 2.) $\sum_{i=1}^{n} y^{i*}(p) \subset y^{agg*}(p)$.

1. Suppose $y \in y^{agg*}(p)$, but, seeking a contradiction, that $y \notin \sum_{i=1}^{n} y^{i*}(p)$. Let $(y^1, \ldots, y^n)$ be an arbitrary profile with $y^i \in Y^i$ for all $i$ and $y = \sum_{i=1}^{n} y^i$. Since $y \notin \sum_{i=1}^{n} y^{i*}(p)$, there is some $j$ and $\hat{y}^j \in Y^j$ such that $p\hat{y}^j > py^j$. Denote by $(\tilde{y}^1, \ldots, \tilde{y}^n)$ the profile with $\tilde{y}^i = y^i$ for $i \neq j$ and $\tilde{y}^j = \hat{y}^j$, and let $\tilde{y} = \sum_{i=1}^{n} \tilde{y}^i$. Sum over $i$ to get
$$p\tilde{y} = p\hat{y}^j + \sum_{i \neq j} py^i > py^j + \sum_{i \neq j} py^i = py,$$
so $y$ does not maximize profits over $Y^{agg}$, a contradiction.

2. Suppose $y \in \sum_{i=1}^{n} y^{i*}(p)$. Let $\hat{y}$ be an arbitrary element of $Y^{agg}$, and let $(\hat{y}^1, \ldots, \hat{y}^n)$ be an arbitrary profile with $\hat{y} = \sum_{i=1}^{n} \hat{y}^i$ and, for all $i$, $\hat{y}^i \in Y^i$.

Since $y \in \sum_{i=1}^n y^{i*}(p)$, there is a decomposition of $y$ into a profile $(y^1, \ldots, y^n)$ with $y = \sum_{i=1}^n y^i$ and, for all $i$, $y^i \in y^{i*}$. But that means, for all $i$, we have $py^i \geq p\hat{y}^i$. Sum over $i$ to get

$$py = \sum_{i=1}^n py^i \geq \sum_{i=1}^n p\hat{y}^i = p\hat{y}.$$

Since $\hat{y}$ was arbitrary, $y \in y^{agg*}(p)$.

■

We will see in the next section that convexity of production sets is a very important property. It is preserved under aggregation.

**Theorem 5.3.** *Suppose each $Y^j$ is convex. Then $Y^{agg}$ is convex.*

**Proof.** Exercise 5.3. ■

Theorem 5.2 is an important part of the background for many applied arguments. A recently prominent example is the standard framework economists use to think about skill-based wage inequality.

Imagine an economy with $n$ workers. $H$ of these worker are high skilled, and $L$ are low skilled. Each of these workers has one unit of labor to supply.

The economy produces output, $y$, from low-skilled labor, $l$, and high-skilled labor, $h$, according to an aggregate production function:

$$y = [(A_l l)^\rho + (A_h h)^\rho]^{1/\rho}.$$

We assume that $A_h > A_l$ and $\rho \leq 1$. The second of these assumptions implies that the production function is concave.

Let's look for a set of prices $(p_y, p_l, p_h) = (1, w_l, w_h)$ such that, at those prices, profit maximization is consistent with full employment: $l = L$ and $h = H$. Since the production function is concave, the necessary and sufficient conditions are the following first-order conditions:

$$\begin{aligned} w_l &= A_l^\rho [A_l^\rho + A_h^\rho (H/L)^\rho]^{(1-\rho)/\rho} \\ w_h &= A_h^\rho [A_h^\rho + A_l^\rho (H/L)^{-\rho}]^{(1-\rho)/\rho} \quad = A_h^\rho [A_h^\rho (H/L)^\rho + A_l^\rho]^{(1-\rho)/\rho} \cdot \left((H/L)^{-\rho}\right)^{(1-\rho)/\rho} \end{aligned}$$

From this we can define the *skill premium,* denoted $\omega$, by

$$\omega = \frac{w_h}{w_l} = \left(\frac{A_h}{A_l}\right)^{\rho} \left(\frac{H}{L}\right)^{-(1-\rho)}.$$

Empirical economists prefer to write this in terms of an object called the *elasticity of substitution.* You can read about what this means in general in other sources. Here, I'll just note two facts. First, for our production function, the elasticity of substitution, $\sigma$, is a simple function of $\rho$:

$$\sigma = \frac{1}{1-\rho}.$$

Second, the empirical consensus is that, for the U.S. economy, $\sigma$ is between 1.5 and 2.

In elasticity of substitution terms, the skill premium is:

$$\omega = \left(\frac{A_h}{A_l}\right)^{(\sigma-1)/\sigma} \left(\frac{H}{L}\right)^{-1/\sigma}$$

This is particularly easy to understand if we take logs:

$$\log \omega = \left(\frac{\sigma - 1}{\sigma}\right) \log\left(\frac{A_h}{A_l}\right) - \frac{1}{\sigma}\log\left(\frac{H}{L}\right)$$

Given the empirical consensus about $\sigma$, we see two implications. First, skill-biased technological change (i.e., technological changes that increase $A_h$ more than $A_l$) will increase the skill premium. Second, increases in the fraction of workers who are high-skilled will decrease the skill premium. Claudia Goldin and Larry Katz have written a lovely book, *The Race Between Education and Technology,* interpreting the history of technological innovation, education policy, and wage inequality over U.S. history in these terms.

From our point of view in this course, the essential point is that all of this only makes sense because of Theorem 5.2. The history recounted by Goldin and Katz is not one in which there was a single, centrally controlled technology. Instead, there were many different firms, each independently trying to maximize profits. Theorem 5.2 tells us that the analysis is nonetheless legitimate.

## 5.4 Prices from Efficiency

Given technology $Y$, the production plan $y^0$ is **production efficient** if there is no $y \in Y$ such that $y > y^0$. In other words, if $y^0$ is production efficient, then there is no other feasible plan that yields at least as much of everything and strictly more of something.

Prices $p$ **support** the production plan $y^0$ if

1. $py^0 \geq py$ for all $y \in Y$, and
2. $py^0 > py$ for all $y \in \operatorname{int} Y$. (Note that this implies only boundary points can be supported.)

From the definition, we see that if there is any supporting price vector for $y^0$, then there are many—if $p$ supports $y^0$ then so does $\lambda p$ for any scalar $\lambda > 0$.

There is a close connection between production efficiency and support by prices.

**Proposition 5.1.** *If $y^0$ is supported by strictly positive prices $p$, then $y^0$ is production efficient.*

**Proof.** If $y > y^0$, then $py > py^0$, since $p$ is strictly positive.

Since $p$ supports $y^0$, we must have $y \notin Y$. ■

The reverse direction, that any efficient plan can be supported by prices, is not true without additional conditions.

**Theorem 5.4** (Supporting Hyperplane Theorem)**.** *Suppose $Y$ is a nonempty, convex subset of $\mathbb{R}^n$, and that $y^0$ is on the boundary of $Y$. Then there is a $p \neq 0$ such that:*

1. *for all $y \in Y$, we have $p \cdot y \leq p \cdot y^0$ and*
2. *for all $y \in \operatorname{int} Y$, we have $p \cdot y < p \cdot y^0$.*

We will derive a special case from the following fundamental fact:

**Lemma 5.1.** *If $g$ is differentiable and $Y = \{y \mid g(y) \geq g(y^0)\}$ is convex, then*

$$y \in Y \implies \mathbf{D}g(y^0) \cdot (y - y^0) \geq 0.$$

**Proof.** Choose any $y \in Y$. By convexity of $Y$,

$$y^\lambda \equiv \lambda y + (1 - \lambda)y^0 \in Y$$

for all $\lambda \in [0, 1]$. Since $Y$ is an upper contour set, this is equivalent to

$$g(y^\lambda) - g(y^0) \geq 0.$$

Define $h(\lambda) = g(y^\lambda) = g(y^0 + \lambda(y - y^0))$. Then

$$\frac{h(\lambda) - h(0)}{\lambda} = \frac{g(y^0 + \lambda(y - y^0)) - g(y^0)}{\lambda}.$$

For any $\lambda \in (0, 1)$, the RHS is $\geq 0$, since $y^\lambda \in Y$. Thus the limit

$$\lim_{\lambda \to 0} \frac{h(\lambda) - h(0)}{\lambda} \geq 0.$$

But that limit is just the derivative of $h$, so $h'(0) \geq 0$.

By the chain rule,

$$\frac{dh}{d\lambda}(\lambda) = \mathbf{D}g(y^0 + \lambda(y - y^0)) \cdot (y - y^0).$$

Set $\lambda = 0$ then gives

$$\mathbf{D}g(y^0)(y - y^0) \geq 0,$$

as desired. ■

**Proof of supporting hyperplane (for the differentiable case).** Assume, in addition to the hypotheses of the theorem, that there is a differentiable function $g$ such that

$$Y = \{y \mid g(y) \geq g(y^0)\},$$

and $\mathbf{D}g(y^0) \neq 0$. By Lemma 5.1, $y \in Y$ implies $\mathbf{D}g(y^0)(y - y^0) \geq 0$. Take $p = -\mathbf{D}g(y^0)$ to get

$$\mathbf{D}g(y^0)(y - y^0) \geq 0 \iff p(y^0 - y) \geq 0 \iff py^0 \geq py.$$

■

To provide a converse to Proposition 5.1, we need to know that the supporting prices are nonnegative. But the Supporting Hyperplane Theorem does not guarantee that. But one more assumption will do it.

**Proposition 5.2.** *Let $Y$ be a non-empty, convex technology that satisfies free disposal. If $y^0 \in Y$ is production efficient, then there is a non-negative price vector $p$ such that $p$ supports $y^0$. If, in addition, $0 \in Y$, then $py^0 \geq 0$.*

**Proof.** Suppose $y^0$ is production efficient. Then there is no $y \in Y$ with $y > y^0$, so $y^0$ is a boundary point of $Y$. By the Supporting Hyperplane Theorem, there is a price vector $p \neq 0$ such that $py^0 \geq py$ for all $y \in Y$.

We need to show that $p > 0$.

Write $e_i$ for the vector whose components are 0 except for the $i$th, which is 1. By free disposal, $y^1 = y^0 - e_i \in Y$. But $py^0 \geq py^1$ implies

$$p(y^0 - y^1) = pe_i = p_i \geq 0.$$

If in addition $0 \in Y$, that $p$ is a supporting price implies

$$py^0 \geq p \cdot 0 = 0.$$

■

**Example 5.4.** Consider again the technology in Example 5.1:

$$Y = \{(q, z) \mid q \leq \sqrt{(-z)} \text{ and } z \leq 0\}.$$

To use our theorems, introduce the function $g(q, z) = \sqrt{(-z)} - q$. Then we can rewrite

$$Y = \{(q, z) \mid g(q, z) \geq 0 \text{ and } z \leq 0\}.$$

Differentiate

$$\frac{d}{dz}\sqrt{(-z)} = -\frac{1}{2\sqrt{(-z)}}$$

$$\begin{aligned}\frac{d^2}{dz^2}\sqrt{(-z)} &= -\frac{1}{4(-z)^{3/2}} \\ &< 0.\end{aligned}$$

Since sums of concave functions are concave, this implies $g$ is concave, hence quasi concave.

A production plan $(q, z)$ is efficient if and only if $q = \sqrt{-z}$. Thus two efficient plans are

$$(q, z) = (1, -1) \quad \text{and} \quad (q, z) = (2, -4).$$

In the first case, the derivative of $g$ is

$$\mathbf{D}g(1,-1) = \begin{pmatrix} -1 \\ -\frac{1}{2} \end{pmatrix},$$

so the supporting prices are $p = -\mathbf{D}(1,-1) = (1, 1/2)$. And our analysis from before shows that the plan $(1,-1)$ in does in fact maximize profits at these prices.

Now consider the other efficient plan. The derivative is

$$\mathbf{D}g(2,-4) = \begin{pmatrix} -1 \\ -\frac{1}{4} \end{pmatrix},$$

so the supporting prices are $p = -\mathbf{D}(2,-4) = (1, 1/4)$.

To check this, consider the maximization problem

$$\max_{(q,z)\in Y} 1 \cdot q + \frac{1}{4}z.$$

Maximization implies not throwing away output, so this is equivalent to

$$\max_{z} \sqrt{(-z)} + \frac{1}{4}z.$$

The FOC is

$$-\frac{1}{2\sqrt{(-z)}} + \frac{1}{4} = 0,$$

which holds only at $z = -4$.

## 5.5 Decentralization via Prices

We have seen that, if technology is convex, prices can guide a producer to an efficient production plan. Now we will consider a (very simple) complete economy, and see that prices work well there also.

There are two agents in the economy, a *consumer* and a *producer.*

The consumer has preferences $\succsim$ defined on $\mathbb{R}^n_+$, and an *endowment* $\omega \in \mathbb{R}^n_+$.

- Throughout this section, assume that $\succsim$ are represented by a concave and continuously differentiable utility function $U$, with $\mathbf{D}U(x) \neq 0$ for all $x$.

- The endowment represents the goods the consumer owns prior to any production. In many examples, the only non-produced good will be labor. Taking labor to be good one, we can then write the endowment as $\omega = (\overline{L}, 0 \ldots, 0)$, where $\overline{L} > 0$ is the amount of time the consumer can work.

The producer has a technology $Y \subset \mathbb{R}^n$ that is:

- Nonempty, closed, and satisfies no free lunch, irreversibility, and possibility of inaction.
- $Y = \{y \mid g(y) \geq g(0)\}$ for some concave function $g$ with $g(0) = 0$.

If the producer chooses the production plan $y$, then final resources in the economy will be $\omega + y$. The consumer must then consume some bundle $x$ with $x \leq \omega + y$.

A first perspective about what this economy should do comes from a benevolent planner, who sits outside the system and seeks to maximize the consumer's utility. This planner will solve:

$$\begin{aligned}
\max_{x,y} \; & U(x) & \\
\text{st} \; & \omega_i + y_i - x_i \geq 0 & \text{for all } i \\
& g(y) \geq 0 & \\
& x_i \geq 0 & \text{for all } i
\end{aligned}$$

Since $U$ and $g$ are both concave, the Kuhn-Tucker conditions are necessary and sufficient for a solution. That is, $(x^*, y^*)$ is a solution if and only if:

$$\begin{aligned}
\frac{\partial U}{\partial x_i}(x^*) - q_i \leq 0 & \qquad \text{with equality if } x_i^* > 0 \\
q_i + \mu \frac{\partial g}{\partial y_i}(y^*) = 0, &
\end{aligned}$$

where $q_i \geq 0$ is the multiplier on the resource constraint for good $i$ and $\mu \geq 0$ is the multiplier on the production constraint.

Something very important has happened. Prices, in the form of the multipliers $q_i$, have just appeared.

First, solve the second FOC to get

$$\frac{1}{\mu} q = -\mathbf{D}g(y^*).$$

That is, $\frac{1}{\mu} q$ supports production plan $y$.

Second, the first FOC has a similar interpretation. Consider an interior solution, so the FOC becomes:

$$\mathbf{D}U(x^*) = q.$$

This says that $-q$ defines a supporting hyperplane to the set of bundles that the consumer prefers to $x^*$. And that means that any bundle better than $x^*$ is also more expensive.

There is another way to reach the same allocation. Suppose that the consumer owns both the initial endowment and the technology. But she does not run the technology herself—instead, it is run by a manager who always maximizes profit. The only way that the consumer and manager interact is through price-mediated transactions. (This is artificial in this simple economy, but the ideas generalize to large numbers of both consumers and producers.)

If prices are $p$, the producer will solve

$$\begin{aligned} &\max_y \; py \\ &\text{st} \;\; g(y) \geq 0. \end{aligned}$$

The FOC for this problem is

$$p = \gamma \mathbf{D}g(\hat{y}).$$

Denote maximized profits by $\pi(p) = p\hat{y}$.

If prices are $p$, the consumer will solve

$$\begin{aligned} &\max_x \; U(x) \\ &\text{st} \;\; p\omega + \pi(p) - px \geq 0. \end{aligned}$$

The FOC for this problem is

$$\mathbf{D}U(\hat{x}) \leq \lambda p.$$

Say that $p$ is a **Walrasian equilibrium price** if markets clear:

$$\underbrace{\hat{x}}_{\text{demand}} = \underbrace{\omega + \hat{y}}_{\text{supply}}.$$

Notice that, at a WE price, the cost of the consumer's demand $px$, is exactly equal to her income $p(\omega + \hat{y}) = p\omega + \pi(p)$.

Finally, notice that, if $p = \frac{1}{\mu}q$, $\lambda = \mu$, and $\gamma = 1$, there is an exact correspondence between the necessary and sufficient conditions for the planners solution and the conditions for the Walrasian equilibrium.

Both approaches lead to the same allocation. This is not a coincidence. There are an important pair of theorems, called the First and Second Welfare theorems, that generalize these results to arbitrary numbers of consumers and producers. We'll see these generalizations soon.

## 5.6 Appendix: Subjective Probability

Although it really has nothing to do with production, this is an opportune spot to clear up a few loose ends from the discussion of subjective probability from Section 3.2. The link to this chapter is the supporting hyperplane theorem.

Recall the setup:

1. There is a set of states of nature, $\Omega$.
2. There is a set of consequences, $X$.
3. An act is a map $a : \Omega \to X$.

We will follow the classical development due to Ramsey and de Finetti, which involves two simplifications. First, assume that $\Omega$ is finite. This assumption is not needed for the kind of result we are aiming for, but it simplifies the mathematics considerably. The second assumption is more substantive—assume that $X = \mathbb{R}$. The interpretation is that $a(\omega)$ is the amount of money won or lost if the state is $\omega$. (When this assumption is in force, we often refer to an act as a *bet.*)

We assume that the DM has complete, transitive, and continuous preferences $\succsim$ over the set of all possible bets. These preferences satisfy the following additional axioms:

**Additivity** For all $x, y, z$, we have $x \succsim y$ if and only if $x + z \succsim y + z$.

**Monotonicity** If $x \geq y$, then $x \succsim y$.

**Non-triviality** There exist $x$ and $y$ such that $x \succ y$.

These assumptions are not entirely satisfactory—the following proposition shows that they imply expected value maximization. More advanced treatments give expected utility maximization instead. Basically, this involves combining the idea of the following proof with the independence axiom and the proof of the von Neumann-Morgenstern Theorem.

**Proposition 5.3.** *Preferences $\succsim$ over bets are complete, transitive, continuous, additive, monotone, and satisfy non-triviality if and only if there exists a probability vector $p$ such that*

$$x \succsim y \text{ if and only if } px \geq py.$$

**Proof.** Necessity is left as an exercise.

By additivity, $x \succsim y$ if and only if $0 \succsim y - x$. Thus we can fully describe preferences over bets by saying which bets are worse than the zero bet, which is naturally interpreted as "no bet". This is the set of **unacceptable bets**,

$$B = \{x \mid 0 \succsim x\}.$$

The key observation is the following:

**Lemma 5.2.** *Suppose $\succsim$ satisfies continuity and additivity. Then the set of unacceptable bets is convex.*

**Sketch of the proof.** Suppose $x, y \in B$ satisfy $x \succsim y$. Let $z = \frac{x+y}{2}$, so

$$z - x = y - z = \frac{y - x}{2} \equiv d.$$

Then additivity gives

$$x \succsim z \Leftrightarrow x + d \succsim z + d \Leftrightarrow z \succsim y.$$

Since $x \succsim y$, the assumption that $z \succ x$ would imply $z \succ y$, contradicting $z \succsim y$. Thus $x \succsim z$. A similar argument gives $z \succsim y$, so we have

$$x \succsim \frac{1}{2}x + \frac{1}{2}y \succsim y.$$

Next consider $x \succsim y$ and $\lambda x + (1 - \lambda)y$, where $\lambda$ is a *dyadic rational*: $\lambda = \frac{k}{2^i}$ for integers $k, i \geq 1$. An inductive version of the previous paragraph yields

$$x \succsim \lambda x + (1 - \lambda)y \succsim y.$$

Continuity then extends the conclusion to any real $\lambda \in [0, 1]$, and transitivity then implies $0 \succsim \lambda x + (1 - \lambda)y$. ■

Now we can use the supporting hyperplane theorem. The set $B$ is nonempty because it contains 0, and we just saw that it is convex. Now we need a boundary point.

By non-triviality, there is a bet $x$ such that $0 \succ x$. By convexity of $A$, for any $\epsilon > 0$, we can take $\|x\| < \epsilon$. But additivity gives $-x \succ 0$, so the ball $\{y \mid \|y\| < \epsilon\}$ contains points inside of and outside of $B$.

Putting all of this together, Theorem 5.4 tells us that there is a $q \neq 0$ such that $x$ is unacceptable only if $q \cdot x \leq q \cdot 0 = 0$.

Now assume that $x$ is not in $B$, so $x \succ 0$, but $q \cdot x < 0$. By additivity, we would have $0 \succ -x$ and $q \cdot (-x) > 0$, a contradiction. Thus we have

- $x \in B \Rightarrow q \cdot x \leq 0$; and
- $q \cdot x < 0 \Rightarrow x \in B$.

But continuity implies that $A$ is closed, so $A$ includes its boundary $q \cdot x = 0$, which closes the gap between the two implications.

Next we argue that each $q_i \geq 0$. Let $e_i$ be the unit vector in direction $i$. By monotonicity, $0 \succsim -e_i$. Thus $q \cdot (-e_i) = -q_i \leq 0$.

Since $q \neq 0$, we can normalize:

$$qx \leq 0 \Leftrightarrow px \equiv \frac{1}{\|q\|}(qx) \leq 0.$$

Together, the last two points imply that $p$ is a vector of probabilities.

Now we just unpack the meaning of unacceptable bet:

$$x \succsim y \Leftrightarrow 0 \succsim y - x \Leftrightarrow p \cdot (y - x) \leq 0 \Leftrightarrow px \geq py.$$

That is, $x \succsim y$ if and only if the expected value of $x$ is greater than the expected value of $y$. ■

## Problems

**Exercise 5.1.** Suppose a technology $Y$ is convex and has the property that, if $y^0$ and $y^1$ are both in $Y$ and $\alpha \in (0, 1)$, then $\alpha y^0 + (1 - \alpha)y^1$ is not production efficient in $Y$. Show that, if prices are strictly positive, there is at most one solution to the profit maximization problem.

**Exercise 5.2** (Binmore)**.** Suppose that a profit-maximizing producer chooses a production plan from a technology $Y$ that is compact and satisfies the property from the previous exercise. Write $s(p)$ for the supply function:

$$s(p) = \operatorname{argmax}_{y \in Y} py.$$

Answer the parenthetical questions in the following "proof" that the supply function is continuous, and point to a flaw in the argument. What can be done to patch up the proof?

> Let $p_k \to p$ as $k \to \infty$. Write $y_k = s(p_k)$. Then, for any $z \in Y$, we have $p_k z \leq p_k y_k$. (Why?) If $y_k \to y$, it follows that, for any $z \in Y$, we have $pz \leq py$. (Why?) Hence $y = s(p)$. (Why?) Thus $s(p_k) \to s(p)$ as $k \to \infty$, and so $s$ is continuous.

**Exercise 5.3.** Prove Theorem 5.3.

**Exercise 5.4** (Rubinstein)**.** An event that could have occurred with probability 0.5 either did or did not occur. A firm must provide a report in the form of "the event occurred" or "the event did not occur". The quality of the report (the firms product), denoted by $q$, is the probability that the report is correct. Each of $k$ experts (input) prepares an independent recommendation that is correct with probability $1 > p > 0.5$. The firm bases its report on the $k$ recommendations in order to maximize $q$.

1. Calculate the production function $q = f(k)$ for (at least) $k = 1, 2, 3$.
2. We say that a discrete production function is concave if the sequence of marginal product is nonincreasing. Is the firms production function concave?

Assume that the firm will get a prize of $M$ if its report is actually correct. Assume that the wage of each worker is $w$.

3. Explain why it is true that if $f$ is concave, the firm chooses $k^*$ so that the $k^*$th worker is the last one for whom marginal revenue exceeds the cost of a single worker.
4. Is this conclusion true in our case?

**Exercise 5.5.** Using a carefully labeled figure, give an example of a production technology $Y$, a production plan $y \in Y$, and prices $p \geq 0$ such that

- $y$ is not efficient in $Y$, but
- $y$ maximizes profits at prices $p$.

**Exercise 5.6.** Draw the hyperplanes and closed half spaces in $\mathbb{R}^2$ determined by each if the following vectors.

1. $p = (1, 2)$
2. $p = (1, -2)$
3. $p = (-1, -2)$

**Exercise 5.7.** Suppose commodity 1 can be used to produce commodity 2 according to the technology

$$Y = \{(y_1, y_2) \mid y_1 \leq 0 \text{ and } g(y) = -y_1 - y_2 - y_2^3 \geq 0\}.$$

1. Show that $Y$ is a convex set.
2. Show that the production plan $\overline{y} = (-10, 2)$ is on the boundary of this set.
3. Calculate the supporting prices for the production plan $\overline{y}$.
4. Verify directly that $\overline{y}$ maximizes profits at the prices you calculated in the previous step.
5. Depict the production set and the supporting line in a neat figure.

**Exercise 5.8** (Riley)**.** Robinson Crusoe lives alone on an island off the coast of New Zealand. He has a production set

$$Y = \{(-z_1, y_2) | y_2 \leq 16 z_1^{1/3}, z_1 \geq 0\}$$

and an endowment vector $\omega = (32, 0)$.

His preferences are represented by the utility function $U(x) = \log x1 + \log x2$ .

1. Solve for his optimal choice of input and hence his optimal production plan and consumption plan $x^*$.
2. Depict the production set and the set $Y + \omega$ in a neat figure and indicate the optimal production and consumption plans. Explain what it means for the optimal production plan to be supported by a price vector $p = (p_1, p_2)$.
3. Solve for the price vector that supports the optimal production plan.

4. Depict this supporting price line, Crusoes budget set, and indifference curve though $x^*$.

5. Hence explain why the supporting price vector is a WE price vector if Robinson Crusoe is a price-taker.

# Chapter 6

# Welfare Economics

We are now going to spend a little time on the standard economist's approach to normative evaluation. Along the way, we'll talk about the very beginnings of the theory of competitive equilibrium, although you'll have to wait until next quarter for a detailed study of markets.

As you go through a public policy education, it is easy to become complacent about the normative commitments of practices like cost-benefit analysis. Much attention must be paid to technical developments. And practical economists like to minimize the contribution ethical stances make to their policy recommendations. It is crucial to step back and see that there are substantive normative claims inextricably mixed with science in economics-based policy advice.

The general setting has a set of **allocations** $A \subset \mathbb{R}^n$, with typical element $a$, and a collection $H$ of individuals, each with a utility function $u^h : A \to \mathbb{R}$.

An important example covers the division of a fixed bundle of commodities.

**Example 6.1.** An **exchange economy** is a tuple

$$\mathcal{E} = \langle H, K, (\succsim^h)_{h \in H}, \omega \rangle,$$

where

- $H$ is the (finite) set of **consumers**,
- $K$ is the (finite) set of **commodities**,
- $\succsim^h$ is consumer $h$'s preference over the **commodity space** $\mathbb{R}^K_+$, and
- $\omega \in \mathbb{R}^K_+$ is the social **endowment**.

We assume that each preference is complete, transitive, continuous, monotonic, and convex, with utility representation $u^h$.

An **allocation** is an array $x = (x^h)_{h \in H}$. An allocation is **feasible** if

1. $x^h \in \mathbb{R}_+^K$ for all $h$, and
2. $\sum_h x^h \leq \omega$.

Write $A$ for the set of feasible allocations.

It will be convenient to stack the utility function as $u : A \to \mathbb{R}^n$, where $a \mapsto \left(u^h(a)\right)_{h \in H}$. Such a vector of utility levels is called a **utility imputation**. The set of all utility imputations, $U = \{x \in \mathbb{R}^n \mid \exists a \text{ st } x = u(a)\}$, is called the **utility possibility set**.

We have been treating preferences as the consumer's reasons for her choices. The exercise we are about to undertake will keep this assumption, and add another: that the consumer's preferences accurately reflect her welfare.

This is not always reasonable—think of children. It seems better for adults, but even there psychologists question the link between choices and welfare (projection bias, weakness of will, etc.). See the article by Kahneman and Varey in Elster and Roemer (eds.) *Interpersonal Comparisons of Well-Being*. In addition to these failures of choices to reflect welfare as understood by the agent, we may even be concerned that the agent's conception of her welfare is different than her true welfare. Sen gives the example of the "tamed housewife", who has no options other than being a housewife and adjusts her aspirations downward.

## 6.1 Normative Concepts for Welfare Economics

Consider a policy analyst, YOU for short. You want to give policy advice based on your own preferences about how this economy should operate. Now, I know very little about you. I certainly don't know what you think about the aims of public policy. What I do know is what assumptions I need to make about you to believe you should subscribe to the standard approach of welfare economics.

**Assumption 1** Your normative stance is fully captured by a complete, transitive preference $\succsim$ over $A$. (In fact, I'll go ahead and assume that your preferences are sufficiently well-behaved to be represented by a utility function $V : A \to \mathbb{R}$.)

In addition, I need to assume that your preferences respect the preferences of the individuals in society. To state the relevant assumption, I will use the following formal notions.

Allocation $x$ **Pareto dominates** $y$ if $x \succsim^h y$ for all $h$ and $x \succ^k y$ for some $k$. Allocation $x$ is **Pareto optimal** if there is no feasible $y$ which Pareto dominates $x$.

We can translate our definitions of Pareto dominance, etc into the space of utilities. Allocation $a$ Pareto dominates $a'$ if $u(a) > u(a')$. The set of utility imputations corresponding to the set of all Pareto optimal alternatives is called the **Pareto frontier**.

Now we can continue with the assumptions.

**Assumption 2** Your preferences respect Pareto dominance: $u(a) > u(a')$ implies $V(a) > V(a')$.

**Assumption 3** Your preferences respect **Pareto indifference**: $u(a) = u(a')$ implies $V(a) = V(a')$.

**Theorem 6.1.** *Fix utility representations $u^h$ for the consumers. Your preferences are represented by a function $V$ that respects Pareto dominance and Pareto indifference if and only if*

$$V(a) = W(u(a))$$

*for some function $W : \mathbb{R}^n \to \mathbb{R}$ that is strictly increasing on $U$.*

**Proof.** First, given a function $W$ that is strictly increasing on $U$, we can define $V$ by $V(a) = W(u(a))$. This obviously satisfies the conditions. Now assume that your preferences satisfy the three assumptions. The first says that there is a $V$ that represents your preferences. Take any $r \in U$. By the definition of $U$, there is an $a \in A$ such that $u(a) = r$. Define $W(r)$ to be $V(a)$ for the associated $a$. This is well-defined by Pareto indifference, and is strictly increasing on $U$ because the preferences respect Pareto dominance. To complete the proof, arbitrarily extend $W$ to all of $\mathbb{R}^n$. ∎

Functions $W$ of the sort introduced in the proposition are called **social welfare functionals**. Some standard examples are:

1. $W(u^1, \ldots, u^n) = \sum_i u^i$, the **utilitarian** social welfare functional.

2. Let $\{\alpha^i\}$ be strictly positive weights, and let $W(u^1, \ldots, u^n) = \sum_i \alpha^i u^i$. This defines a **weighted utilitarian** or **Bergsonian** social welfare functional.

3. $W(u^1, \ldots, u^n) = \min\{u^i \mid i \in N\}$, the **maximin** social welfare functional. (This is sometimes called the **Rawlsian** social welfare functional, although that is not entirely fair to John Rawls.)

This $W$ is not quite a social welfare functional, since it is not strictly increasing. If it represents your preferences, then you require a strict version of Pareto dominance before you conclude $V(a) > V(a')$.

It's easy to get confused about the meaning of $W$. Kreps writes:

> Note well that the function $W$ defined on $\mathbb{R}^n$ depends crucially on the particular $u^i$ that are chosen to represent the preferences of the individual consumers. We could, for example, replace $u^1$ with a function $u'^1$ given by $u'^1(a) = \left(u^1(a) + 1000\right)^3$ (which is a strictly increasing transformation), and then we would have to change how $W$ responds to its first argument. [p. 159, notation adjusted to conform to our notation]

The point is that you do *not* have preferences directly over utility imputations. Instead, you have preferences over $A$, and use $W$ only to give a convenient representation of those preferences.

We now have all the pieces needed to discuss how social welfare functionals are used in the characterization of Pareto optimal allocations and in the foundations of cost-benefit analysis. But before we go into that, I want to highlight how strong the assumptions really are.

- Assumption 1 says that your preferences are defined directly on the set of allocations. As a result, you cannot distinguish between two identical allocations that are arrived at in different ways. A particular allocation might be arrived at as the result of the free choices of individuals; the same allocation might be reached by the fiat of a dictator. You are blind to the difference. You are similarly blind to things like equality of opportunity and desert.

- If these omissions bother you, you could try augmenting the description of a social state to include details about process in addition to the allocation itself. But the Pareto assumptions will only allow this fix if the consumers value the process the same way you do. Amartya Sen gave the following example to make this problem vivid. There are three social states, $x$, $y$, and $z$, and there are two people, 1 and 2. Their utilities are

| | $u^1$ | $u^2$ |
|---|---|---|
| $x$ | 4 | 10 |
| $y$ | 7 | 8 |
| $z$ | 7 | 8 |

In state $x$, 1 is hungry and 2 has enough to eat; in state $y$, some food has been transferred from 2 to 1; and in state $z$, the food allocation is as in $x$, but 1, a sadist, is whipping 2. Pareto indifference says that you are indifferent between $y$ and $z$. I conclude that *my* preferences do not satisfy Pareto indifference.

- The previous point shows that Pareto indifference may be problematic. But it is just the principle of respect for Pareto dominance extend to ensure continuity of your evaluation function $V$. As such, problems with Pareto indifference suggest that Pareto dominance itself might be problematic as a normative guide. In Sen's example, would you feel better about the whipping if person 2 held back just enough that person 1 very, very slightly preferred being whipped to the loss of resources?

- Once you've ruled out anything but allocations and preferences as subjects for evaluation, it seems natural to take preference satisfaction as the ultimate goal. And that is what is happening in Theorem 6.1. Sen's example illustrates one reason to back away from preference satisfaction as the goal: some preferences might themselves be normatively objectionable.

  There are other reasons to be wary about preference satisfaction as the ultimate goal. One is lack of information. Assume that a person prefers that no one use cell phones because of he believes it creates a large risk of cancer. He is wrong about the science. Must we treat that preference the same way he treat his preference, based on his subjective sense of tastiness, for oranges over bananas?

- Another problem with the preference-satisfaction view arises when preferences can change over time. Someone, let's call him Scott, might prefer to exercise tomorrow rather than not exercise. But, when tomorrow rolls around, his preferences change, and he prefers not exercising over exercising. Which preference is normatively relevant?

- The problem of changing tastes is even more pressing when the polices being considered will lead to different preferences. Whether or not behavioral economics pans out, these policy-dependent preferences will continue to be a problem for people interested in child development, education, and related policy areas.

- An even knottier problem of which preferences should be satisfied arises when different policies lead to different populations. For example, consider two development assistance policies, just one of which has the effect of reducing fertility. What does it even mean to talk about the preferences of someone who only exists under one policy?

## 6.2 Characterizing Efficient Allocations

### 6.2.1 Bergen-Samuleson Social Welfare Functionals

It's immediate from our assumptions that any maximizer of a social welfare functional is Pareto optimal. It turns out that we cannot further refine the set of possible social welfare optima.

**Theorem 6.2.** *Suppose that $A$ is convex and that each $u^h$ is concave. If $a^*$ is Pareto optimal, then there exists a set of nonnegative weights $\{\alpha^i\}$, at least one positive, such that*

$$a^* \in \arg\max_{a \in A} \sum_i \alpha^i u^i(a).$$

(Note that the theorem requires weakening the definition of a social welfare functional in the same way that the maximin rule did.)

**Proof.** The result is an application of the supporting hyperplane theorem. To start, we need to define the convex set we will work with. Let $\hat{U}$ be the extension of $U$ to satisfy free disposal:

$$\hat{U} = \{x \in \mathbb{R}^n \mid \exists a \text{ st } x \leq u(a)\}.$$

I claim that $\hat{U}$ is convex. Given that, suppose $u^*$ is the utility imputation of a Pareto optimal allocation. The Supporting Hyperplane Theorem says that there is a vector $\alpha \neq 0$ such that, for all $u \in \hat{U}$,

$$\sum_h \alpha^h (u^*)^h = \alpha u^* \geq \alpha u = \sum_h \alpha^h u^h,$$

with strict inequality for $u \in \operatorname{int} \hat{U}$. Since $\hat{U}$ has free disposal, $u \in U$ and $\delta \geq 0$ imply that $u_\delta = u - \delta \in U$. By the implication of the supporting hyperplane,

$$\alpha(u^* - u^*_\delta) = \alpha\delta \geq 0$$

for all $\delta \geq 0$. If $\alpha^h$ were negative, taking $\delta = -e^h$ would then give a contradiction.[1]

All that remains is to prove the claim. Let $u$ and $u'$ be two utility imputations in $\hat{U}$. This implies that there are allocations $a$ and $a'$ with $u \leq u(a)$ and $u' \leq u(a')$. Convexity of $A$ implies that, for any $\lambda \in [0, 1]$, the convex combination $\hat{a} = \lambda a + (1 - \lambda)a'$ is also in $A$.

[1] Notation: $e^k$ is the unit vector in direction $k$.

By concavity of $u^h$, we have

$$\lambda u^h(a) + (1-\lambda)u^h(a') \leq u^h(\lambda a + (1-\lambda)a') = u^h(\hat{a})$$

for each $h$. Stacking these inequalities gives

$$\lambda u + (1-\lambda)u' \leq \lambda u(a) + (1-\lambda)u(a') \leq u(\hat{a}).$$

■

In some sense, this is a disappointing result. It says that assumptions 1–3, for all that they rule out, are not enough to get beyond Pareto dominance as a collectively shared normative standard. But there is also a positive side. The result gives us a way to completely characterize the implications of Pareto optimality. This is, in fact, how it is most often used.

### 6.2.2 Efficient Allocations of Commodities

Let's see this in our exchange economy example. Recall that an allocation is feasible if

1. $x^h \in \mathbb{R}_+^K$ for all $h$, and
2. $\sum_h x^h \leq \omega$.

These clearly define a convex set. And an allocation is Pareto efficient if it maximizes some weighted sum of (concave) utility representations. So the Pareto optimal allocations are the solutions to

$$\max_{(x^h)_{h\in H}} \sum_h \alpha^h u^h(x^h) \tag{6.1}$$

$$\text{st } x_j^h \geq 0 \qquad \text{for all } h \text{ and } j \tag{6.2}$$

$$\omega - \sum_h x_j^h \geq 0 \qquad \text{for all } j \tag{6.3}$$

Let $q_j$ be the shadow price of the adding up constraint for good $j$. The Kuhn-Tucker conditions are

$$\alpha^h \frac{\partial}{\partial x_j} u^h(x^h) - q_j \leq 0 \qquad \text{with equality if } x_j^h > 0, \text{ for all } h \text{ and } j \tag{6.4}$$

$$\omega_j - \sum_h x_j^h \geq 0 \text{ and } q_j \geq 0 \qquad \text{with complementary slackness, for all } j \tag{6.5}$$

Since the utilities are concave, these conditions are necessary and sufficient for an allocation to solve the optimization problem.

If we assume that $\mathbf{D}u^h \gg 0$, then all of the resource constraints must hold with equality, and their shadow prices must be positive. If we further restrict attention to interior allocations, we get a simple condition:

$$\alpha^h \mathbf{D}u^h(x^{*h}) = q \ \text{ for all } h.$$

Now take some consumers $h$ and two goods, $j$ and $k$. We have, as part of the above,

$$\alpha^h \frac{\partial}{\partial x_j} u^h(x^h) = q_j \tag{6.6}$$

$$\alpha^h \frac{\partial}{\partial x_k} u^h(x^h) = q_k. \tag{6.7}$$

Divide to get

$$\frac{\frac{\partial}{\partial x_j} u^h(x^h)}{\frac{\partial}{\partial x_k} u^h(x^h)} = \frac{q_j}{q_k},$$

which says that consumer $h$'s MRS between $j$ and $k$ equals the ratio of shadow prices for the resource constraints of $j$ and $k$. And since $h$, $j$, and $k$ were arbitrary, this implies that all consumers have equal MRS's for all pairs of goods.

So if we find (at an interior allocation) that two consumers do not have the same MRS for some pair of goods, we know there exists a Pareto improvement.

All of this should remind you very much of consumer theory, suggesting that markets might work well. We'll turn to that in a moment. But first, there are some caveats that should always be kept in mind when applying these ideas to policy analysis.

- The presumption that Pareto inefficient allocations are bad is often over interpreted. What might be uncontroversial is that a Pareto dominated allocation is bad because *one of the allocations that Pareto dominates it is better.* This is very different than claiming that any Pareto optimal allocation is better than any Pareto inefficient allocation. *That* claim is very hard to justify.

- One justification often given for the second claim is that, after moving from a Pareto inefficient allocation to a Pareto optimal one, it is then possible to make transfers from winners to losers so that we have a genuine Pareto improvement. One problem with this is that it is not true. (It is true in an exchange economy. But the claim can fail in more complicated problems.)

Another problem is that it is, without further argument, a pretty crap justification. If the transfers are not made, why should their existence in some other possible world be of any relevance to our actual world? Perhaps a (philosophical) utilitarian could buy this argument, but that is exactly the kind of substantive moral point of view we are trying to avoid in the move to Pareto concepts.

## 6.3 Further Directions

From the point of view of this course, which is focused on individual agents, we could stop here. But we have come so close to two major topics in the multi-person part of the theory that it would be a shame not to at least preview them. First, we might like to find reasonable ways for several agents to compromise and arrive at a collective ranking. This is the purview of **social choice theory**, covered in Subsection 6.3.1. Second, we are now in a position to discuss the purely price-theoretic aspects of markets. This is the purview of **general equilibrium theory**, covered in Subsection 6.3.2.[2] Finally, we can come full circle and use the results about general equilibrium to unpack the assumptions behind cost-benefit analysis. This is done in Subsection 6.3.3.

### 6.3.1 Arrow's Impossibility Theorem

The previous sections considered one person (you) evaluating allocations. If you have preferences over allocations that are rational, respect Pareto dominance, and respect Pareto indifference, then your preferences are represented by the composition of a social welfare functional and a profile of utility representations.

If you were a dictator who got to decide on allocations, then this would be all there was to say. But you are (thankfully!) not a dictator. There are typically many rational preferences over allocations consistent with Pareto dominance and Pareto indifference, so different people can can have quite different rankings of allocations, even if they all have preferences represented by social welfare functionals. And there is even more scope for disagreement if some people do not accept all of the assumptions of the previous section. What can we do then?

One thing we might do is ask if there is some procedure that accepts the rankings of several people, and returns a reasonable compromise. This is the setting of a famous, and disappointing, result due to Ken Arrow.

[2] A comprehensive treatment of markets really requires game theory, and is left for next quarter.

Arrow setup has a finite set $A$ of alternatives and a finite set $N = \{1, \ldots, n\}$ of citizens. Citizen $i$'s preferences over $A$ are given by a complete and transitive binary relation $\succsim_i$. We want a rule which determines a preference relation for each specification of the citizens' preferences.

This setup is quite flexible.

1. $A$ is the set of allocations in some economy, and each $i \in N$ is a policy analyst, as in the previous section.

2. A decathlon is a sporting event in which athletes compete in ten separate events. We can use Arrow's framework to think about scoring systems, with $A$ the set of competitors and each $i \in N$ as an event. The complete and transitive relation $\succsim_i$ is the order of finish in event $i$.

Let $\mathcal{R}$ be the set of all rational preferences on $A$. A **social welfare function** is a map $f : \mathcal{R}^n \to \mathcal{R}$. Notice two things:

1. *Any* profile of rational preferences is allowed.

2. The collective preference must also be rational.

This combination of assumptions is called **universal domain**.

Most people's first thought about how to compromise in such a setting is to use majority rule. But we saw way back in Chapter 1 that majority rule can lead to intransitivity. Thus it does not define a social welfare function.

Here is one social welfare function: fix some rational preference, $\succsim^*$, and use it for the collective preference whatever the citizens' preferences are. This swf is **imposed**—it is insensitive to individual preferences. This is unsatisfying. We want collective preferences to track individual preferences in at least the following sense: The social welfare function $f$ satisfies **weak Pareto** if $x \succ_i y$ for all $i$ implies $x\, f(\succsim)\, y$.

Arrow imposed one more assumption of social welfare functions. Say that $f$ satisfies **Independence of Irrelevant Alternatives** (IIA) if

$$x \succsim_i y \quad \text{if and only if} \quad x \succsim_i' y$$

implies

$$x\, f(\succsim)\, y \quad \text{if and only if} \quad x\, f(\succsim')\, y.$$

In words, the collective preference between $x$ and $y$ depends only on the individual preferences between $x$ and $y$—comparisons to third alternatives are irrelevant.

Not all rules satisfy IIA. Consider the **Borda count**: Each citizen assigns numbers to alternatives: 1 to the top ranked alternative, 2 to the second ranked, etc. (For simplicity, assume everyone has strict preferences.) For each alternative, sum the numbers assigned to that alternative by the citizens. The alternative with the lowest score is top-ranked socially, etc. This rule is obviously weakly Paretian, and it satisfies universal domain because the total scores make up a social utility function.

However, the Borda count does have an unattractive property. Let's see what it does on the Condorcet triple.

| 1 | 2 | 3 |
|---|---|---|
| $x$ 1 | $y$ 1 | $z$ 1 |
| $y$ 2 | $z$ 2 | $x$ 2 |
| $z$ 3 | $x$ 3 | $y$ 3 |

Each alternative gets a score of 6, so the social preference is $x \sim_S y \sim_S z$.

Now replace 3's preference by $x \succ z \succ y$. The Borda count gives scores of 5 to $x$, 6 to $y$, and 7 to $z$. (Check this!) All of the individual preferences over $y$ and $z$ are the same as in the Condorcet triple, but the social preference over $y$ and $z$ has changed.

Are there any rules that satisfy universal domain, weak Pareto, and IIA?

Yes. Pick some individual $i$, and declare her preference, whatever it is, to be the collective preference. Formally, $i$ is a **dictator** if $x \succ_i y$ implies $x$ is strictly collectively preferred to $y$. (Note that the definition is weaker than the example—the collective preference does not need to respect the dictator's indifference.)

**Theorem 6.3** (Arrow)**.** *Suppose $A$ contains at least three alternatives. If $f$ be a social welfare function satisfying universal domain, weak Pareto, and IIA, then $f$ has a dictator.*

A complete proof of this Theorem would be to much of a digression for our purposes. But a simple argument for the two citizens and three alternatives shows the heart of it.

Consider two citizens and three alternatives, $a$, $b$, and $c$. Assume that $a \succ_1 b$ and $b \succ_2 a$. Society must go one way or the other; assume it is $a \succ^S b$. Now consider the profile:

$$a \succ_1 b \succ_1 c$$
$$b \succ_2 c \succ_2 a$$

We have $a \succ^S b$ by IIA, $b \succ^S c$ by weak Pareto, and $a \succ^S c$ by transitivity. Since $c$ was arbitrary, we have shown that the rule must resolve all disagreements in favor of 1.

### 6.3.2 Prices and Walrasian Equilibrium

A **private ownership exchange economy** is a tuple

$$\mathcal{E}^P = \langle H, K, (\succsim^h)_{h\in H}, (\omega^h)_{h\in H}\rangle.$$

This is simply an exchange economy in which the social endowment has been divided into a private endowment $\omega^h$ for each consumer $h$. All of our previous definitions hold with $\omega = \sum_{h\in H}\omega^h$.

A **Walrasian equilibrium** (WE) is a pair $(p, x)$ where $p \in \mathbb{R}^K_+$ is a **price system** and $x$ is a feasible allocation such that each consumer optimizes:

$$x^h \text{ is } \succsim^h\text{—maximal in the set } B(p, \omega^h) = \{y \mid py \le p\omega^h\}$$

and markets clear: for each good $j$,

$$\sum_h x^h_j \le \sum_h \omega^h_j \quad \text{with equality if } p_j > 0.$$

The budget constraint here is a bit different than in the canonical consumer theory model: income depends on prices (as the value of the endowment), rather than being an exogenous number.

It will often be convenient to break this idea into two parts. Say that $p$ is a **Walrasian equilibrium price** if there is an allocation $x$ such that $(p, x)$ is a WE. Similarly, say that $x$ is a **Walrasian equilibrium allocation** if there is a price system $p$ such that $(p, x)$ is a WE.

Some comments:

1. The consumers are all **price takers**.

2. Equilibrium requires that all markets clear simultaneously.

3. If $(p, x)$ is a WE of $\mathcal{E}^P$, then so is $(\lambda p, x)$ for any real $\lambda > 0$.

There is a remarkable connection between Walrasian equilibria and Pareto optima. Assume that each consumer has locally-nonsatiated preferences. Then each consumer will spend her entire wealth: $px^h = p\omega^h$ for all $h$. Sum over consumers to get

$$p\sum_h x^h = p\sum \omega^h.$$

This is called **Walras's Law**.

One simple implication of Walras's law comes from rearranging:

$$p\left(\sum_h x^h - \omega^h\right) = 0.$$

We have restricted prices to be non-negative; the market clearing condition implies that $\sum_h (x_j^h - \omega_j^h) \leq 0$ for all goods $j$ at a Walrasian equilibrium. Together, these imply a condition that looks a lot like complementary slackness:

$$p_j\left(\sum_h x_j^h - \omega_j^h\right) = 0$$

for all $j$. In english, any good with excess supply at equilibrium has price 0.

A much deeper implication of Walras's Law is the following:

**Theorem 6.4** (First Fundamental Theorem of Welfare Economics)**.** *Suppose all consumers have locally-nonsatiated preferences and that $(p, x)$ be a WE. Then the allocation $x$ is Pareto optimal.*

**Proof.** Let $(p, x)$ be a WE and let $y$ be a feasible allocation that Pareto dominates $x$. Then $y^h \succsim^h x^h$ for all $h$ and $y^{h'} \succ^{h'} x^{h'}$ for at least one $h'$. This and consumer optimization imply $py^h \geq px^h$ for all $h$ and $py^{h'} > px^{h'}$. Sum over consumers to get

$$p\sum_h y^h = \sum_h py^h > \sum_h px^h = p\sum_h x^h = p\sum_h \omega^h.$$

But feasibility implies that $\sum_h y^h \leq \sum_h \omega^h$, a contradiction. ■

The First Welfare Theorem tells us that any Walrasian equilibrium allocation is Pareto optimal. The next result gives a kind of converse.

**Theorem 6.5** (Second Fundamental Theorem of Welfare Economics)**.** *Let $x^*$ be a Pareto optimal allocation, and assume that there is a Walrasian equilibrium when endowments are $\omega^h = x^{*h}$. Then there is a price system $p^*$ such that $(p^*, x^*)$ is a Walrasian equilibrium.*

**Proof.** Let $(p', x')$ be a WE of the economy $\langle H, K, (\succsim^h)_{h\in H}, (x^{*h})_{i\in H}\rangle$. By construction, $x^{*h}$ is in $B(p', x^{*h})$. Then optimization implies $x'^h \succsim^h x^{*h}$, and Pareto optimality of $x^*$ then implies $x'^h \sim^h x^{*h}$ for all $h$. Thus $x^{*h}$ is also optimal for $h$, and $(p', x^*)$ is a WE. ■

This theorem tells us that, under assumptions on primitives that guarantee existence, any Pareto optimum can be supported as a WE, and we sometimes say that $p^*$ is a vector of **supporting prices**.

Existence did not directly appear in the First Welfare Theorem, but that Theorem is pretty useless when there is no WE. As discussed in the appendix to this chapter, this is where convexity enters the story.

We can link these results more closely to our earlier work on welfare optima if we use calculus. To make our lives easier, we're going to go ahead and make some strong assumptions. Nothing that we say later really depends on these, but I want to focus on the core economic ideas, rather than math.

1. Each consumer $h$ has strictly positive endowment: $\omega^h \gg 0$.
2. Each consumer $h$ has a utility function $u^h$ that is
   (a) continuously differentiable,
   (b) strongly monotone: $\mathbf{D}u^h(x) \gg 0$,
   (c) strictly concave, and
   (d) has infinite marginal utility at zero consumption.

The first step is a characterization of WE based on first-order conditions. At an equilibrium $(p^*, x^*)$, each consumer $i$ is maximizing her utility function $u^i$ over the set $B(p^*, \omega^i)$. That is, she solves

$$\max_h u^h(x) \tag{6.8}$$

$$\text{st} \sum_j p_j x_j \leq \sum_j p_j \omega_j^h \tag{6.9}$$

Letting $\lambda^h$ be the shadow price of the budget constraint, the KT conditions are

$$\mathbf{D}u^h(x^{h^*}) = \lambda^h p,$$

along with the budget constraint. (The infinite marginal utility on the boundary condition means we do not have to worry about corner solutions.)

These conditions should look very familiar. In the case of interior Pareto optima, the FOCs are

$$\alpha^h \mathbf{D}u^h(x^{h^*}) = q.$$

These conditions are the same if we take $p^* = q$ and $\alpha^i = 1/\lambda^i$.

We can use this equivalence to shed more light on the Second Fundamental Welfare Theorem. Let $x^*$ be a Pareto optimum characterized by

$$\mathbf{D}u^h(x^{*h}) = \frac{1}{\alpha^h} q \quad \text{for} \quad h \in H.$$

If we choose $q$ as the price system and $(x^{*h})$ as the initial endowments, then consumer $h$ faces the budget $\{x^h \mid qx^h \leq qx^{*h}\}$. If $u^h$ is concave, then we can bound the utility of any bundle $x^h$:

$$u^h(x^h) \leq u^h(x^{*h}) + \mathbf{D}u^h(x^{*h})(x^h - x^{*h}) = u^h(x^{*h}) + \frac{1}{\alpha^h} q(x^h - x^{*h}) \leq u^h(x^{*h}).$$

Thus the consumer is willing to choose the correct bundle.

Again, we have arrived at results that almost beg to be over-interpreted.

- The theorems do *not* say that markets are better than other forms of allocation. They say that perfect markets are unbeatable by the Pareto criterion. That only means markets are better than something else if you know something about the other institution. If you do, you did not learn it from the welfare theorems.

- The second theorem says that a policy maker who can make lump-sum transfers between consumers and can operate perfect markets can separate the problems of distribution and efficiency. This says nothing about the possibility of such a separation in general.

### 6.3.3 Towards Cost-Benefit Analysis

Assume that YOU have rational and benevolent preferences over allocations, and that your preferences are jointly represented by the swfl $W$ and the profile of utility functions $(u^h)_{h \in H}$.

Then your favorite allocation solves

$$\max_{(x^h)_{h \in H}} \; W(u(x)) \tag{6.10}$$

$$\text{st} \quad \sum_h x^h \leq \sum_h \omega^h. \tag{6.11}$$

Restrict attention to interior optima, $x^* \gg 0$. Letting $q$ be the vector of shadow prices, the FOC are

$$\frac{\partial}{\partial u^h} W(u(x^*)) \cdot \frac{\partial}{\partial x_j} u^h(x^*) = q_j \quad \text{for all } h \text{ and } j.$$

We can now obviously complete the analysis as before, taking the weight $\alpha^h$ to be the derivative of $W$ with respect to consumer $h$'s utility.

The benefit of this slightly more general approach is that is shows our analysis is actually consistent with the full range of social preferences (assuming rationality and Pareto indifference, at least). And that extra generality is useful when we try to understand cost-benefit analysis.

Suppose we are at a WE $(p, x)$ and are considering moving to the allocation $x'$. (Assume that the project uses a non-market technology that has just become available.)

If $x$ and $x'$ are close together (the project is "small"), then we can approximate the change in welfare as

$$W(u(x')) - W(u(x)) \approx \sum_h \frac{\partial}{\partial u^h} W(u(x)) \mathbf{D}u^h(x) \cdot ((x^h)' - x^h).$$

Since $(p, x)$ is a WE,

$$\sum_h \frac{\partial}{\partial u^h} W(u(x) \mathbf{D}u^h(x^h) \cdot (x'^h - x^h) = \sum_h \frac{\partial}{\partial u^h} W(u(x)) \lambda^h p(x'^h - x^h).$$

This says that sum over consumers the change in the market value of their consumptions, using as weights

$$\sum_h \frac{\partial}{\partial u^h} W(u(x)) \lambda^h.$$

This is intuitive: the derivative factor is the marginal value of $h$'s utility to social welfare, and $\lambda^h$ is consumer $h$'s marginal utility of extra consumption (the same for all goods).

If $x^h$ is a welfare optimum according to the swfl $W$, then $\frac{\partial}{\partial u^h} W(u(x)) = 1/\lambda^h$, and the approximate change in welfare is

$$\sum_h p(x'^h - x^h).$$

Thus, in the neighborhood of a welfare optimal WE, a small project is welfare enhancing if it increases national income at the market prices.

This analysis highlights two limitations of standard cost-benefit analysis, on top of all the caveats we have already issued.

- If the WE is not a welfare optimum, then the distributional factors $\frac{\partial}{\partial u^h} W(u(x)) \lambda^h$ do not drop out, and we must account for distributional consequences of the project.

- We have implicitly assumed that all relevant goods are traded. If the project affects consumption levels of non-traded goods, then we can't use only market prices to value the change.

Now, I don't want to leave you with the impression that cost-benefit experts are unaware of these problems. Indeed, if you take cost-benefit, much of your time in that course will be spent dealing with these two difficulties. But in practice, these problems loom large, as can be seen by comparing the guidelines for CBA published by, say, the EPA to the syllabus of your cost-benefit class.

## 6.4 Appendix: Technical Details

### 6.4.1 Proof of Arrow's Theorem

To simplify the proof, I assume that every citizen has strict preferences. We begin by showing that all social rankings are made the same way.

**Lemma 6.1.** *Consider two pairs of alternatives,* $(a, b)$ *and* $(\alpha, \beta)$*. If each citizen has the same relative ranking of* $(a, b)$ *and* $(\alpha, \beta)$*, then the social preference over* $(a, b)$ *is the same as the social preference over* $(\alpha, \beta)$*, and both preferences are strict.*

PROOF OF LEMMA Assume the pair $(a, b)$ is distinct from the pair $(\alpha, \beta)$. (We can assume this because there are at least three alternatives.) Assume WLOG that $a \succsim_S b$. Consider a new profile in which $\alpha$ is ranked just above $a$ for every citizen (if $a \neq \alpha$), $\beta$ is ranked just below $b$ for every citizen (if $b \neq \beta$), and the rankings of $a$ and $b$ are the same as in the original profile. IIA implies that the social preferences between $a$ and $b$ and between $\alpha$ and $\beta$ are the same as in the original profile. By the weak Pareto property, $\alpha \succ_S a$ (if $a \neq \alpha$) and $b \succ_S \beta$ (if $b \neq \beta$). Since $a \succsim_S b$, transitivity of the social preference implies that $\alpha \succ_S \beta$. Finally, we can reverse the roles of $(a, b)$ and $(\alpha, \beta)$ in the above argument to conclude that, in fact, $a \succ_S b$. □

Next we will find a citizen who is pivotal between two alternatives $a$ and $b$. This citizen will turn out to be a dictator.

Consider two alternatives $a$ and $b$. Start with a profile in which $b \succ_i a$ for all $i$. By the weak Pareto property, $b \succ_S a$. Now let each citizen successively move $a$ above $b$, starting with $i = 1$. The weak Pareto property implies that we will eventually have $a \succ_S b$, and the lemma implies that $a$ becomes strictly better than $b$ as soon as it moves up at all. Thus there is a citizen $i^*$ such that $b \succ_S a$ if $b \succ_i a$ if and only if $i \geq i^*$ and $a \succ_S b$ if $a \succ_i b$ if and only if $i \leq i^*$.

Finally, we show that $i^*$ is a dictator. Consider an arbitrary pair of alternatives $\alpha$ and $\beta$, and assume $\alpha \succ_{i^*} \beta$. The rankings of $\alpha$ and $\beta$ by the other citizens are arbitrary. Take some alternative, $c$ distinct from both $\alpha$ and $\beta$. Consider a profile in which $c \succ_i \alpha$ and $c \succ_i \beta$ for all $i < i^*$, $\alpha \succ_i c$ and $\beta \succ_i c$ for all $i > i^*$, and $\alpha \succ_{i^*} c \succ_{i^*} \beta$. The lemma and the profile in the previous paragraph imply that $\alpha \succ_S c$ and $c \succ_S \beta$, so transitivity implies that $\alpha \succ_S \beta$. IIA implies that $\alpha \succ_S \beta$ whenever the individual rankings of $\alpha$ and $\beta$ are the same as in the profile we constructed—but all that was specified about this profile was that $i^*$ preferred $\alpha$ to $\beta$. By the lemma, this implies $i^*$ is a dictator.

### 6.4.2 Existence and Uniqueness of Walrasian Equilibrium

The first thing we need to do is verify that our solution concept is non-vacuous. This can be done in great generality, using techniques you will see (or perhaps already have seen) in Political Economy. The details are in chapter 5 of Riley.

The following construction is key. If prices are $p$, write $x^h(p)$ for consumer $h$'s demand. Since $h$ already owns $\omega^h$, the net trade she wants to make at prices $p$ is $z^h(p) = x^h(p) - \omega^h$. This is called $h$'s **excess demand**. The **aggregate excess demand** is the sum of these excess demands across all consumers:

$$z(p) = \sum_h z^h(p) = \sum_h x^h(p) - \sum_h \omega^h.$$

The market for good $j$ clears is excess demand for $j$ is zero, or if excess demand for $j$ is negative and the price of good $j$ is zero. That is, market clearing requires $z_j(p) \leq 0$ and $p_j z_j(p) = 0$.

A price $p \geq 0$ is a WE price if all markets clear at $p$.

**Lemma 6.2** (Walras's Law)**.** *Assume each consumer has locally non-satiated preferences. At any price vector $p$, the market value of excess demand is 0: $pz(p) = 0$.*

**Proof.**

$$pz(p) = p\left(\sum_h x^h(p) - \omega^h\right) = \sum_h \left(px^h(p) - p\omega^h\right).$$

Local non-satiation implies that each consumer spends her entire wealth:

$$px^h(p) = p\omega^h \quad \text{for all } h.$$

■

Since the market value of excess demand is always zero, we know that market $j$ clears if the other markets all clear. That is, we can ignore one market-clearing condition. Since only relative prices matter, we can always normalize prices so that $\sum_j p_j = 1$. Together, these allow us to reduce the dimensionality of the problem by one.

That is particularly powerful when there are only two goods, since it implies that at any strictly positive price vector that is not a WE price, one good is in excess supply and the other is in excess demand. For the rest of this discussion, we focus on this two-good case.

With only two goods, prices can be normalized to $(p_1, 1-p_1)$. By Walras's Law, it is enough to find a $p_1$ that clears the market for good 1: $z(p_1) = 0$.

Assume preferences are continuous and strictly convex. Then $z^h$ is a single-valued and continuous for all strictly positive prices. Thus aggregate excess demand in single-valued and continuous.

Continuity when one price is zero is more delicate. Assume the boundary marginal utility condition, so $\lim_{p_1 \to 0} z_1(p_1) = \infty$ and $\lim_{p_1 \to 1} z_2(p_1) = \infty$.

Then there are $\underline{p}$ close to 0 and $\overline{p}$ close to 1 such that $z_1(\underline{p}) > 0$ and $z_2(\overline{p}) > 0$. Now take any $\hat{p}$ with $\underline{p} < \hat{p} < \overline{p}$. If $\hat{p}$ is not a WE price, then either $z_1(\hat{p})$ or $z_2(\hat{p})$ is negative, by Walras's Law. If $z_1(\hat{p}) < 0$, then by continuity and the intermediate value theorem, there is a $\hat{\hat{p}} \in (\underline{p}, \hat{p})$ with $z_1(\hat{\hat{p}}) = 0$. A similar argument works if $z_2(\hat{p}) < 0$, so in any case there is a $p^* \in (\underline{p}, \overline{p})$ that is a WE price.

It would be nice to complement this existence result with a uniqueness one. But uniqueness is not at all a general property of WE.

## Problems

**Exercise 6.1.** Suppose that you have utility representations $u^h$ for the $n$ consumers in some society, and that your preferences over allocations are represented by a weighted utilitarian social welfare functional with weights $(\alpha^h)_{h=1}^n$. Show that you can change to an equal-weighted utilitarian social welfare functional without changing your ranking of allocations by changing at the same time the utility representations for the consumers.

**Exercise 6.2.** Consider the problem of allocating consumptions of two goods across two consumers. The two goods are called tillip and quillip, and the two consumers are called 1 and 2. Consumer 1 has utility function $u^1(t,q) = 6 + .4 \log t + .6 \log q$ (where $t$ is the amount of tillip 1 consumes and $q$ is the amount of quillip). Consumer 2 has utility function $u^2(t,q) = 8 + \log t + \log q$. The social endowment consists of 15 units of tillip and 20 units of quillip.

1. Suppose that, relative to these utility representations, your preferences over allocations are represented by a social welfare functional of the following form: Social welfare, as a function of $(u^1, u^2)$, is a weighted sum with weight 2 on the lesser of $u^1$ and $u^2$ and weight 1 on the greater of the two. What is your optimal allocation?

2. What is the set of all Pareto optimal allocations for this economy?

3. Assume that the social endowment is divided between the consumers, with consumer 1 getting 10 units of each good, and consumer 2 getting 5 units of tillip and 10 units of quillip. What is the Walrasian equilibrium of this economy?

**Exercise 6.3.** Consider an exchange economy with two goods in which all consumers have quasilinear utility: $u^h(x_1^h, x_2^h) = x_1^h + v^h(x_2^h)$. Assume that each $v^h$ is twice continuously differentiable with

$$\frac{\partial v^h}{\partial x_2^h}(x_2^h) > 0 \quad \text{and} \quad \frac{\partial^2 v^h}{\partial (x_2^h)^2}(x_2^h) < 0.$$

Let the social endowment be $\omega = (\omega_1, \omega_2)$.

This problem will walk you through the argument for the following result:

> Allocation $x^*$ with $(x^*)_1^h > 0$ for all $h$ is Pareto optimal if and only if the vector $[(x^*)_2^h]_{h \in H}$ solves
>
> $$\begin{aligned} \max_{x_2^h} \ & \sum_h v^h(x_2^h) \\ \text{st} \ & \sum_h x_2^h \le \omega_2. \end{aligned}$$

1. Use the equivalence of Pareto optimality and maximization of a monotonic social welfare functional to explain why any solution to the maximization in the claim in fact determines Pareto optimal allocations.

2. To start the other direction, derive the first-order conditions that characterize Pareto optimality.

3. Show that, an allocation is a Pareto optimum with $x_1^h > 0$ for all $h$ only if there is a constant $\lambda$ such that, for all $h$,

   $$\frac{\partial v^h}{\partial x_2^h}(x_2^h) \le \lambda,$$

   with equality for $x_2^h > 0$.

4. Why does this complete the proof?

**Exercise 6.4.** The concept of Pareto optimality defined in the notes is sometimes called *strong Pareto optimality.* An outcome is **weakly Pareto optimal** is there is no alternative feasible allocation that makes all individuals strictly better off.

1. Show that if an allocation is strongly Pareto optimal, then it is also weakly Pareto optimal.

2. Consider an exchange economy in which every consumer has continuous and strictly monotone preferences. Show that weak and strong Pareto optimality are equivalent for interior allocations.

3. What can go wrong without interiority?

**Exercise 6.5** (Rubinstein)**.** Consider the following social choice problem: a group has 2 members who must choose from the set $\{A, B, L\}$, where $A$ and $B$ are prizes and $L$ is the lottery that gives each prize with equal probability. Each citizen has strict preferences that satisfy the vNM axioms. Show that there is a nondictatorial swf that satisfies IIA and WP. Reconcile this fact with Arrow's theorem.

**Exercise 6.6.** Consider a social choice problem on the domain of all possible strict preferences on $A$. Plurality rule is the SWF which gives each alternative one point for each citizen who has that alternative top ranked and zero points otherwise, and then ranks the alternatives in order of the points. Which of Arrow's conditions does plurality rule satisfy?

**Exercise 6.7.** A particular policy maker I know is very big on mellow consumers. Specifically, she hopes to prevent consumers from envying each other. To this end, she defines an *envy-free* allocation as one in which no consumer would rather have the consumption bundle assigned to another consumer rather than his or her own. She also wishes the allocation to be efficient.

This policy maker is also lazy. She isn't willing to figure out the utility functions of the consumers. (She does have a list of all of their endowments.) She is blessed with an economy that functions well as an exchange economy—however she rearranges endowments, the economy finds a Walrasian equilibrium.

Can you help out this policy maker? Specifically, describe how to reallocate endowments so that the resulting Walrasian equilibrium is guaranteed to be both efficient and envy-free. (Hint: the trick is to find some way to redistribute endowments so that, at every set of prices, consumers all begin with the same wealth to spend on consumption.)

**Exercise 6.8.** Imagine a three-consumer economy in which the first commodity is gardening services, consumption of which makes one's yard more beautiful, and the second good is food. Imagine that two of the consumers live in adjacent houses, while the third lives on the other side of a particularly large mountain. Consumption by the third consumer of gardening services generates no externality for the other consumers, but each of the others generates a positive externality for her neighbor through the consumption of gardening services. To be precise, imagine that consumers 1 and 2 have utility functions of the form

$$u^h(x) = w(x_1^1) + w(x_1^2) + x_2^h,$$

where $w$ is a strictly increasing, strictly concave, and differentiable function. Note well that consumers 1 and 2 get just as much utility out of their neighbor's yard as the do out of their own, and their utility for food is linear. Also imagine that consumer 3 has utility

$$u^3(x) = w(x_1^3) + x_2^3.$$

There is a social endowment of gardening services and food.

1. Suppose the social endowment is initially allocated evenly among the three consumers. What will be the corresponding Walrasian equilibrium?

2. Characterize the set of Pareto optimal allocations of the social endowment. Is the equilibrium allocation in part (a) Pareto optimal?

# Chapter 7

# The Envelope Theorem

So far, almost all of our results have relied, one way or another, on one mathematical tool—the supporting hyperplane theorem. But this is not the only mathematical tool we need for price theory. This short chapter introduces the second big mathematical hammer, and the next two show how useful it can be.

Let's start with a concrete example. Consider a profit maximizing firm with a very simple technology: it can produce $q$ units of good 2 ("output") using at least $\frac{1}{2}q^2$ units of good 1 ("input"). The price of good one is fixed at 1; the price of good 2 will vary and is denoted $p$.

If the firm can choose between output quantities $q \in Q$ for some set $Q$, it will solve

$$\max_{q \in Q} \; pq - \frac{1}{2}g^2.$$

We will be particularly concerned with the way the solutions to problems like this are related to the value of the objective at the solution. To be concrete about this, define the function $\pi : \mathbb{R}_+ \to \mathbb{R}$ by

$$\pi(p) = \max_{q \in Q} \; pq - \frac{1}{2}q^2.$$

And let the optimal choice(s) be $q^*$, from

$$q^*(p) = \arg\max_{q \in Q} \; pq - \frac{1}{2}q^2.$$

Our question is: how are $\pi$ and $q^*$ related?

Let's start with a discrete choice version: $Q = \{1, 2\}$. If the firm chooses $q = 1$, profit is $p - \frac{1}{2}$. If it chooses $q = 2$, profit is $2p - 2$. Choosing 2 maximizes profit if

$$2p - 2 \geq p - \frac{1}{2} \quad \Leftrightarrow \quad p \geq \frac{3}{2}.$$

(Notice that the two plans give equal profit for $p = \frac{3}{2}$.) Thus

$$q^*(p) = \begin{cases} 2 & \text{if } p > \frac{3}{2} \\ \{1, 2\} & \text{if } p = \frac{3}{2} \\ 1 & \text{if } p < \frac{3}{2} \end{cases},$$

and

$$\pi(p) = \begin{cases} 2p - 2 & \text{if } p \geq \frac{3}{2} \\ p - \frac{1}{2} & \text{if } p < \frac{3}{2} \end{cases}.$$

Now a relationship between $\pi$ and $q^*$ jumps out at you:

- for $p > \frac{3}{2}$, we see $q^*(p) = 2 = \frac{d\pi}{dp}(p)$, and
- for $p < \frac{3}{2}$, we see $q^*(p) = 1 = \frac{d\pi}{dp}(p)$.

At any price where the profit maximization problem has a unique solution, that solution is the derivative of the maximized profit function.

You might think that something special is going on here, due to the discrete choice nature of the problem. After all, the optimal choice is constant on the intervals not including $p = \frac{3}{2}$. If that is your intuition, you will be surprised by what happens next.

Let $Q = \mathbb{R}_+$. The firm solves

$$\max_{q \geq 0} \; pq - \frac{1}{2}q^2.$$

The objective function is strictly concave, so the first-order condition gives

$$q^*(p) = p$$

as the unique solution. Substitute this into the expression for profit to get

$$\pi(p) = pq^*(p) - \frac{1}{2}(q^*(p))^2 = \frac{1}{2}p^2.$$

Again, we have

$$q^*(p) = p = \frac{d\pi}{dp}(p).$$

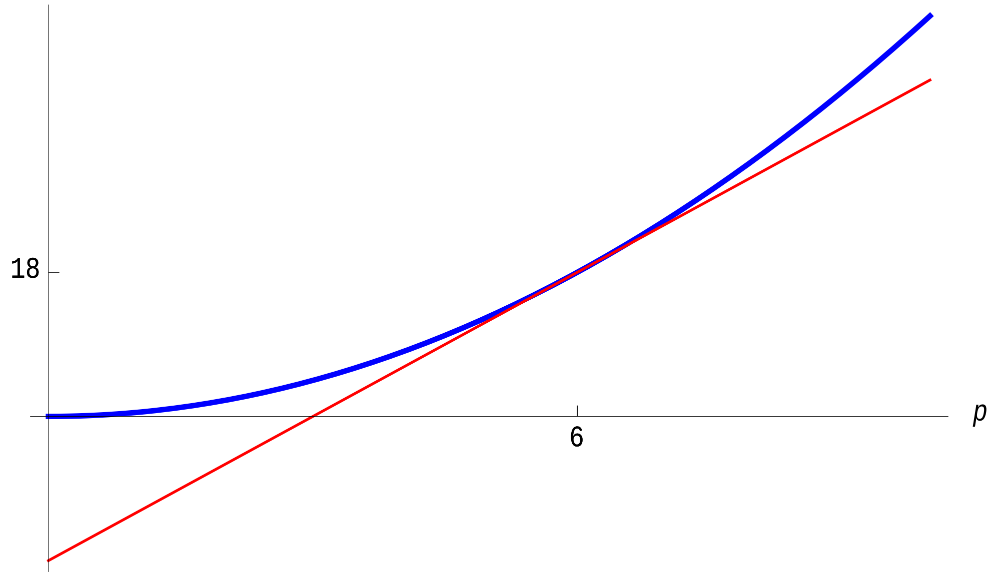


Figure 7.1: Comparison of profit with output fixed at 6 (in red, light) and maximized profit (in blue, heavy), as functions of the output price.

Intuitively, thinking about the change in price with output held fixed gives the right answer even when output varies optimally. In jargon you will hear repeatedly, the direct effect of price on profit is *first-order*, while the indirect effect working through the optimal adjustment of output to price is *second-order,* at least near an optimum.

The best way to understand what is going on is through Figure 7.1. It plots two different functions of the output price. In red, we have the function $p \mapsto 6p - 18$. This is the profit the firm earns if it produces the optimal output for price $p = 6$ no matter what the price is. In blue, we have $p \mapsto \frac{1}{2}p^2$. This is the maximum profit possible for each price.

The figure illustrated two important facts. First, at $p = 6$, choosing output 6 is actually optimal, and leads to a profit of 18. Thus both curves go through the point $(6, 18)$. Second, the blue curve can never be below the red. If it were, then the maximal level of profit would be less than the profit to be earned by choosing $q = 6$. That's impossible, since 6 is a feasible output. Together, these observations show that the blue curve is tangent to the red line.

## 7.1 A Formal Statement and Application

There is one way in which the previous example is misleading. The idea we are developing here has nothing to do with linearity or convexity. (This is makes it a particularly useful complement to the supporting hyperplane theorem.) The general form of the problem we are interested in is as follows. There is a set of feasible choices, $X$, and a set of possible parameters, $\Theta$. Given an **objective function** $f : X \times \Theta \to \mathbb{R}$, we consider the family of optimization problems given by:

$$\max_{x \in X} f(x, \theta).$$

Solving this problem for all $\theta$ defines two objects. The **solution correspondence** is the set of maximizers as a function of $\theta$:

$$x^*(\theta) = \arg\max_{x \in X} f(x, \theta).$$

The **value function** tells us the maximized value of the objective for any parameter:

$$V(\theta) = \max_{x \in X} f(x, \theta).$$

(To be really careful, that max should be a sup. But we will only be concerned with problems that actually have solutions.)

Figure 7.2 shows how the argument about the firm's profit generalizes to this more general context. Suppose that $x^*$ is a singleton for $\theta = 0.5$. The red curve is the graph of the function $\theta \mapsto f(x^*(0.5), \theta)$. That is, it is the value obtained by choosing $x^*(0.5)$ no matter what. The blue curve is the graph of the value function $V$. An argument just like the one for the firm before shows that the two curves are tangent at $\theta = 0.5$. In other words, we have

$$V'(\theta) = \frac{\partial f}{\partial \theta}(x^*(\theta), \theta).$$

This is an instance of a kind of result called an **envelope theorem**. There are many ways to make the idea precise, and which way you want will depend on what problem you are considering. Here is a version that clarifies the economic content without asking for too much background in real analysis.

**Theorem 7.1.** *Suppose*

1. *$X$ is compact and $\Theta$ is open,*

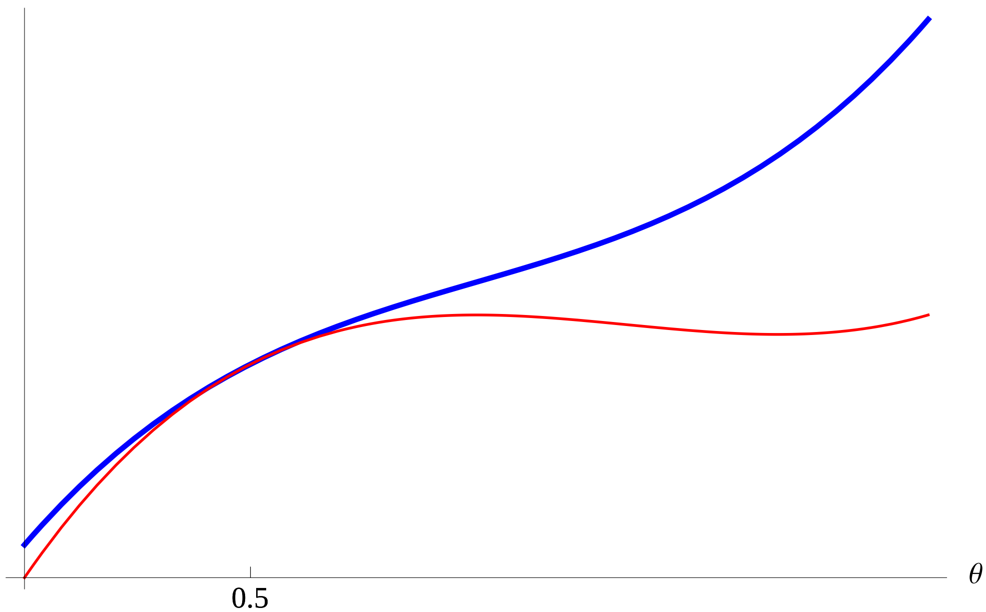


Figure 7.2: Comparison of an objective function $f(x^*(0.5), \theta)$ (in red, light) and the associated value function $V(\theta)$ (in blue, heavy), as functions of $\theta$.

2. *$f$ is continuous in $x$ and $\theta$ and is continuously differentiable in $\theta$, and*

3. *the derivative $\frac{\partial f}{\partial \theta}$ is continuous in $x$ and $\theta$.*

*Then at any $\theta$ for which $x^*(\theta)$ is a singleton, the value function is differentiable with*

$$V'(\theta) = \frac{\partial f}{\partial \theta}(x^*(\theta), \theta).$$

**Remark 7.1.** Multidimensional problems don't introduce any new complications. Just replace the derivative in assumption 3 with the assumption that the gradient

$$\mathbf{D}_\theta f(a, \theta) = \begin{pmatrix} \frac{\partial f}{\partial \theta_1} \\ \vdots \\ \frac{\partial f}{\partial \theta_m} \end{pmatrix}$$

is continuous, and replace the displayed equation in the conclusion with

$$\frac{\partial V}{\partial \theta_j}(\theta) = \frac{\partial f}{\partial \theta_j}(x^*(\theta), \theta) \quad \text{for all } j.$$

### 7.1.1 Cost Minimization

In the next chapter, we will get a lot of mileage out of this result applied to the following (rather strange) approach to consumer theory. Consider a consumer of the sort we studied in Chapter 4—she has preferences over bundles in $\mathbb{R}^n_+$ represented by a utility function $U$, and she faces linear prices $p$. But instead of asking her to maximize utility given $p$, we fix a value for utility, $\overline{U}$, and ask how much the consumer must spend to attain that utility.

Formally, the consumer's expenditure minimization problem given prices $p$ and utility level $\overline{U}$ is

$$\min_{x \in \mathbb{R}^n_+} \; px$$
$$\text{st } \; U(x) \geq \overline{U}$$

Let $x^c(p, \overline{U})$ be the solution to the expenditure minimization problem. It is called the **compensated demand**. Let $M(p, \overline{U})$ be the minimized value of this problem. It is called the **expenditure function**.

**Theorem 7.2** (Shepard's Lemma)**.** *Suppose that the consumer's expenditure minimization problem has a unique solution at prices $p$ and utility level $\overline{U}$. Then the expenditure function is differentiable in $p$ at $(p, \overline{U}$ with*

$$\mathbf{D}M(p, \overline{U}) = x^c(p, \overline{U}).$$

**Proof.** This is an almost automatic corollary of Theorem 7.1. The only complication is that that theorem is about maximization problems rather than minimization problems. But going back and forth is easy. Define $g(x, p) = -px$. Then the expenditure minimization problem is identical to the problem

$$\begin{aligned} \max_{x \in \mathbb{R}^n_+} \; & g(x, p) \\ \text{st} \; & U(x) \geq \overline{U}. \end{aligned}$$

If $V$ is the value function of this problem, the envelope theorem says

$$\mathbf{D}V(p, \overline{U}) = -x^c(p, \overline{U}).$$

But $V(p, \overline{U}) = -M(p, \overline{U})$. ■

You will really understand what is going on when you can replicate the analysis of Figure 7.1 for the case of expenditure minimization, without transforming the problem into a maximization.

**Remark 7.2.** Shepard's Lemma can be applied almost word for word in producer theory. Consider a firm that produces a single output according to the production function $f$. Assume that the firm is a price-taker on the input markets (but not necessarily on outputs). Fix input prices at $r$. Then the **cost function** $C(r, q)$ is the value function of the following problem:

$$\min_z \; rz \tag{7.1}$$

$$\text{st} \; f(z) \geq q. \tag{7.2}$$

In this context, Shepard's Lemma says that the gradient of the cost function gives the input demands, holding target output fixed as prices vary.

## 7.2 Some Formal Details and Extentions

This section will expand on the previous section's discussion of the envelope theorem in three ways. First, I will say a bit about the proof of Theorem 7.1. Second, I will informally describe a sense in which the conclusion of that Theorem can be beefed up without making any more assumptions. Third, I will discuss the extension to the case where the feasible set depends on the parameter.

1. The best way to think about the argument for Theorem 7.1 is the simple application of revealed preference analysis, combined with a lot of mathematical analysis to ensure that the revealed preference argument can get started. I'll discuss some of the mathematical throat-clearing in the guise of discussing the hypotheses of the Theorem.

    - We assume $\Theta$ is open because open sets are the natural domains for talking about derivatives.
    - A solution exists for each $\theta$ because $X$ is compact and $f$ is continuous in $x$.
    - The solution correspondence is upper hemicontinuous because $X$ is compact and $f$ is continuous. (In case you've never seen this terminology, it means that if $\theta_n \to \theta$, $x_n \in x^*(\theta_n)$ for all $n$, and $x_n \to x$, then $x \in x^*(\theta)$. If $x^*$ is singleton-valued for all $\theta$, it reduces to continuity of the function $x^*$.)

    The heart of the proof is the following argument. For simplicity, assume the solution is unique on an interval containing $\theta$. By optimization, the value function satisfies the inequalities

    $$V(\theta) = f(x^*(\theta), \theta) \geq f(x^*(\theta'), \theta) \quad \text{and} \quad V(\theta') = f(x^*(\theta'), \theta'),$$

    which imply

    $$V(\theta) - V(\theta') \geq f(x^*(\theta'), \theta) - f(x^*(\theta'), \theta').$$

    Similarly,

    $$V(\theta') - V(\theta) \geq f(x^*(\theta), \theta') - f(x^*(\theta), \theta).$$

    Now, fix $\theta > \theta'$. Combine the two inequalities above to get

    $$f(x^*(\theta'), \theta) - f(x^*(\theta'), \theta') \leq V(\theta) - V(\theta') \leq f(x^*(\theta'), \theta') - f(x^*(\theta'), \theta).$$

Divide by $\theta - \theta'$ to get

$$\frac{f(x^*(\theta'),\theta) - f(x^*(\theta'),\theta')}{\theta - \theta'} \leq \frac{V(\theta) - V(\theta')}{\theta - \theta'} \leq \frac{f(x^*(\theta'),\theta') - f(x^*(\theta'),\theta)}{\theta - \theta'}.$$

As $\theta' \to \theta$, both expressions on the outside approach

$$\frac{\partial f}{\partial \theta}(x^*(\theta),\theta),$$

by the definition of the derivative and the fact that the derivatives of $f$ are continuous. Since they sandwich the quotient defining the derivative of $V$, the theorem follows.

2. The argument outlined above is enough to establish Theorem 7.1. But much more can be derived from those same assumptions. In particular, they ensure that the value function $V$ satisfies a property from real analysis called *absolute continuity.* I won't give a formal definition of that here, but I do want to point out a couple of implications.

   First, an absolutely continuous function is almost everywhere differentiable. Intuitively, this means that if you were to pick parameters at random from a uniform distribution on the parameter space, the probability you'd pick values at which the function is not differentiable is zero.

   Second, an absolutely continuous function is equal to the integral of its almost-everywhere derivative. In our application, we have

   $$V(\theta) - V(\theta') = \int_{\theta'}^{\theta} \frac{\partial f}{\partial \theta}(x^*(\theta),\theta)\,d\theta.$$

   This integral form of the envelope theorem is just the tool we will need later to discuss consumer surplus, and is also very important for the theories of auctions and mechanism design.

3. So far, we have restricted attention to problems in which the feasible set did not depend on the parameters. This is restrictive—it rules out the classic consumer's problem, to take just one example. But the result does not depend on that restriction.

Consider the problem

$$\max_x f(x,\theta)$$
$$\text{st } g(x,\theta) \geq 0.$$

We can apply the previous envelope theorems to the Lagrangian,

$$\mathcal{L}(x,\theta,\lambda) = f(x,\theta) + \lambda \cdot g(x,\theta).$$

This turns out to be key to interpreting Kuhn-Tucker multipliers. Consider the special case:

$$\max_x f(x)$$
$$\text{st } h(x) \leq \theta.$$

The Lagrangian is

$$\mathcal{L}(x,\theta,\lambda) = f(x) + \lambda \cdot (\theta - h(x)).$$

The value function $V(\theta)$ tells us the maximum value possible given resources $\theta$. And the envelope theorem says that $V'(\theta) = \lambda$. Thus the multiplier is exactly the marginal benefit of relaxing the constraint.

### 7.2.1 The Second-Price Auction

In Political Economy, we considered the following auction setup. The seller has a single unit of some good. Each of $n$ bidders has valuation $v_i \in \mathbb{R}_+$ for the good. In a second-price auction, each bidder submits a sealed bid, $b_i \in \mathbb{R}_+$. The bidder who submits the highest bid wins the good. (Ties are broken by a uniform randomization.) The winning bidder pays the seller the amount equal to the highest bid submitted by a non-winner.

The amazing fact about this auction is that each bidder has a weakly dominant strategy, namely, bid truthfully. That is, $b_i = v_i$. In the language of mechanism design, we say that the second-price auction is **dominant-strategy incentive compatible** (DSIC).

DSIC is an attractive property for at least two reasons. First, it means the auction is easy to play. Bidders do not have to have much sophistication or knowledge of the other bidders in order to bid well. Second, the auction is robust. Since the optimal strategy is independent of any facts about the distributions of valuations, the seller does not have to know such facts to predict the bidders' strategies.

Since DSIC is such a nice property, we would like to know if any other reasonable auction can satisfy it. For our purposes, we will take **reasonable** to mean two things.

1. A bidder with valuation $v_i = 0$ gets payoff 0.

2. The good is allocated to a bidder with the highest valuation (i.e., the auction is efficient).

And now we get the payoff: A reasonable auction is DSIC if and only if it is payoff equivalent to the second-price auction.

The envelope theorem is at the heart of the argument for this result. To see how to apply it, some notation is helpful. Write a profile of bids for everyone other than bidder $i$ as $b_{-i}$. If bidder $i$ bids $b_i$, write the probability she gets the good as $x(b_i, b_{-i})$ and the amount she pays as $p(b_i, b_{-i})$. In this notation, for a fixed $b_{-i}$, bidder $i$ solves

$$\max_b x(b, b_{-i}) \cdot v_i - p(b, b_{-i}),$$

and the value function of bidder $i$ is

$$U_i(v_i, b_{-i}) = \max_b x(b, b_{-i}) \cdot v_i - p(b, b_{-i}).$$

A few observations will make this really useful. First, in any DSIC auction, each of the other bidders submits their true valuation as their bid: $b_j = v_J$ for $j \neq i$. Second, the efficiency part of reasonableness implies that

$$x(b, b_{-i}) = \begin{cases} 1 & \text{if } b > \max b_{-i} \\ 0 & \text{if } b < \max b_{-i} \end{cases}. \tag{7.3}$$

(There are many ways to be reasonable when $b = \max b_{-i}$. The choice does not matter.)

So, if we are entitled to use the integral form of the envelope theorem, we have

$$U_i(v_i, b_{-i}) - U_i(0, b_{-i}) = \int_0^{v_i} x(v, b_{-i})\, dv.$$

From the first condition of reasonableness, we have $U_i(0, b_{-i}) = 0$. Using that and Equation 7.3 yields

$$U_i(v_i, b_{-i}) = \begin{cases} v_i - \max b_{-i} & \text{if } v_i > \max b_{-i} \\ 0 & \text{if } v_i \leq \max b_{-i} \end{cases}.$$

But this is exactly the payoff bidder $i$ gets in the second-price auction!

Indeed, if bidder $i$ bids $b = v_i$ and wins the auction, we have

$$\begin{aligned} U_i(v_i, b_{-i}) &= v_i - \max b_{-i} \\ &= v_i - p(b, b_{-i}), \end{aligned}$$

so $p(b, b_{-i}) = \max b_{-i}$ when $i$ is the winner.

There is one loose end—are we entitled to use the envelope theorem here? This is not guaranteed by Theorem 7.1, since that result assume the payoff function is continuous. It turns out this is not a problem. Theorem 2 of Milgrom and Segal's "Envelope Theorems for Arbitrary Choice Sets" (*Econometerica,* 2002) does cover the relatively mild discontinuity of the auction setting. That paper is where you should turn if you know what absolute continuity is and you want to see the most general form of the envelope theorem.

## Problems

**Exercise 7.1.** A profit-maximizing firm must decide where to locate a retail outlet. The set of possible locations is the interval $[0, 1]$. All locations cost the same; what distinguishes locations is how many consumers are nearby. Profit at location $x$ is given by a function $R(x - \theta)$, where $R$ is differentiable, strictly concave, and maximized at 0. Here, $\theta$ is a parameter related to the distribution of customers along the interval. It satisfies $0 < \theta < 1$.

1. What is the firm's optimal location? Justify your answer with reference to the first-order condition.

2. Use the envelope theorem to deduce how the maximized profit varies with $\theta$.

3. Now suppose that the firm can only locate at one of the endpoints of the interval: $x \in \{0, 1\}$. What does the envelope theorem say in this case?

**Exercise 7.2.** Consider the auction environment from the text, with one change. Each bidder has a budget $B_i$, and can afford to pay only amounts less than or equal to $B_i$. Consider the allocation rule that, given bids $(b_1, \ldots, b_n)$, awards the good to the bidder with the largest value of $\min(b_i, B_i)$. (Ties are broken arbitrarily but consistently.)

Derive the payment rule that makes this auction DSIC. (Note: the integral form of the envelope theorem applies to this problem.)

# Part III

# Specialty Topics

# Chapter 8

# Consumer Theory: A Deeper Look

So far, we haven't been able to say much that is both general and interesting about demand. The problem, that you will recall from intermediate micro, is that income effects are completely unrestricted, and price effects are "contaminated" by income effects.

Changing focus a bit will let us develop some general theory. We will:

1. Formally decompose price effects into income and substitution effects.

2. Give a complete analysis of substitution effects.

3. Learn how to make quantitative statements about consumer welfare.

All three steps make essential use of the expenditure function.

## 8.1 Duality in Consumer Theory

Consider a consumer who chooses a bundle of consumption from $\mathbb{R}^n_+$ subject to the budget constraint $px \leq I$. The consumer has locally insatiable preferences represented by utility function $U$. Assume that the consumer has a unique, interior demand for any vector of positive prices. Further assume that the demand function is a differentiable function of prices and income.

> **Remark 8.1.** In terms of primitives, a set of sufficient assumptions is that $U$ is strictly quasiconcave, twice continuously differentiable, and satisfies both the boundary assumption from Chapter 4 and a technical restriction involving something called the *bordered Hessian* of $U$.

The **consumer's utility maximization problem** is:

$$\max_{x \in \mathbb{R}^n_+} U(x)$$
$$\text{st } px \leq I.$$

The solution is the demand function $x^*(p, I)$, and the value function is

$$V(p, I) = U(x^*(p, I)).$$

In this context, the value function is called the **indirect utility function.**

We will learn about this problem indirectly, by studying the consumer's expenditure minimization problem. As defined in the last chapter, this is

$$\min_{x \in \mathbb{R}^n_+} px$$
$$\text{st } U(x) \geq \overline{U}.$$

The solution is the compensated demand function $x^c(p, \overline{U})$, and the value function is the expenditure function

$$M(p, \overline{U}) = px^c(p, \overline{U}).$$

The expenditure minimization problem is not a problem any consumer actually faces. But it is closely related to the actual consumer's problem. Figure 8.1 illustrates the relationship. (This use of *duality* is common in optimization theory.)

**Lemma 8.1** (Duality Lemma)**.** *Suppose $U$ is a continuous utility representation of locally non-satiated preferences on $\mathbb{R}^n_+$ and that the price vector is $p \gg 0$.*

1. *If $x^*$ solves the utility maximization problem with income $I$, then $x^*$ is solves the expenditure maximization problem when $\overline{U} = U(x^*)$, and the minimized value of expenditure is $I$.*

2. *If $x^*$ solves the expenditure minimization problem with $\overline{U} > U(0)$, then $x^*$ solves the utility maximization problem with income $px^*$, and the maximized value of utility is $\overline{U}$.*

**Proof.**

1. Suppose $x^*$ solves the utility maximization problem but does not minimize expenditure with target utility $U(x^*)$. Then there is an $x'$ such that $U(x') \geq U(x^*)$ and $px' <$

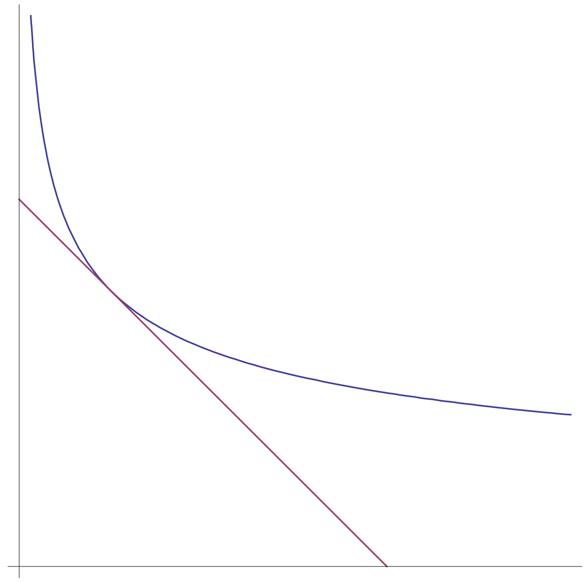

Figure 8.1: Is this an indifference curve tangent to a budget constraint or an iso-expenditure line tangent to a utility constraint? Lemma 8.1 says yes..

$px^* \leq I$. Local non satiation implies that there is some $x''$ very close to $x'$ with $U(x'') > U(x^*)$ and $px'' < I$. But this contradicts the optimality of $x^*$ in the utility maximization problem. Thus $x^*$ minimizes expenditure for target utility $U(x^*)$, and minimized expenditure is $px^*$, which equals $I$ by local nonsatiation.

2. Since $\overline{U} > U(0)$, the expenditure minimizing $x^*$ must be $x^* \neq 0$, and so $px^* > 0$. Suppose $x^*$ solves the expenditure minimization problem but does not maximize utility with income $px^*$. Then there is an $x'$ with $U(x') > U(x^*)$ and $px' \leq px^*$. Consider the bundle $x'' = \alpha x'$, where $\alpha < 1$ is a real number. If $\alpha$ is small enough, continuity of $U$ implies $U(x'') > U(x^*)$, while $\alpha < 1$ implies $px'' < px' \leq px^*$. But this contradicts optimality of $x^*$ in the expenditure minimization problem. Thus $x^*$ maximizes utility for income $px^*$.

   Next suppose $U(x^*) > \overline{U}$. Consider the bundle $x' = \alpha x'$, where $\alpha < 1$ is a real number. If $\alpha$ is small enough, continuity of $U$ implies $U(x') > \overline{U}$, while $\alpha < 1$ implies $px' < px^*$, a contradiction.

■

Our first step in using duality to understand the consumer's utility maximization problem will be to derive a statement in the spirit of Shepard's Lemma, but connecting indirect utility and demand. Start with the identity

$$V(p, M(p, \overline{U})) = \overline{U}.$$

(This is one part of part 2 of Lemma 8.1.) Differentiate with respect to $p_j$ to get

$$\frac{\partial V}{\partial p_j}(p, M(p, \overline{U})) + \frac{\partial V}{\partial I}(p, M(p, \overline{U}))\frac{\partial M}{\partial p_j}(p, \overline{U}) = 0.$$

Shepard's Lemma says

$$\frac{\partial M}{\partial p_j}(p, \overline{U}) = x_j^c(p, \overline{U}),$$

and the Duality Lemma says

$$x_j^c(p, \overline{U}) = x_j^*(p, M(p, \overline{U})).$$

Make these substitutions and rearrange to get **Roy's Identity**:

$$x_j^*(p, I) = -\frac{\frac{\partial V}{\partial p_j}(p, I)}{\frac{\partial V}{\partial I}(p, I)}.$$

**Remark 8.2.** It is possible to derive Roy's identity without going through duality. The envelope theorem lets us calculate the two derivatives of the indirect utility function:

$$\frac{\partial V}{\partial I}(p, I) = \lambda \quad \text{and} \quad \frac{\partial V}{\partial p_j}(p, I) = -\lambda x_j^*(p, I),$$

where $\lambda$ is the multiplier on the budget constraint. Eliminate $\lambda$ to get Roy's Identity.

Now for the real magic. By the duality lemma,

$$x_j^c(p, \overline{U}) = x_j^*(p, M(p, \overline{U})$$

for all $p$. Differentiate both sides to get

$$\frac{\partial x_j^c}{\partial p_j} = \frac{\partial x_j^*}{\partial p_j} + \frac{\partial x_j^*}{\partial I}\frac{\partial M}{\partial p_j}.$$

Substitute from the envelope theorem and rearrange to get

$$\frac{\partial x_j^*}{\partial p_j} = \frac{\partial x_j^c}{\partial p_j} - x_j \frac{\partial x_j^*}{\partial I}.$$

This is the **Slutsky equation.**

The price effect is decomposed in the **substitution effect** (price effect on compensated demand) and the **income effect**.

We don't have to limit attention to own-price effects. The exact same argument gives cross-price effects: We have, as before,

$$x_j^c(p, \overline{U}) = x_j^*(p, M(p, \overline{U})$$

Differentiate with respect to $p_k$ to get

$$\frac{\partial x_j^c}{\partial p_k} = \frac{\partial x_j^*}{\partial p_k} + \frac{\partial x_j^*}{\partial I}\frac{\partial M}{\partial p_k}.$$

Substitute from the envelope theorem and rearrange to get

$$\frac{\partial x_j^*}{\partial p_k} = \frac{\partial x_j^c}{\partial p_k} - x_k \frac{\partial x_j^*}{\partial I}.$$

Why is this progress? Because we can definitively sign substitution effects.

## 8.2 Comparative Statics of Compensated Demand

Let's start with a direct approach to showing that $x_j^c$ is decreasing in $p_j$.

Take two price vectors, $p$ and $p'$. Let $x$ and $x'$ be the associated compensated demands: $x = x^c(p, \overline{U})$ and $x' = x^c(p', \overline{U})$. Since $x$ is cost minimizing at $p$,

$$px' \geq px.$$

Similarly,

$$p'x \geq p'x'.$$

Sum these two inequalities to get

$$px' + p'x \geq px + p'x'.$$

Rearrange to get

$$0 \geq px - px' - p'x + p'x'.$$

Factor

$$0 \geq p(x - x') - p'(x - x'),$$

and again

$$0 \geq (p - p')(x - x').$$

If $p$ and $p'$ differ only in terms of the price of good $j$, this says

$$0 \geq (p_j - p'_j)(x_j - x'_j),$$

and the compensated demand for good $j$ moves in the opposite direction of the price change.

> **Remark 8.3.** This style of proof is sometimes called a *revealed preference proof.* These proofs are conceptually quite satisfying, since they appeal directly to the underlying assumption of optimization. This helps us "see" the decision maker's point of view, and it tends to avoid extraneous assumptions.

The following result gives slightly stronger information, and uses another very useful style of proof.

**Proposition 8.1.** *The expenditure function is a concave function of prices.*

**Proof.** Remember what concavity means: for any $p$, $p'$, and $\lambda \in [0, 1]$, we have

$$M(\lambda p + (1 - \lambda)p', \overline{U}) \geq \lambda M(p, \overline{U}) + (1 - \lambda)M(p', \overline{U}).$$

To conserve on writing, let $\hat{p} = \lambda p + (1 - \lambda)p'$. With this we can use the definition of the expenditure function to rewrite the concavity condition as

$$\hat{p}x^c(\hat{p}, \overline{U}) \geq \lambda(px^c(p, \overline{U})) + (1 - \lambda)(p'x^c(p', \overline{U})).$$

What do we know? Well, whatever is consumed when prices are $\hat{p}$ must give utility at least $\overline{U}$. That means that $x^c(\hat{p}, \overline{U})$ is feasible when prices are $p$, so

$$px^c(\hat{p}, \overline{U}) \geq px^c(p, \overline{U}) = M(p, \overline{U}),$$

or else $x^c(p,\overline{U})$ would not be cost minimizing.

Similarly,

$$p'x^c(\hat{p},\overline{U}) \geq p'x^c(p',\overline{U}) = M(p',\overline{U}).$$

Together, these inequalities imply

$$\lambda(px^c(\hat{p},\overline{U})) + (1-\lambda)(p'x^c(\hat{p},\overline{U})) \geq \lambda M(p,\overline{U}) + (1-\lambda)M(p',\overline{U}).$$

But the LHS is just

$$[\lambda p + (1-\lambda)p']x^c(\hat{p},\overline{U}) = M(\hat{p},\overline{U}).$$

Thus

$$M(\lambda p + (1-\lambda)p',\overline{U}) \geq \lambda M(p,\overline{U}) + (1-\lambda)M(p',\overline{U}),$$

as required. ■

To see what this says about substitution effects, recall that Shepard's Lemma says

$$\mathbf{D}M(p,\overline{U}) = x^c(p,\overline{U}).$$

But that means that the second derivative of the expenditure function:

$$D^2M(p,\overline{U}) = \left(\frac{\partial^2}{\partial p_j \partial p_k} M(p,\overline{U})\right)$$

is equal to the first derivative of the (vector-valued) compensated demand:

$$Dx^c(p,\overline{U}) = \left(\frac{\partial}{\partial p_j} x_k^c(p,\overline{U})\right).$$

$D^2M$ is symmetric (as a second derivative) and is negative semi-definite (as $M$ is concave).

Thus the derivative of the compensated demand is symmetric and negative semi-definite. This has several implications. First, a negative semi-definite matrix has non-positive diagonal terms. In other words, for all $j$, the own price effect is negative:

$$\frac{\partial x_j^c}{\partial p_j}(p,\overline{U}) \leq 0.$$

Negative semi-definiteness says a little about cross-price effects. For example, all matrices of the form

$$\begin{pmatrix} \frac{\partial x_j^c}{\partial p_j}(p,\overline{U}) & \frac{\partial x_j^c}{\partial p_k}(p,\overline{U}) \\ \frac{\partial x_k^c}{\partial p_j}(p,\overline{U}) & \frac{\partial x_k^c}{\partial p_k}(p,\overline{U}) \end{pmatrix}$$

have nonnegative determinant. This implies

$$\frac{\partial x_j^c}{\partial p_j}(p,\overline{U})\frac{\partial x_k^c}{\partial p_k}(p,\overline{U}) \geq \frac{\partial x_j^c}{\partial p_k}(p,\overline{U})\frac{\partial x_k^c}{\partial p_j}(p,\overline{U})$$

for all $j$ and $k \neq j$. This is a restriction on cross-price effects, but not a terribly useful one.

Symmetry is more interesting. It says that, for all $j$ and $k \neq j$, we have

$$\frac{\partial x_k^c}{\partial p_j}(p,\overline{U}) = \frac{\partial x_j^c}{\partial p_k}(p,\overline{U}).$$

So, e.g., the effect of the price of gasoline on compensated demand for chocolate is the same as the effect of the price of chocolate on compensated demand for gasoline. I doubt anyone ever would have come up with that thought without going through all of this math.

The main benefit of symmetry is that it allows us to unambiguously define complements and substitutes. Say that goods $j$ and $k$ are **complements** if $\frac{\partial x_j^c}{\partial p_k}(p,\overline{U}) < 0$, and say that goods $j$ and $k$ are **substitutes** if $\frac{\partial x_j^c}{\partial p_k}(p,\overline{U}) > 0$. When it is useful to avoid ambiguity, these are called *net* complements and *net* substitutes, with the modifier *gross* used to indicate a definition in terms of ordinary demand. But be warned—gross complementarity and gross substitutability are trickier concepts, since income effects make it possible that $\frac{\partial x_j^*}{\partial p_k}(p,\overline{U}) > 0$ but $\frac{\partial x_k^*}{\partial p_j}(p,\overline{U}) < 0$.

By the Slutsky equation, the **Slutsky substitution matrix**,

$$\left(\frac{\partial x_k^*}{\partial p_j} + x_j\frac{\partial x_k^*}{\partial I}\right)$$

is also symmetric and negative semi-definite.

This is the entire empirical content of the hypothesis of utility maximization in consumer choice. That is, any function that is homogeneous degree 0 in prices and has symmetric, negative semi-definite Slutsky substitution matrix is the demand function derived from maximizing some utility function. The formal statement of this result is called the **Integrability Theorem.**

## 8.3 Welfare Measures

Now we turn to the question of how demand can be used to help quantify a consumer's gains or losses from some policy change. Assume that the status quo has prices $p$ and consumer income $I$. A proposed policy will result in new prices $p'$. How much does this affect the consumer's welfare?

In intermediate micro, this question is answered with *consumer's surplus,* measured as the area under the demand curve. We are now in a position to understand this as an approximation to a well-founded theory based on compensated demand.

Let $v(p, I)$ be an indirect utility function at prices $p$ and income $I$. I say "an" because the precise choice depends the utility representation. We can eliminate the dependence of choice of a utility representation by defining concepts in terms of expenditure.

Let $\overline{p} \gg 0$ be an arbitrary price vector. The function

$$(p, I) \mapsto M(\overline{p}, v(p, I))$$

is another indirect utility function representing the same preferences. (Think about how I know this.) This indirect utility function is called a **money metric** indirect utility function. It represents the amount of income the consumer needs to reach utility $v(p, I)$ when prices are $\overline{p}$.

A money metric indirect utility function answers the question about the change from $p$ to $p'$ with $M(\overline{p}, v(p', I)) - M(\overline{p}, v(p, I))$.

Two choices are considered particularly natural for $\overline{p}$, namely, $\overline{p} = p$ and $\overline{p} = p'$. To simplify the expressions, write $u = v(p, I)$ and $u' = v(p', I)$. Then our two natural choices for $\overline{p}$ yield:

$$EV(p, p', I) = M(p, u') - M(p, u) = M(p, u') - I$$

and

$$CV(p, p', I) = M(p', u') - M(p', u) = I - M(p', u).$$

These are the **equivalent variation** and the **compensating variation**, respectively.

The equivalent variation is the amount of extra income that, at the original prices, has the same welfare effect as the policy change. The compensating variation is the amount of extra income that, at the new prices, returns welfare to the pre-change level. The compensating variation privileges the status quo. The equivalent variation privileges the new policy.

If a single commodity changes price, the equivalent variation is easy to write in terms of compensated demand. Assume that $p$ and $p'$ differ only in the price of good $j$. Then:

$$\begin{aligned} EV(p,p',I) &= M(p,u') - I \\ &= M(p,u') - M(p',u') \\ &= \int_{p'_j}^{p_j} x_j^c\left((\tilde{p}_j, p_{-j}), u'\right)\, d\tilde{p}_j, \end{aligned}$$

where the last equality uses the integral form of the envelope theorem.

A similar argument works for the compensating variation:

$$CV = \int_{p'_j}^{p_j} x_j^c\left((\tilde{p}_j, p_{-j}), u\right)\, d\tilde{p}_j.$$

Multiple price changes are no problem for the $EV$ or $CV$. For simplicity, consider a world with just two commodities. Then

$$\begin{aligned} EV(p,p',I) &= M(p,u') - M(p',u') \\ &= M((p_1,p_2),u') - M((p'_1,p_2),u') + M((p'_1,p_2),u') - M((p'_1,p'_2),u') \\ &= \int_{p'_1}^{p_1} x_2^c((\tilde{p}_1,p_2),u')\, d\tilde{p}_1 + \int_{p'_2}^{p_2} x_1^c((p'_1,\tilde{p}_2),u')\, d\tilde{p}_2. \end{aligned}$$

An analogous argument works for the $EV$.

Consumers' surplus as measured by ordinary demand is not so simple, as the answer can depend on the order in which you consider price changes.

Still, the area under the ordinary demand curve is an important part of our story—that is what is potentially observable. The Slutsky equation says

$$\frac{\partial x_j^c}{\partial p_j} = \frac{\partial x_j^*}{\partial p_j} + x_j \frac{\partial x_j^*}{\partial I}.$$

Thus is income effects are small, ordinary and compensated demands are close together, and the change in the area under the demand curve is close to both the equivalent and compensating variations. In the limiting case of quasi-linear utility, there are no income effects, and we have that the compensating and equivalent variations are equal, and they are both equal to the change in area under the demand curve. Much more generally, when price changes are small, the three measures will be close together. See Willig's "Consumer Surplus Without Appology" (*AER*, 1976).

**Example 8.1.** Assume that we start with prices $p$ and income $I$. A tax of $t$ per unit is then applied to good 1. Write $p' = p + te_1$ for the new price vector, and write $U'$ for the utility attained at prices $p'$.

The tax revenue collected will be $T \equiv tx_1^*(p', I) = tx_i^c(p', u')$. If we instead took $T$ as a lump-sum from the consumer, without changing prices, would the consumer be better or worse off?

We can answer by comparing the lump-sum to the equivalent variation. The consumer is better off with lump-sum taxation if $I - T > M(p, u')$. Thus it makes sense to measure the deadweight loss of taxation as $I - T - M(p, u') = -T - EV(p, p', I)$. We have

$$\begin{aligned}
-T - EV(p,p',I) &= M(p',u') - M(p,u') - T \\
&= \int_{p_1}^{p_1+t} x_1^c((\tilde{p}_1, p_{-1}), u')\, d\tilde{p}_1 - tx_i^c((p_1+t, p_{-1}), u') \\
&= \int_{p_1}^{p_1+t} x_1^c((\tilde{p}_1, p_{-1}), u')\, d\tilde{p}_1 - \int_{p_1}^{p_1+t} x_i^c((p_1+t, p_{-1}), u')\, d\tilde{p}_1 \\
&= \int_{p_1}^{p_1+t} \left( x_1^c((\tilde{p}_1, p_{-1}), u') - x_i^c((p_1+t, p_{-1}), u') \right) d\tilde{p}_1 \\
&\geq 0,
\end{aligned}$$

where the inequality is from the law of demand. (It is strict if compensated demand is strictly decreasing.)

## Problems

**Exercise 8.1.** Alice consumes two goods, $x_1$ and $x_2$. Her expenditure function is

$$M(p_1, p_2, \overline{u}) = 2\overline{u}\sqrt{p_1 \cdot p_2}.$$

What are her compensated demands?

**Exercise 8.2.** The $n$-good Cobb-Douglas utility function is

$$u(x) = A \prod_{j=1}^{n} x_j^{\alpha_j},$$

where $A > 1$ and $\sum_j \alpha_j = 1$.

1. Derive the demand function.

2. Derive the indirect utility function.

3. Compute the expenditure function.

4. Compute the compensated demand.

Note: If you recall the solution to part (a) from lecture a few weeks ago, you should be able to do this problem without solving any constrained optimization problem at all.

**Exercise 8.3.** A firm has technology $Y$. It chooses production plan $y \in Y$ to maximize profits given prices $p$.

Let $y^*(p)$ be the profit maximizing production plan, and let $\pi(p)$ be the *profit function* $p \cdot y^*(p)$. (Assume that $y^*$ is single-valued.)

1. Show that $\pi$ is a convex function of $p$.

2. Explain how to calculate $y^*$ given only knowledge of $\pi$.

3. Show that supply is upward-sloping: $\frac{\partial y_i^*}{\partial p_i}(p) \geq 0$.

**Exercise 8.4.** Stay with the setup of the previous problem, but consider the following timing: First prices are drawn from some distribution, and then the firm observes prices and chooses $y$. If Congress considers a reform that eliminates price uncertainty, fixing prices at their expected values for sure, will the firm support the reform?

**Exercise 8.5.** Consider the following discrete-choice problem. A consumer has an endowment of $I$ units of good 1 ("money"). She can consume either 0 or 1 units of good 2 (a car). Write her consumption bundle as $(m, c)$, where $m \in \mathbb{R}$ is money and $c \in \{0, 1\}$ is cars. Her utility is

$$U(m, c) = m + vc,$$

where $v > 0$ is her willingness to pay for a car.

1. Suppose the price of money is fixed at 1. What is the consumer's demand when the price of a car is $p$?

2. What is the consumer's indirect utility function?

3. Does Roy's identity hold for this consumer?

**Exercise 8.6.** Economists use the following model to think about labor supply. An individual values two goods, consumption ($c$) and leisure ($\ell$), according to the strictly monotone, strictly concave utility function $U(c, \ell)$. She has $\overline{L}$ units of time that can be divided between leisure and work for wage $w > 0$. Thus, her budget constraint is $c \leq w(\overline{L} - \ell) + I$, where $I$ is non-labor income.

Mimic, as much as you can, our development leading up to the Slutsky equation to study this problem. How can your analysis account for the empirical fact that (at least for "prime-aged" men) hours worked have remained roughly constant in the face of dramatic increases in real wages?

**Exercise 8.7.** Consider an individual who is concerned about monetary payoffs in the states of nature $s = 1, \ldots, S$ which may occur tomorrow. Denote the dollar payoff in state $s$ by $x_s$ and the probability of state $s$ by $p_s$. The individual chooses $\mathbf{x} = (x_1, \ldots, x_S)$ to maximize the discounted, expected value of monetary payoff, with discount factor $\delta > 0$. That is, the individual's utility from payoff vector $\mathbf{x}$ is $\delta(\mathbf{p} \cdot \mathbf{x})$.

The set of possible payoff vectors, denoted by $X$, is nonempty and compact.

1. Write down the individual's maximization problem for a fixed vector of probabilities $\mathbf{p} = (p_1, \ldots, p_S)$, and formally define the value function $v(\mathbf{p}, \delta)$.

2. Show that $v(\mathbf{p}, \delta)$ is homogeneous of degree 1 in $\delta$.

3. Show that $v(\mathbf{p}, \delta)$ is convex in $\mathbf{p}$.

Now suppose that the individual has the option of getting additional information before making her decision. Specifically, she knows that probabilities of the states are either $\mathbf{p}_0$ or $\mathbf{p}_1$, with each possibility equally likely. Thus she assesses the probabilities as $\mathbf{p} = \frac{1}{2}\mathbf{p}_0 + \frac{1}{2}\mathbf{p}_1$. There is an expert, who knows which of $\mathbf{p}_0$ and $\mathbf{p}_1$ are the correct probabilities. This expert is known to always tell the truth if asked to report on the probabilities.

1. Use part ($c$) to prove that the individual weakly prefers to decide after hearing the expert's report than to decide without the report.

2. What can you say about when the preference in part ($d$) will be strict?

# Chapter 9

# The Second Best

In Chapter 6, we looked at normative analysis in terms of choices of allocations. But policy-makers rarely have the power to choose allocations directly. Instead, a policy-maker will have access to a set of policy instruments, like taxes or regulations that forbid certain choices by private actors. These policy instruments lead indirectly to allocations through their impact on what private choices through feasible sets and incentives.

The question of optimal policy when you can choose allocations directly, limited only by the economies endowment of resources and technology, is called **first-best analysis**. The question of optimal policy when there are further limits on policies is called **second-best analysis**. The additional constraints of second-best analysis have several sources. Legal or institutional constraints may limit choices, as when a constitutional rule of equal treatment forces the policy-maker to tax distinct private actors at the same rate. A more fundamental source of second-best constraint comes from limited information. If private actors know things that the policy-maker does not, policies must be crafted to give private actors incentives to reveal the information.

Second-best analysis generates two important and general lessons:

1. With limited policy instruments, you generally do not want to satisfy optimality conditions derived from looking directly at allocations.

2. With limited policy instruments, you generally cannot separate efficiency and equity

## 9.1 First-Best: Price Regulation

Consider an economy with 3 goods: money, electricity, and gas heat. There is a single consumer, with endowment $(m, e, g) = (\overline{m}, 0, 0)$. This consumer's preferences are represented by the utility function:

$$u(m, e, g) = m + v(e) + w(g)$$

where $v$ and $w$ are increasing, strictly concave and continuously differentiable. To avoid corner solutions later on, assume that

$$\lim_{e \to 0} v'(e) = \lim_{g \to 0} w'(g) = \infty \quad \text{and} \quad \lim_{e \to \frac{\overline{m}}{2}} v'(e) = \lim_{g \to \frac{\overline{m}}{2}} w'(g) = 0.$$

(The second set of limits suggest that $\overline{m}$ is "very large".)

There is a technology that can transform m into e and g. It is represented by the set of netput vectors:

$$Y = \{z_m, z_e, z_g \mid z_m + F + c_1 z_e + c_2 z_g \leq 0 \text{ and } z_m \leq 0\}.$$

Since there is a single consumer, Pareto optimality just means maximizing her utility, given the constraints of the endowment and technology. So consider the program

$$\begin{aligned}
\max_{m,e,g} \; & m + v(e) + w(g) \\
\text{st} \; & -z_m - F - c_1 z_e - c_2 z_g \geq 0 \\
& \overline{m} - m + z_m \geq 0 \\
& -z_m \geq 0 \\
& z_e - e \geq 0 \\
& z_g - g \geq 0.
\end{aligned}$$

It is clear that no good should be wasted, so we can eliminate the last three constraints:

$$\begin{aligned}
\max_{m,e,g} \; & m + v(e) + w(g) \\
\text{st} \; & (\overline{m} - m) - F - c_1 e - c_2 g \geq 0.
\end{aligned}$$

At an interior solution, the Kuhn-Tucker conditions are

$$\begin{aligned}
1 - \lambda &= 0 \\
v'(e) - \lambda c_1 &= 0 \\
w'(g) - \lambda c_2 &= 0,
\end{aligned}$$

where $\lambda$ is the shadow price.

Solving these, the first-best allocation $(m^*, e^*, g^*)$ is characterized by

$$v'(e^*) = c_1 \quad w'(g^*) = c_2 \quad m^* = \overline{m} - F - c_1 e^* - c_2 g^*.$$

So far, this is just an abstract statement about allocations. To see how it related to policy, we need to think about institutions. Let's assume that the technology is operated by a regulated firm. The regulator has three policy instruments. She sets prices for each good produced by the firm, $p_1$ and $p_2$. In addition, she can choose a lump-sum transfer, $T$, of money from the consumer to the firm. The consumer then chooses how much of each produced good to buy. The firm's production must be financed by revenue from sales of the two produced goods and the transfer.

This set of policy instruments is very powerful. In particular, the regulator can induce any consumption allocation consistent with the social feasibility constraints.

As a preliminary step, let's consider the consumer's problem given the regulator's choice of prices and transfer:

$$\max_{m,e,g} m + v(e) + w(g)$$
$$\text{st } \overline{m} - m - T - p_1 e - p_2 g \geq 0.$$

The Kuhn-Tucker conditions simplify to

$$v'(e) = p_1 \quad \text{and} \quad w'(g) = p_2.$$

Now suppose the regulator wants to implement the feasible allocation $(\hat{m}, \hat{e}, \hat{g})$. Strict concavity of $v$ and $w$ imply that their derivatives are invertible, and by continuity of $v'$ and $w'$ and the Inada conditions, the intermediate value theorem implies that the equations $\hat{e} = (v')^{-1}(p_1)$ and $\hat{g} = (w')^{-1}(p_2)$ have unique solutions, $\hat{p}_1$ and $\hat{p}_2$.

At these prices, the firm will have revenue $\hat{p}_1\hat{e} + \hat{p}_2\hat{g}$. It's input requirement is $F + c_1\hat{e} + c_2\hat{g}$. Thus the firm has a deficit of

$$\Delta = F + (c_1 - \hat{p}_1)\,\hat{e} + (c_2 - \hat{p}_2)\,\hat{g}.$$

Consider the lump-sum transfer that exactly offsets the deficit: $\tilde{T} = \Delta$. By construction, the firm can meet its production with this transfer. All that's left to check is that this transfer leaves the consumer with enough money. She has:

$$\begin{aligned}\overline{m} - \Delta - \hat{p}_1\hat{e} - \hat{p}_2\hat{g} &= \overline{m} - F - (c_1 - \hat{p}_1)\,\hat{e} - (c_2 - \hat{p}_2)\,\hat{g} - \hat{p}_1\hat{e} - \hat{p}_2\hat{g} \\ &= \overline{m} - F - c_1\hat{e} - c_2\hat{g}.\end{aligned}$$

But feasibility of the allocation implies

$$\hat{m} \leq \overline{m} - F - c_1\hat{e} - c_2\hat{g}.$$

So the only thing that can go wrong is that we have left too much! To fix this, augment the transfer by the amount of waste in the target allocation:

$$\hat{T} = \Delta + (\overline{m} - \hat{m} - F - c_1\hat{e} - c_2\hat{g})\,.$$

The policy $(\hat{p}_1, \hat{p}_2, \hat{T})$ implements the allocation $(\hat{m}, \hat{e}, \hat{g})$.

The upshot of this discussion is that, when it comes to implementing allocations, the regulator can do anything consistent with the constraints of technology and resources. If the regulator is benevolent, seeking only to maximize the consumer's welfare, this makes his problem easy: The best allocation is $(m^*, e^*, g^*)$, and the analysis we've done shows that that allocation results from the policy $(T^*, p_1^*, p_2^*) = (F, c_1, c_2)$. So, just as you would expect from intermediate micro, the optimal policy is *marginal cost pricing.*

## 9.2 The Second-Best: Ramsey Pricing

The conclusion that marginal-cost pricing is optimal is very sensitive to the assumption that the regulator can use lump-sum transfers to cover the fixed cost of production. Another, more realistic, institutional assumption is that the regulator can set prices for electricity and gas, and the firm's entire production must be financed out of revenue from the sale of those two goods. With this institution, marginal cost pricing leads to a deficit, with no way of making it up.

This doesn't mean we can't give any guidance on what to do. The regulator can still use the procedure of choosing his policy instruments to maximize the consumer's welfare, given all of the constraints. One constraint will be that the firm must be able to produce the desired amount without any subsidy:

$$p_1 e + p_2 g - F - c_1 e - c_2 g \geq 0.$$

Another pair of constraints come from the fact that the consumer will decide how much of each good to buy to maximize her utility:

$$v'(e) = p_1 \quad \text{and} \quad w'(g) = p_2.$$

The problem set will walk you through a particular example of this procedure. The rest of this section will look at two simpler problems, to illustrate the two main generalizable implications of the theory of the second-best.

First, let's collect a few facts about quasilinear utility. Suppose the consumer maximizes $u(m, e, g) = m + v(e, g)$. From the Inada conditions and the analysis in Example 4.2, we know that demands for $e$ and $g$, denoted $e^*$ and $g^*$, are independent of $\overline{m}$. Thus we can write the indirect utility function as

$$\begin{aligned} V(p_1, p_2, \overline{m}) &= \overline{m} - p_1 e^*(p_1, p_2) - p_2 g^*(p_1, p_2) + v(e^*(p_1, p_2), g^*(p_1, p_2)) \\ &\equiv \overline{m} + w(p_1, p_2) \end{aligned}$$

for some function $w$.

A similar argument gives an additively separable indirect utility when the $v$ is additively separable.

But then the derivative of $V$ with respect to income is 1, and Roy's identity tells us that

$$e^*(p_1, p_2) = -\frac{\partial}{\partial p_1} w(p_1, p_2) \quad \text{and } g^*(p_1, p_2) = -\frac{\partial}{\partial p_2} w(p_1, p_2).$$

Now we can set up the Ramsey problem. For simplicity, assume additive separability. The regulator chooses $p_1$ and $p_2$ to solve

$$\begin{aligned} &\max_{p_1, p_2} w(p_1, p_2) \\ &\quad \text{st } (p_1 - c_1) e^*(p_1) + (p_2 - c_2) g^*(p_2) - F \geq 0. \end{aligned}$$

The FOCs for an interior solution are:

$$\begin{aligned} &\frac{\partial}{\partial p_1} w(p_1, p_2) + \lambda \left( e^*(p_1) + (p_1 - c_1)(e^*)'(p_1) \right) = 0 \\ &\frac{\partial}{\partial p_2} w(p_1, p_2) + \lambda \left( g^*(p_2) + (p_2 - c_2)(g^*)'(p_2) \right) = 0. \end{aligned}$$

Use Roy's identity and rearrange to get:

$$\lambda\left((p_1 - c_1)(e^*)'(p_1)\right) = (1-\lambda)e^*(p_1)$$
$$\lambda\left((p_2 - c_2)(g^*)'(p_2)\right) = (1-\lambda)g^*(p_2).$$

Now define the elasticity of demand for good $i$ as

$$\epsilon_i(p_i) = (x_i^*)'(p_i) \cdot \frac{p_i}{x_i^*(p_i)}.$$

Using this, we can reexpress the FOCs in terms of the tax rates on good $i$, $\tau_i = \frac{p_i - c_i}{p_i}$, as follows:

$$\tau_i = \frac{1-\lambda}{\lambda} \cdot \frac{1}{\epsilon_i(p_i)}$$

This is the *inverse elasticity rule.*

## 9.3 Two More Applications

### 9.3.1 An Ineliminable Distortion

Consider a variant of the model we used to study marginal cost pricing. There are two differences. First, there is no fixed cost of production and both marginal costs are 1, so

$$Y = \{z_m + z_e + z_g \leq 0 \text{ and } z_m \leq 0\}.$$

Second, the utility function is not separable between $e$ and $g$:

$$u(m, e, g) = m + v(e, g).$$

The analysis of the marginal-cost pricing example can be easily modified to show that the first-best can be implemented with prices $p_1 = p_2 = 1$, with no subsidy to the firm.

Assume now that the $p_1$ is fixed at $\overline{p} > 1$, and that the policy-maker cannot affect it. All she can do is choose $p_2$.

Now the productive sector will make profits, denoted $\Pi$. Assume that these profits are returned to the consumer. But also assume that the consumer ignores the effect of her own decisions on these profits. (Think of this as a "competitive" assumption, one that would be justified in a model with many identical consumers.)

The consumer will solve:

$$\max_{m,e,g} m + \Pi + v(e, g)$$
$$\text{st } \overline{m} - m - \overline{p}e - p_2 g \geq 0.$$

Write the demands as $e^*(\overline{p}, p_2)$ and $g^*(\overline{p}, p_2)$.

The second-best problem is:

$$\max_{p_2} w(\overline{p}, p_2) + (\overline{p} - 1)e^*(\overline{p}, p_2) + (p_2 - 1)g^*(\overline{p}, p_2)$$

Notice that here, we do not ignore the effect of prices on profit.

The FOC is

$$\frac{\partial}{\partial p_2} w(\overline{p}, p_2) + (\overline{p} - 1)\frac{\partial}{\partial p_2} e^* + g^* + (p_2 - 1)\frac{\partial}{\partial p_2} g^* = 0.$$

By Roy's identity, this simplifies to

$$(\overline{p} - 1)\frac{\partial}{\partial p_2} e^* + (p_2 - 1)\frac{\partial}{\partial p_2} g^* = 0$$

If $e^*$ is independent of $p_2$, then $p_2 = 1$ is a solution.

But the case without additive separability is more interesting. In this case, $p_2 = 1$ would imply

$$((\overline{p} - 1)\frac{\partial}{\partial p_2} e^*(\overline{p}, p_2) = 0.$$

### 9.3.2 Equity and Efficiency

I mentioned that the separation of efficiency and distribution promised by the second welfare theorem did not hold in general. Here I illustrate this in a simple economy.

Consider a society with two people, Alice and Bob. Both value consumption of good 1, but the social endowment contains none of that good. Instead, Alice can produce $e$ units of the good at private cost $\frac{1}{2}e^2$. Bob, who is disabled, cannot produce any of the consumption good.

An allocation in this economy is a triple $(x^A, x^B, e)$, where $x^h$ is person $h$'s consumption of good 1 and $e$ is Alice's production. An allocation is feasible if

$$(x^A, x^B, e) \geq 0 \quad \text{and} \quad x^A + x^B \leq e.$$

Each person has quasilinear utility:

$$u^A(x^A, x^B, e) = x^A - \frac{1}{2}e^2 \quad \text{and} \quad u^B(x^A, x^B, e) = x^B.$$

The set of Pareto optimal allocations is given by the solutions to:

$$\begin{aligned}
\max_{x^A, x^B, e} \ & \alpha^A \left( x^A - \frac{1}{2}e^2 \right) + \alpha^B x^B \\
\text{st } \ & e - x^A - x^B \geq 0 \\
& (x^A, x^B, e) \geq 0
\end{aligned}$$

The first-order conditions are:

$$\begin{aligned}
\alpha^A - \lambda &\leq 0 \\
\alpha^B - \lambda &\leq 0 \\
-\alpha^A e + \lambda &\leq 0,
\end{aligned}$$

all with complementary slackness. If Alice has positive consumption, the first and third FOCs hold with equality, implying $e^* = 1$. Call this the *first-best level of effort.*

Assume that this society restricts attention to a simple family of tax-transfer policies: Alice pays fraction $\tau$ of whatever she produces to Bob, and keeps the rest for herself. Given this, Alice will solve

$$\max_e (1 - \tau)e - \frac{1}{2}e^2,$$

which is solved at $\hat{e}(\tau) = 1 - \tau$.

Given a policy $\tau$, the associated utility imputation is

$$u^A(\tau) = \frac{1}{2}(1 - \tau)^2 \quad \text{and} \quad u^B(\tau) = \tau(1 - \tau).$$

There is only one $\tau$ that leads to the first-best level of effort, namely $\tau = 0$. But that certainly does not mean any $\tau > 0$ is Pareto dominated by $\tau = 0$. Bob is better off at, say, $\tau = 1/3$, where his utility is $2/9$, instead of the 0 he gets at $\tau = 0$.

That's not to say Pareto considerations tell us nothing. Alice's utility is decreasing in $\tau$. Bob's is strictly concave in $\tau$, with maximum at $\tau = 1/2$. Thus, if $\tau > 1/2$, both people can be made better off by a move to $\tau = 1/2$.

Finally, we can return to the question of whether or not focusing on allocations is appropriate. When I described the example, I said Bob was "disabled", but I didn't give any details. Here are two different stories, each consistent with all of the math I just did.

1. Bob used to be a hard worker, but a machine crushed his legs. He is no longer capable of doing any productive work, although he is still able to enjoy consumption.

2. Bob is actually perfectly healthy, and is capable of doing even more than Alice. But he is lazy, and will not work no matter how much reward he is offered.

The standard approach assumes you are have the same policy preferences, whichever story is true. Do you?

# Chapter 10

# Monotone Comparative Statics

## 10.1 Comparative Statics of the Firm: The Traditional Approach

Consider a single-output firm with cost function $c$. Assume that the firm is a price taker on outputs, and that the set of feasible outputs is $Q \subset \mathbb{R}_+$. Then the profit maximization problem is

$$\max_{q \in Q} pq - c(q).$$

Call the solution $q^*(p)$.

We'd like conditions under which we can prove that $q^*$ is (weakly) increasing. A traditional set is:

1. $Q = \mathbb{R}_+$.

2. $c$ is twice continuously differentiable with $c' > 0$ and $c'' > 0$.

3. $c'(0) = 0$ and $\lim_{q \to \infty} c'(q) = \infty$.

With these assumptions, we can argue as follows.

Since the objective function is strictly concave (since $c'' > 0$) and the feasible set is convex (since $Q = \mathbb{R}_+$). Thus, by Kuhn-Tucker, a necessary and sufficient condition for optimization is the FOC:

$$p - c'(q) \leq 0 \quad \text{with equality if } q > 0.$$

For a positive price $p$, we have $p > c'(0) = 0$, so there cannot be a corner solution, and we need a solution to the equation

$$p = c'(q).$$

By the limit condition, there is a $\overline{q}$ such that $p < c'(\overline{q})$. But then continuity of $c'$ and the intermediate value theorem imply there is a $q^*$ with $p = c'(q^*)$. And $c'$ strictly increasing implies that there is only one such $q^*$.

Thus, for all $p > 0$, the profit-maximizing quantity $q^*(p)$ solves

$$p - c'(q^*(p)) = 0.$$

Since $c'(q^*(p)) \neq 0$, the implicit function theorem implies that $q^*$ is differentiable with derivative

$$\frac{d}{dp}q^*(p) = \frac{1}{c''(q^*(p))}.$$

Since $c'' > 0$, we can (finally!) conclude that $q^*$ is increasing in $p$: supply curves slope up.

In practice, the argument would be given much more briefly. I was pedantic here to make it clear exactly what each assumption does. Sometimes, the assumptions are doing things like ensuring a solution exists. But most of the work being done by the assumptions is to ensure that the FOC characterizes the optimum. And the only reason we're doing that is so we can use the implicit function theorem.

There are two problems with this.

1. Our intuition suggests a much simpler argument: a higher price increases the marginal benefit of output, without affecting marginal cost. Thus the firm should produce more. This argument has nothing to do with the first-order condition being necessary and sufficient for maximization. Is there a mathematical tool that better reflects the economics?

2. The convexity assumptions are not at all innocuous (even thought they are "standard".) We saw before a result that ensures convex costs, but it assumed convex production technology (ruling out increasing returns over any range of inputs) and price-taking in input markets. Is the result that supply is upward sloping really contingent on such assumptions?

It turns out that none of assumptions 1–3 are needed to show that supply is nondecreasing, and only a small subset of them are needed to show strict monotonicity. The next section develops the needed mathematics.

## 10.2 The Main Theorems

Let $f : \mathbb{R}^2 \to \mathbb{R}$ and $X \subset \mathbb{R}$. The function $f$ has **increasing differences** in $x$ and $\theta$ if, for all $\overline{x} > \underline{x}$ and $\overline{\theta} > \underline{\theta}$, we have

$$f(\overline{x}, \overline{\theta}) - f(\underline{x}, \overline{\theta}) \geq f(\overline{x}, \underline{\theta}) - f(\underline{x}, \underline{\theta}).$$

If the inequality is strict, $f$ has **strictly increasing differences** in $x$ and $\theta$. The interpretation is that the incremental benefit of increasing $x$ from $\underline{x}$ to $\overline{x}$ is increasing in $\theta$.

Several things to notice:

- If $f(x, \theta)$ has increasing differences, so does $g(x, \theta) = f(x, \theta) + h_1(x) + h_2(\theta)$ any choices of $h_1$ and $h_2$. This gives the results to follow robustness against mispecification that does not involve interaction between $x$ and $\theta$.

- If $f$ is continuously differentiable in $x$, the fundamental theorem of calculus implies that $f$ has increasing differences if and only if $\frac{\partial}{\partial x} f(x, \theta)$ is nondecreasing in $\theta$.

  If $f$ is twice continuously differentiable, then $f$ has increasing differences if and only if $\frac{\partial^2}{\partial \theta \partial x} f(x, \theta) \geq 0$.

  Either of these derivative conditions imply strict increasing differences if the inequalities are strict except perhaps at isolated values of $(x, \theta)$.

It will be useful to look at an example where increasing differences can appealed to directly.

**Example 10.1** (Becker's Theory of Marriage)**.** A group of $n$ men and $n$ women are to be paired off into $N$ couples. Each person has "productivity": $m_i$ for man $i$ and $w_j$ for woman $j$. Order the groups so that

$$m_1 < m_2 < \cdots < m_n \quad \text{and} \quad w_1 < w_2 < \cdots < w_n.$$

A pair $(m_i, w_j)$ produce surplus $f(m_i, w_i)$ when matched together.

Using a game-theoretic solution concept called the core, Becker proved that the equilibrium matching must maximize the sum $\sum f(m_i, w_j)$ over all possible pairings. (After you study chapter 8 of Osborne's game theory textbook, this will be an easy exercise.) Becker was interested in when this implied *assortative matching*—the pairs being $(m_1, w_1)$, $(m_2, w_2)$, and so on.

The key condition turns out to be this: $f$ has strictly increasing differences. To see this, consider some matching that is not assortative. Then there are pairs $(m_i, w_j)$ and $(m_k, w_\ell)$ with $m_i > m_k$ and $w_\ell > w_j$. The contribution of these two pairs to total surplus is

$$f(m_i, w_j) + f(m_k, w_\ell).$$

But strict increasing differences implies that this is less than

$$f(m_i, w_\ell) + f(m_k, w_j).$$

Thus swapping partners increases total surplus, and the original, non-assortative matching did not maximize surplus.

This argument does not require differentiability of $f$ or monotonicity of $f$ in either argument. (Becker's original presentation assumed both.) Only complementarity, in the form of strict increasing differences, is needed.

Now, back to the main plot. We are interested in the maximization problem

$$\max_{x \in X} f(x, \theta).$$

The set of maximizers is $X^*(\theta)$, and a particular maximizer is $x^*(\theta) \in X^*(\theta)$.

When is $x^*$ nondecreasing? (And what exactly does monotonicity mean when $X^*$ is not a singleton?) What is needed to strengthen the conclusion to strict monotonicity?

**Theorem 10.1** (Topkis)**.** *Assume that $f$ has strictly increasing differences. Fix parameters $\overline{\theta} > \underline{\theta}$. If $\overline{x} \in X^*(\overline{\theta})$ and $\underline{x} \in X^*(\underline{\theta})$, then $\overline{x} \geq \underline{x}$.*

**Proof.** By optimality, we have

$$f(\overline{x}, \overline{\theta}) \geq f(\underline{x}, \overline{\theta}) \quad \text{and} \quad f(\underline{x}, \underline{\theta}) \geq f(\overline{x}, \underline{\theta}).$$

Sum these inequalities and rearrange to get

$$f(\overline{x}, \overline{\theta}) - f(\underline{x}, \overline{\theta}) - f(\overline{x}, \underline{\theta}) + f(\underline{x}, \underline{\theta}) \geq 0.$$

If $\underline{x} > \overline{x}$, then strict increasing differences implies

$$f(\overline{x},\overline{\theta}) - f(\underline{x},\overline{\theta}) - f(\overline{x},\underline{\theta}) + f(\underline{x},\underline{\theta}) < 0,$$

a contradiction. Thus $\overline{x} \geq \underline{x}$. ■

This proof starts just like our revealed preference proofs of comparative statics for compensated demand and supply—the inequalities defining optimization. But here the nest step is an appeal to increasing marginal benefits, rather than to the linear structure of the problem facing a price-taking DM.

Topkis's theorem only gives weak monotonicity. That's really all that we can ask for, given that we allow for finite feasible sets.

**Theorem 10.2** (Edlin-Shannon)**.** *Assume that $f$ is continuously differentiable in $x$ with $\frac{\partial}{\partial x} f(x,\theta)$ strictly increasing in $\theta$. Fix parameters $\overline{\theta} > \underline{\theta}$. If $\overline{x} \in X^*(\overline{\theta})$, $\underline{x} \in X^*(\underline{\theta})$, and at least one of $\overline{x}$ or $\underline{x}$ is in the interior of $X$, then $\overline{x} > \underline{x}$.*

**Proof.** The strictly increasing derivative implies strict increasing differences, so Topkis's theorem implies $\overline{x} \geq \underline{x}$. So we just need to rule out $\overline{x} = \underline{x}$.

If $\underline{x} \in \operatorname{int} X$, then the FOC must hold:

$$\frac{\partial}{\partial x} f(\underline{x},\underline{\theta}) = 0.$$

By increasing partial derivatives,

$$\frac{\partial}{\partial x} f(\underline{x},\overline{\theta}) > 0,$$

and $\underline{x}$ is not optimal for $\overline{\theta}$.

A similar argument works for $\overline{x} \in \operatorname{int} X$. Thus $\overline{x} \neq \underline{x}$. ■

**Example 10.2.** Let us return to the firm choosing a quantity to supply. Let $Q$ be the set of possible outputs, and let $c$ be the cost function. The firm's profit at price $p$ is

$$\pi(q,p) = pq - c(q).$$

The cross-partial derivative is 1, so this objective function has strictly increasing differences. Thus Topkis tells us that supply is weakly increasing. This is true without any assumptions about $Q$ or $c$. (We do need solutions to exist, but nothing more).

Furthermore, if $Q$ is an interval, then supply is strictly increasing on any neighborhood of an interior supply.

I don't know about you, but that strikes me as a huge improvement over the traditional analysis.

We are now going to focus on the profit maximization problem of a single-output firm. Along the way, we will see many tricks that help make Topkis's theorem really useful.

So consider a firm that chooses inputs $(x_1, \ldots, x_n)$ to solve

$$\max_{x \in X} p \cdot f(x) - w \cdot x.$$

When we want a more compact notation, we write $\pi(x; p, w) = p \cdot f(x) - w \cdot x$.

Here $X$ is the set of possible input combinations. We assume that it is a sublattice—the firm's purchases of one input are unconstrained by it's other purchases. But we do not assume $X$ is convex—some inputs might be available only in discrete amounts. This is an important concern for the all-or-nothing decision to buy a large piece of machinery, or if union contracts limit flexibility in hiring workers with unusual numbers of working hours. Without convexity of $X$, we are not going to be able to use the FOCs to get comparative statics.

We are also not going to assume that $X$ is compact—the most classical application of this model assumes that $X = \mathbb{R}^n_+$, and we want to cover that as well.

Even though we cannot use FOCs, we are still going to avail ourselves of calculus. So assume that $f$ is twice-continuously differentiable. But we do not make any assumptions about concavity. So we allow for fixed cost and increasing returns to scale.

We also want to avoid the hassle of potential non-existence of solutions. So assume that for all prices $(p, w) \gg 0$, the set of solutions $x^*(p, w)$ is nonempty.

**Remark 10.1.** We can give primitive conditions that imply this. Assume that, for all $x_{-i}$, we have

$$\lim_{x_i \to \infty} \frac{\partial}{\partial x_i} f(x_i, x_{-i}) = 0.$$

Then, since $w_i > 0$, profits diverge to minus infinity as $x_i$ increase without bound. This implies we can restrict attention to input combinations in the intersection of $X$ and some compact rectangle. Since our differentiability assumptions imply continuity, existence follows by Wierstrass.

What can we say about such problems with multiple decision variables? For this, we need to make more assumptions. There are two approaches:

1. Assume that the objective function is separable into a part where a single choice variable $x_i$ interacts with some parameter, and another part where there are no interactions with the parameter.

2. Assume that there are complementarities between all of the choice variables.

### 10.2.1 The Method of Aggregation

Let's kick things off with a discussion of own-price effects. How does the optimal choice of input $i$ vary with the factor price $w_i$?

Fix input $i$ at the level $x_i$, and consider the problem

$$\max_{x_1,\ldots,x_{i-1},x_{i+1},\ldots x_n} p \cdot f(x_1,\ldots,x_n) - \sum_{k\neq i} w_k x_k.$$

This is the problem of maximizing profit with $x_i$ fixed. Write $\pi(x_i)$ for the value function of this problem. Then we can write maximized profit as a function of $x_1$ as

$$-w_i x_i + \pi(x_1).$$

Differentiate to get

$$\frac{\partial^2}{\partial w_i \partial x_i}\pi(x;p,w) = -1.$$

Our general results on comparative statics refer to positive cross-partials. But this negative cross-partial is no problem. Just reexpress the problem as one with choice $x_i$ and parameter $-w_i$. Then we get strict increasing differences, and conclude that $x_i^*$ is weakly increasing in $-w_i$. But that is the same as: $x_i^*$ is weakly decreasing in $w_i$ itself.

**Remark 10.2.** This approach, artfully choosing minus signs, is sometimes glorified with the name *the method of reordering.*

Consider an abstract optimization problem

$$\begin{aligned} \max_{z,y}\ & g(z,y,\theta) \\ \text{st}\ & z\in\mathbb{R}, y\in\mathbb{R}^k \\ & (z,y)\in W. \end{aligned}$$

The optimal set of $z$s is also the solution to

$$\max_z h(z,\theta),$$

where $h$ is the maximized value of the program

$$\begin{aligned} \max_{y}\ & g(z,y,\theta) \\ \text{st}\ & y\in R^k \\ & (z,y)\in W. \end{aligned}$$

This (obvious!) equivalence is sometimes called the *method of aggregation.*

The important thing for us is that sometimes $h$ has increasing differences even though the original problem might not satisfy all of the conditions for comparative statics directly. This happens when, as in our example, the problem has natural additive separabilities.

Notice that this approach gives us no information about choices other than $x_i$.

### 10.2.2 Supermodularity

Let $X = X_1 \times \cdots X_m$, where each $X_i \subset \mathbb{R}$. Also let $\Theta = \Theta_1 \times \Theta_n$, where each $\Theta_j \subset \mathbb{R}$. We will consider the problem, for each $\theta \in \Theta$,

$$\max_{x\in X} f(x,\theta).$$

We want to know when the solution sets $X^*$ are (weakly) increasing in $\theta$.

First we should remind ourselves exactly what increasing means in this multi-dimensional context. We are looking for a result of the form "if ???, then $\overline{\theta} > \underline{\theta}$ implies that, for any selection $\overline{x} \in X^*(\overline{\theta})$ and $\underline{x} \in X^*(\underline{\theta}$ satisfy $\overline{x} \geq \underline{x}$". By our convention on vector inequalities, this means that increasing any nonempty subset of the parameters, while decreasing none, leads to a new optimum in which no decision variable is lower.

What do we need to fill in for "???"?

Based on our results for unidimensional problems, it should be no surprise that we need strictly increasing differences for all pairs $(x_i, \theta_j)$. But this will not be enough. It ensures that an increase in some $\theta$ makes the DM wants to increase $x_1$, holding the other $x$'s fixed. And it implies that the DM wants to increase $x_2$, holding the other $x$'s fixed. But she might not want to increase both $x_1$ and $x_2$: increasing $x_1$ might well itself lower the marginal benefit to $x_2$.

We can avoid this problem if $f$ also has increasing differences in $(x_i, x_k)$ for all $i$ and $k \neq i$. (We do not need strictly increasing differences here—a weak inequality is enough to rule out the problem pointed out in the previous paragraph.)

We need one more assumption, one that we have already build into our statement of the problem. We have assumed that $X$ is "rectangular" in the sense that which values of $x_i$ are feasible is independent of the values we assign to the other $x$'s. This rules out, e.g., the budget sets from consumer theory. It should be clear that such sets would cause a problem: starting from the budget line, we cannot increase all choices at once.

Here is the theorem as stated by Van Zandt:

**Theorem 10.3** (Topkis)**.** *Assume that*

1. *$f$ has strictly increasing differences in $(x_i, \theta_j)$ for all $i$ and $j$, and*

2. *$f$ has increasing differences in $(x_i, x_k)$ for all $i$ and $k \neq i$.*

*If $\overline{\theta} > \underline{\theta}$, $x \in X^*(\overline{\theta})$, and $x' \in X^*(\underline{\theta})$, then $x \geq x'$.*

The proof is optional. Along the way, it presents some additional definitions that you will encounter again if you are the kind of person who reads optional proofs. In addition, looking closely at the proof shows that the assumptions are slightly stronger than needed—all we really need is that $f$ has strict increasing differences in $(x_i, \theta_j)$ for some $x_i$ and some $\theta_j$ that changes between $\theta$ and $\theta'$; weak increasing differences suffice for the other pairs.

Now let's go back to the problem of the profit-maximizing firm.

First consider the case of all inputs complements:

$$\frac{\partial^2}{\partial x_i \partial x_j} f(x) \geq 0.$$

Then Topkis's multivariate theorem implies that solutions $x$ are non-decreasing in the (re-ordered) price vector $(p, -w)$. Thus

- Input $i$ is weakly decreasing in the price of input $j$

- All inputs are weakly increasing in the output price.
- Output is decreasing in every input price.

Next consider the case of two inputs, that are substitutes:

$$\frac{\partial^2}{\partial x_1 \partial x_2} f(x) \leq 0.$$

Here we have to make a less trivial application of reordering. Define the new variables $z_1 = x_1$ and $z_2 = -x_2$. Also define

$$\hat{f}(z_1, z_2) = f(z_1, -z_2)(= f(x_1, x_2)).$$

Then the function

$$\hat{\pi}(z_1, z_2; p, w_1, w_2) = p\hat{f}(z_1, z_2) - w_1 z_1 + w_2 z_2$$

is identical to the actual profits $\pi(x_1, x_2; p, w_1, w_2)$.

$\hat{\pi}$ has monotone comparative statics in $(p, -w_1, w_2)$, and is supermodular in $(z_1, z_2)$. Thus $z_1$ is increasing in $w_2$ and $z_2$ is increasing in $-w_1$. Translating, $x_1$ is increasing in $w_2$. Similarly, $-x_2$ is increasing in $-w_1$, or more simply, $x_2$ is increasing in $w_1$.

So with two, substitute inputs, cross-price effects are positive.

With more than two goods, we can't say much about the substitute case. Occasionally aggregation can help. (Indeed, you often see two-input models written as aggregates of capital and labor.)

### 10.2.3 Proof of Theorem 10.3

The strategy of the proof is just like that used in the one-dimensional version of Topkis's theorem: Assume that solutions are not ordered as in the conclusion of the theorem. Then write down the inequalities that define optimization, and use the increasing differences conditions to derive a contradiction.

Before we do that, it will be useful to introduce a few more concepts. Let $x$ and $x'$ be two vectors in $\mathbb{R}^n$. The **join** of $x$ and $x'$ is the vector $x \vee x'$ with components

$$(x \vee x')_i = \max(x_i, x'_i).$$

The **meet** of $x$ and $x'$ is the vector $x \wedge x'$ with components

$$(x \wedge x')_i = \min(x_i, x'_i).$$

A subset $D$ of $\mathbb{R}^n$ is a **sublattice** if $x, x' \in D$ imply $x \vee x' \in D$ and $x \wedge x' \in D$. If $D$ is a sublattice and $h : D \to \mathbb{R}$ satisfies

$$h(x \vee x') - h(x) \geq h(x') - h(x \wedge x'),$$

then $h$ is **supermodular**. If the inequality is strict, then $h$ is **strictly supermodular**.

Notice that our feasible set, $X$, is a sublattice.

**Lemma 10.1.** *Let $D = D_1 \times \cdots \times D_n$ with $D_i \subset \mathbb{R}$ for all $i$, and let $\mathcal{I} \subset \{1, \ldots, n\} \setminus \{i\}$. Assume $h$ has (strict) increasing differences in $(x_i, x_j)$ for all $j \in \mathcal{I}$. If $x^1$, $x^2$, $x^3$, and $x^4$ satisfy*

- $x^1_i = x^3_i > x^2_i = x^4_i$,
- $x^1_j = x^2_j > x^3_j = x^4_j$ *for all* $j \in \mathcal{I}$, *and*
- $x^1_k = x^2_k = x^3_k = x^4_k$ *for all* $k \notin \mathcal{I} \cup \{i\}$,

*then*

$$f(x^1) - f(x^2) \geq (>) f(x^3) - f(x^4).$$

**Proof.** Increase the components in $\mathcal{I}$ one by one. At each step, increasing differences in $x_i$ and the component being increased implies the difference does not decrease, and strict increasing differences implies it increases. ■

This has two important consequences for our standard optimization problem. First, increasing differences between each $x_i$ and $\theta_j$ implies that we can extend increasing differences beyond pairs.

**Lemma 10.2.** *Assume $f$ has (strict) increasing differences in $(x_i, \theta_j)$ for all $i$ and $j$. For any $x \geq x'$ and $\theta \geq \theta'$,*

$$f(x, \theta) - f(x', \theta) \geq (>) f(x, \theta') - f(x', \theta').$$

**Proof.** Write the difference as a telescoping sum:

$$f(x, \theta) - f(x', \theta) = \sum_i f(x_1, \ldots, x_{i-1}, x_i, x'_{i+1}, x_n, \theta) - f(x_1, \ldots, x_{i-1}, x'_i, x'_{i+1}, x_n, \theta).$$

Each term in the sum is increasing in $\theta$, by Lemma 10.1. ■

Second, pairwise increasing differences in all of the $x$'s implies supermodularity in $x$.

**Lemma 10.3.** *If $f : X \times \Theta \to \mathbb{R}$ has increasing differences in $(x_i, x_k)$ for all $i$ and $k \neq i$, then $f(\cdot, \theta)$ is supermodular in $x$ for all $\theta$.*

(The converse is also true, although we will not use that result here.)

**Proof.** Write $w = x \vee x'$ and $z = x \wedge x'$. We have

$$\begin{aligned} f(w,\theta) - f(x,\theta) &= \sum_i f(w_1,\ldots,w_{i-1},w_i,x_{i+1},\ldots,x_m,\theta) - f(w_1,\ldots,w_{i-1},x_i,x_{i+1},\ldots,x_m,\theta) \\ &\geq \sum_i f(x'_1,\ldots,x'_{i-1},x'_i,z_{i+1},\ldots,z_m,\theta) - f(x'_1,\ldots,x'_{i-1},z_i,z_{i+1},\ldots,z_m,\theta) \\ &= f(x',\theta) - f(z,\theta). \end{aligned}$$

The two inequalities follow because the sums are telescoping. What about the inequality? For some $i$, we have $w_i = x_i \geq x'_i = z_i$. For those $i$, the differences in the two sums are zero, and the weak inequality is justified. For other $i$, we have $w_i = x'_i > x_i = z_i$. In this case, the differences

$$f(w_1,\ldots,w_{i-1},w_i,x_{i+1},\ldots,x_m,\theta) - f(w_1,\ldots,w_{i-1},x_i,x_{i+1},\ldots,x_m,\theta)$$

and

$$f(x'_1,\ldots,x'_{i-1},x'_i,z_{i+1},\ldots,z_m,\theta) - f(x'_1,\ldots,x'_{i-1},z_i,z_{i+1},\ldots,z_m,\theta)$$

involve the same values in the $i$th argument. So Lemma 10.1 implies that $f(\cdot, w_i, \cdot, \theta) - f(\cdot, x_i, \cdot, \theta)$ is weakly increasing. Since $w_k \geq x'_k$ and $x_k \geq z_k$ for all $k$, this gives the weak inequality. ■

Finally, we can return to the basic idea of the proof in the one-dimensional case. Fix $\theta > \theta'$, and assume that $x \in X^*(\theta)$ and $x' \in X^*(\theta')$ but $x \not\geq x'$. The failure of the last inequality implies $x \vee x' > x$ and $x' > x \wedge x'$. We have:

$$\begin{aligned} 0 &\geq f(x \vee x', \theta) - f(x,\theta) && \text{since } x \text{ is optimal at } \theta \\ &> f(x \vee x', \theta') - f(x,\theta') && \text{by Lemma 10.2} \\ &\geq f(x', \theta') - f(x \wedge x',\theta') && \text{by Lemma 10.3} \\ &\geq 0 && \text{since } x' \text{ is optimal at } \theta' \end{aligned}$$

Together, these inequalities imply $0 > 0$, a contradiction.

## 10.3 Applications of Complementarity

### 10.3.1 Short-run vs. Long-run Responses

We will use the regular jargon from intermediate micro: in the long-run, the firm can vary all inputs freely; while in the short-run, some inputs are fixed.

A common intuition is that short-run responses to price changes are smaller than long-run reponses. The idea is that the greater flexibility promotes greater response.

The intuition is not true in general. Consider a firm with technology

$$Y = \{(0, 0, 0), (1, -2, 0), (1, -1, -1)\}.$$

- At prices $p = (2, .7, .8)$, the profit maximizing plan is $(1, -2, 0)$.
- Assume prices change to $p' = (2, 1.1, .8)$, and the second input cannot be increased in the short run.
- Now the plan $(1, -2, 0)$ makes a loss, so the profit maximizing plan is $(0, 0, 0)$
- Thus short-run response is for input 1 to fall by 2
- When all inputs can be adjusted, profit maximizing plan is $(1, -1, -1)$
- Thus long-run response is for input 1 to fall by 1

To recover the intuition, we need inputs to be either complements or substitutes, globally.

We start with the general result. Let $X$ and $Y$ be sublattices, and let

$$x(y, \theta) = \arg\max_{x \in X} g(x, y, \theta)$$

and

$$y(\theta) = \arg\max_{y \in Y} g(x(y, \theta), y, \theta).$$

Interpretation: $x(y, \theta)$ is the short-run optimal choice of $x$ when $y$ is fixed. $y(\theta)$ is the long run optimal level of $y$ (by the method of aggregation).

**Theorem 10.4** (Milgrom-Roberts)**.** *Suppose $g$ has strict increasing differences in all pairs of arguments, that $\theta \geq \theta'$, and that the maximizers described below are uniques for $\theta$ and $\theta'$. Then:*

$$x(y(\theta), \theta) \geq x(y(\theta'), \theta) \geq x(y(\theta'), \theta')$$

*and*

$$x(y(\theta), \theta) \geq x(y(\theta), \theta') \geq x(y(\theta'), \theta').$$

**Proof.** By Topkis, $y$ is monotone. Since $\theta \geq \theta'$, we have $y(\theta) \geq y(\theta')$.

Also by Topkis, $x$ is monotone. The claims follow from that and the previous paragraph. ■

Note that the conclusion is just the statement that long-run responses are larger than short-run repsonses.

We can apply this result to a profit maximizing firm that uses labor ($x_1$) and capital ($x_2$) to produce a single output. Assume that capital is fixed in the short run, and consider a change in wages ($w_1$).

Consider two cases:

1. Capital and labor are complements: $\frac{\partial^2}{\partial x_1 \partial x_2} f(x_1, x_2) \geq 0$. The profits $pf(x_1, x_2) - w_1 x_1 - w_2 x_2$ have strictly increasing differences in every pair from $(x_1, x_2, -w_1)$.

2. Capital and labor are substitutes: $\frac{\partial^2}{\partial x_1 \partial x_2} f(x_1, x_2) \leq 0$. The profits $pf(x_1, x_2) - w_1 x_1 - w_2 x_2$ have strictly increasing differences in every pair from $(x_1, -x_2, -w_1)$.

Thus we have:

**Corollary 10.1** (LeChatelier Principle)**.** *Suppose production is given by $f(x_1, x_2)$, where $f$ has either complements or substitutes. Then if $w_1$ increases, the firm's demand for input 1 decreases, and the decrease will be larger in the long-run than in the short-run.*

### 10.3.2 The Firm as an Incentive System

Consider a principal who employs a single agent to choose a vector $e = (e_1, \ldots, e_n)$ of efforts. The principal's benefit from these efforts is $b \cdot e$ for some weights $b \gg 0$. The agent's cost is quadratic:

$$c(e) = e^\top C e$$

for some positive-definite matrix $C$.

Neither the efforts nor the principal's benefits are contractable. Instead, a contract can only be based on a vector of performance measures: $x_i = e_i + \epsilon_i$, where

$$\epsilon_i \sim \mathcal{N}(0, \sigma_i^2).$$

A contract must be an affine function of these performance measures—the agent is paid a salary $s$ plus bonus $p \cdot x$. (Think of $p_i$ as the *piece rate* for outcome $i$.)

Given such a contract, if the agent chooses $e$, she will receive compensation that is normally distributed with mean $s + pe$ and variance $\sum_i p_i^2 \sigma_i^2$. Assume that her certainty-equivalent payoff in this case is

$$s + \sum_i p_i e_i - \frac{r}{2} \sum_i p_i^2 \sigma_i^2 - c(e).$$

(This could be derived as the certainty equivalent from a Bernoulli utility function with constant absolute risk aversion $r$.)

The principle is risk-neutral, and gets payoff $b \cdot e - s - p \cdot x$. The certainty-equivalent payoff is then $\sum_i b_i e_i - s - \sum_i p_i e_i$.

Since both certainty equivalents are quasi-linear in $s$, any Pareto efficient contract must maximize the sum of the certainty equivalents:

$$TCE(e, p, \sigma^2) = \sum_i b_i e_i - c(e) - \frac{r}{2} \sum_i p_i^2 \sigma_i^2.$$

To be feasible, a contract must satisfy the incentive-comparability constraints. Given a contract, the agent will solve

$$\max_e s + \sum_i p_i e_i - \frac{r}{2} \sum_i p_i^2 \sigma_i^2 - c(e).$$

This payoff is strictly concave in $e$, so a vector $e$ solves the maximization if and only if $p_i = \frac{\partial c}{\partial e_i}(e)$ for all $i$. Call the solution $\hat{e}(p)$ the *effort supply function.*

Since the cost function is quadratic, the effort supply function is defined as the solution to a set of linear equations. This means that $\hat{e}$ itself is linear in $p$.

The principal will design a contract that maximizes total certainty equivalent, subject to the IC constraints. Substitute these into the $TCE$ function to get the indirect payoff function:

$$\begin{aligned} \pi(p, \sigma^2) &= TCE(\hat{e}(p), p, \sigma^2) \\ &= \sum_i b_i \hat{e}_i(p) - c(\hat{e}) - \frac{r}{2} \sum_i p_i^2 \sigma_i^2. \end{aligned}$$

**Proposition 10.1.** *Suppose efforts are substitutes for the agent, in the sense that $\frac{\partial \hat{e}_i}{\partial p_j} \leq 0$ for $i \neq j$. Then $\pi$ is supermodular in $(p, -\sigma^2)$.*

**Proof.** Clearly each $\frac{\partial \pi}{\partial p_i}$ is decreasing in $\sigma_j^2$, strictly so if $i = j$. To calculate the cross-partial with respect to $p_i$ and $p_j$, notice that

$$\begin{aligned}\frac{\partial \pi}{\partial p_j} &= \sum_k \left(b_k - \frac{\partial c}{\partial e_k}\right) \frac{\partial \hat{e}_k}{\partial p_j} - r p_j \sigma_j^2 \\ &= \sum_k (b_k - p_k) \frac{\partial \hat{e}_k}{\partial p_j} - r p_j \sigma_j^2,\end{aligned}$$

where the second equality is the IC constraint. Now for $i \neq j$ we have

$$\begin{aligned}\frac{\partial^2 \pi}{\partial p_i \partial p_j} &= -\frac{\partial \hat{e}_i}{\partial p_j} + \sum_k (b_k - p_k) \frac{\partial^2 \hat{e}_k}{\partial p_i \partial p_j} \\ &= -\frac{\partial \hat{e}_i}{\partial p_j} \\ &\geq 0,\end{aligned}$$

where the second equality is linearity of $\hat{e}$ and the inequality is substitutability of efforts. ■

This result implies that, if efforts are substitutes, then incentives are complements. Moreover, an increase in the precision with which we can measure one dimension of effort will lead to higher-powered incentives for all dimensions.

## Problems

**Exercise 10.1.** Let $f(x, \theta) = h(x - \theta)$, where $h$ is twice-continuously differentiable. Show that $f$ has increasing differences if and only if $h$ is concave.

**Exercise 10.2.** A consumer has utility function $u(x_1, \dots, x_n)$.

1. State the consumer's expenditure minimization problem.

2. Use the method of aggregation and Topkis's theorem to prove that the compensated demand for good 1 is decreasing in the price of good 1.

**Exercise 10.3.** Suppose that there are two inputs to production, technology and the skill level of a single worker who uses the technology. There are two technologies available for production, one that uses computers and another that uses old-fashioned technology. If the firm chooses the computer-driven technology, its production function is $G(s)$, while the production function is $H(s)$ if it chooses the old technology. The total cost of skill $s$ is $c(s)$ on the market.

1. What are sufficient conditions on $G$ and $H$ to guarantee that profit-maximizing firms who use the computer-driven technology buy more skill than firms that use the old-fashioned technology? If we wish the result to hold without any structure on $c(s)$, is there a weaker condition that will do?

   [Hint: Define a function $f(s, \theta)$ where $f(s, \theta^H) = G(s)$ and $f(s, \theta^L) = H(s)$.]

2. Suppose that the cost of computer-driven technology falls smoothly over time, starting at a very high level and eventually reaching zero, and that firms choose their level of technology and skill at the same time. Assume whatever conditions you found in part (a). Do we expect the skill level selected by the firms to change smoothly? [Your answer here can be informal.]

**Exercise 10.4.** This problem gives you a preview of Michael Spense's famous model of labor-market signaling. Suppose a worker's cost of going to school is $c(x, \theta)$, where $x$ is the amount of school and $\theta$ is the worker's ability/productivity.

1. Suppose $c$ is twice differentiable and $\frac{\partial^2}{\partial x \partial \theta} c(x, \theta) \leq 0$. Interpret this condition in words in the context of the model.

2. Now suppose that firms can observe education but not ability, and thus offer wages $w(x)$ that depend only on education. The worker's utility is given by $w(x) - c(x, \theta)$. Is there any wage function that will induce higher ability workers to choose lower levels of education?

3. Now suppose that $\frac{\partial^2}{\partial x \partial \theta} c(x, \theta) < 0$. Based on your answers above, what is a sufficient condition on $w(x)$ such that two workers of different abilities choose different levels of education?

4. Conclude from this that even is education is unproductive, firms may be willing to pay higher wages for higher levels of education.